Collaboration-Based Learning Environments Using Augmented Reality

by

Sebastian Gil Parga

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Authored by: Sebastian Gil Parga
School of Future Environments, Department of Creative Technologies
November 30, 2025

Certified by: Stefan Marks
Associate Professor, Thesis Supervisor

Certified by: Jairo Gutierrez
Professor, Thesis Supervisor

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ABSTRACT

This research project is an exploration of the use of Augmented Reality (AR) for Collaborative Learning (CL). Current literature shows a growing interest in the use of interactive technologies for teaching and the overall impact of Technology Enhanced Learning (TEL) projects experimenting with the implementation of AR. This technology offers interesting visualisation tools to merge digital information with the physical context of the user, creating unique interactive experiences and boosting engagement for the students, but it had been tested mostly on single-user settings.

The proposal of this thesis is that the unique visualisation capabilities of AR and the connectivity offered by the mobile hardware often used for its deployment can provide an advantage for effective CL. For this purpose, *WorkshopAR* was developed following a user-centred design and implemented within the context of a project-based course in construction design. This application offers the students different tools to help them structure and manage their collaborative work.

The Industry Project course was used as a case study to observe the impact that the introduction of *WorkshopAR* produced in the behaviour of the students during their classroom activities. The main objective was to promote positive changes through the application, helping the students develop and apply important skills for effective collaboration and for a successful development of their project. The risks and uncertainties of such a strong intervention with technology in social, collaborative activities were also taken into consideration, and the observation process also identified negative or undesirable changes in behaviour that could be acknowledged through the iterative development process used.

The case study also offered an opportunity to document the design and implementation of the software. The development process integrated the functional requirements of the tool with the learning goals of the course and tied the iterative development with what was observed in the classroom. This allowed a flexible and adaptable design that boosted the interesting approaches that students and teaching staff were proposing while also helped in redesign those elements that were promoting problematic behaviours.

The results obtained showed that students benefited the most with a strong sense of structure when initially establishing relationships with the group, and appreciated more flexible activities once the work dynamics were clearly defined and producing results. The visual and interactive elements proposed by *WorkshopAR* helped the students understand

and be aware of the need of those structures, promoting discussions and guiding work using more impactful collaborative approaches, but failed to integrated in a more substantial way in the workflows that students had already created using a variety of other tools that were better integrated into each other. *WorkshopAR* was able to pivot when aspects of its design were promoting behaviours that were not in line with the learning goals of the course or the research, and several opportunities in the creation of collaborative spaces and communication mediated by AR can be worked upon in future research.

Thesis supervisor: Stefan Marks
Title: Associate Professor

Thesis supervisor: Jairo Gutierrez
Title: Professor

Acknowledgments

Write your acknowledgments here.

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List of Acronyms

VR Virtual Reality

AR Augmented Reality

TEL Technology Enhanced Learning

MR Mixed Reality

CSCW Computer Software for Cooperative Work

CL Collaborative Learning

CoP Community of Practice

VLE Virtual Learning Environment

LMS Learning Management System

BRT Bloom's Revised Taxonomy

SLR Systematic Literature Review

CR Critical Realism

AUT Auckland University of Technology

AE Architectural Engineer

CE Construction Engineer

CM Construction Manager

QS Quantity Surveyor

BYOD Bring Your Own Device

Chapter 1

Introduction

In the Netflix animated series Cyberpunk: Edgerunners (Imaishi et al., 2022), the main character attends courses at a prestigious academy that uses advanced technology to boost education and performance. The students gather in a room with Virtual Reality (VR) headsets and connect to the lesson, which is taught by a holographic (and probably powered by artificial intelligence) tutor. The episode does not show the lesson itself (the entire system fails catastrophically when attacked by malware), but the brief scene highlights the vision many people have of a futuristic classroom, one guided by advanced technology and automatization.

Forsler et al. (2024) propose the term "post-digital classroom" to identify the characteristics and trends in education present in a world beyond the adoption of digital technologies. The purpose of this classroom is not anymore to introduce new technologies to students, but to implement them as an essential factor of the learning process. The post-digital classroom is interconnected, social and global. In this scenario, learning goes beyond the physical classroom because it is based on creating relationships between concepts and developing valuable skills rather than acquiring knowledge. It is also a classroom where information is detached from the physical learning institute, and access to facts and sources is, ideally, immediate, ubiquitous and democratic.

The classroom has also become hybrid. There is little distinction between network-based lessons and the face-to-face spaces associated with schools and universities (Goodyear et al., 2004). Hybrid and blended approaches blur the distinctions between learning in an online setting and learning in a classroom. In both instances, information is at hand in the network, students can easily connect to peers and mentors, distance and time constrains are less meaningful and knowledge is presented in a multimedia format that needs a technological backbone to be created and shared (Rivoltella, 2008; Shank, 2005).

This thesis explores the use of technology in such post-digital classrooms, specifically, AR within the context of mobile computing. Moreover, this research will analyse the creation of effective learning spaces, those places where students can build, personalise and control all the activities they engage with (Gourlay & Oliver, 2016). Learning spaces are interesting tools at the disposal of students because these are spaces that can exist outside the norm of a typical classroom, they are more fluid, less structured and can easily connect to other similar spaces (Nørgård & Hilli, 2022). The proposal of this research is that AR can be used to boost the way students build and interact with their own learning spaces and can also

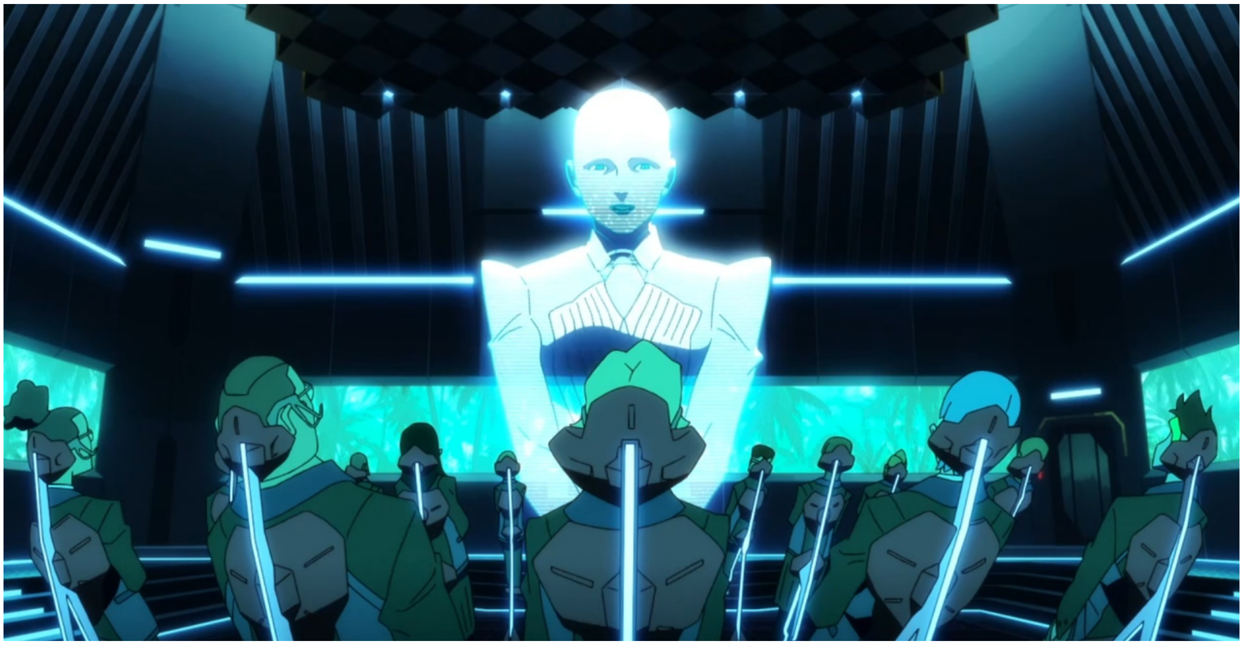

Figure 1.1: Virtual Classroom in Cyberpunk: Edgerunners

offer educators another option to be integrated in the construction of a lesson that capitalises the characteristics of the aforementioned post-digital classrooms.

Within the field of TEL, AR and related technologies were chosen as a focus of interest because of their intrinsic relationship with space. The initial example scene of Cyberpunk: Edgerunners serves to illustrate the collective idea of what could be achieved in the future with technologies like VR and AR: the possibility of creating an entire new space, suited for the needs of the classroom or the students. It is a space that is shared, in which information and interactions flow between users and that can be controlled at will by students and teachers. In reality, the technology is not quite there yet, especially in terms of fidelity and immersion (Checa & Bustillo, 2020; Hamad & Jia, 2022). What can be explored right now is the creation of interesting synthetic environments that facilitate or allow experiences that are not possible in the common classroom. This process is limited more by the creativity and expertise of the developer rather than by the technology, and therefore can be explored easier and in more detail.

AR, compared to VR, has a more direct relationship with the *real*, physical space and with the context of the user. Instead of an isolated immersion, AR creates a connection with the environment based on image detection, data and visualizations. This natural expression of the technology is where the value of AR can be explored in relation to the creation of innovative learning spaces. The technology not only has interesting capabilities for visualization and interaction, it also allows an easier implementation of shared, social components that can be used to build activities based on cooperative simulations, sharing knowledge and the creation of communities, all crucial elements for the construction of effective learning spaces (Bligh & Crook, 2017).

The research is positioned then in the conjunction of these three fields: Learning spaces, TEL and AR. The guiding goal is to understand how students are using technology to learn, and how AR can be used to boost that process. AR shines at creating immersive experiences that can be shared with others and that consider the physical context of the user. I want to use this collaborative space as an educational tool that takes advantage of the inherent social aspects of learning, building and sharing knowledge.

No modern classroom is free of technology, the simple act of taking notes on a paper already implies a set of technological developments needed to put knowledge into written form. Networking and information technologies are just another set of tools at the disposition of any student. A personal learning space can incorporate different software tools, many of which are now ubiquitous and almost indispensable in any learning activity: text processors, multimedia editing tools, web browsers and document readers, plus any specialised tool that the learning subject requires. But that learning space is also built around more elements, some of them equally indispensable for some students. People need a space to work, access to different amenities, access to textbooks, information and teachers. Some may need a quite space to concentrate, or, on the contrary, may need access to music, to snacks or to a separate place to periodically take a break. The construction of a learning space involves many different elements, some more technological than others, and its effectiveness is based more on the integration of those elements than anything else (Lu & Churchill, 2013; Reinius et al., 2021). This research aims at identifying how to properly integrate AR in the construction of a learning space, which is already composed by a complex web of many other objects, tools and circumstances.

As a researcher. I have explored in the past the way different technologies can be used as part of those learning spaces, either with tools for online learning scenarios or using VR to create a digital space to explore multimedia content (Gil Parga, 2015). This current project can be seen as a way to expand on the work done before, identifying ways to adapt a different technology, to apply lessons learned and to explore new avenues of research. In particular, it is an opportunity to expand from personalized learning environments into social learning spaces, and explore the role of the community in teaching and learning.

It was clear in the work done previously and through some notable literature like Richards (2023) and Long and Ehrmann (2005) that the construction of learning spaces is always mediated by the community around us, even when creating a space as personal as possible. Even nowadays that we live in a post-pandemic world where we seek remote, personalised and secure ways of learning, it has become clear that the social aspect of learning is a crucial element in the equation (Baber, 2021; De Felice et al., 2023). The aim of this research is to delve deeper into the social elements that influence the construction of a learning space, how the students negotiate the process of learning with others and the benefits of collaboration, as well as the challenges it conveys.

This is also a technological enquiry. Having explored VR in the past, it was clear that this family of technologies offer interesting advantages by using complex visuals, new forms of presenting information and more natural interactions suited for the exploration and experimentation with 3D spaces. These technologies also present curious challenges, especially in the usability front. For instance, the public has not yet developed a strong relation or consistent literacy with the technology (Özgen et al., 2021), which creates accessibility problems. And those technologies that leverage common and accessible platforms, like mobile devices, risk carrying with them problems associated with the excessive and pervasive presence of such technologies (Forsler & Guyard, 2025; Petrucco, 2021).

The following introductory chapter will expand this brief exploration of the context and motivation for the research proposed, outlining the main research goal and the reasoning behind the formulation of the selected research questions. There will be a brief exposition of the methodology followed to build possible answers, the significance of this research and why the answers to the questions proposed can provide important insights about the way we use technology to learn and how we design technological tools for the classroom. It will be followed by a summary of the results obtained and the contributions achieved, as well as the scope and limitations of the work done. The chapter will conclude with a description of the structure of this thesis document.

1.1 Context

AR technologies are positioned somewhere in the middle of the reality-simulation spectrum proposed by Milgram et al. (1995), a model for display technologies used to show digital or physical data in virtual or synthetic realities. The main characteristic of AR in this spectrum is the incorporation of the real world as part of the experience, making use of different image recognition and composition techniques to identify features in the physical space and overlay them with digital information.

Schmalstieg (2016) adds another layer of analysis to the description of AR-like technologies by using the ubiquitous computing spectrum proposed by Weiser (1991). The model positions different implementations of AR depending on the use of more static or more mobile deployment platforms. In one end of the spectrum are applications that can only be used in singular static terminals, while the other side contemplate platform-independent mobile technologies and ubiquitous systems. Figure 1.2 shows examples of this distribution as proposed by the authors.

In this bi-directional spectrum we can find mobile AR as a family of technologies that offer an in-between mixture of virtual and physical elements in which location, place and context are additional inputs used for interaction with the synthetic reality. Any technology in the spectrum will offer a variety of characteristics that align with one or more descriptions at a time. Mixed Reality (MR), for example, are seen as an extension of both VR and AR, offering seamless interaction and connectivity between the real and the simulated world. It is a blurry line that often must be analysed in a case-by-case approach. No matter what label is used to describe a technology, some agreed upon characteristics must be present for it to be categorized as AR, as stated by Azuma (1997):

- A combination of real space and digital information.
- Interaction with the simulation in real time.
- The simulation is done in a 3D space.

With these descriptors it is possible to position the used of the technology in different fields, independent of the labels used to identify it. Based on the themes introduced in the opening statement of the chapter, the goal is to analyse AR as CSCW. For this task, a useful analytical tool is the classification of CSCW technologies proposed by Rodden (1991)

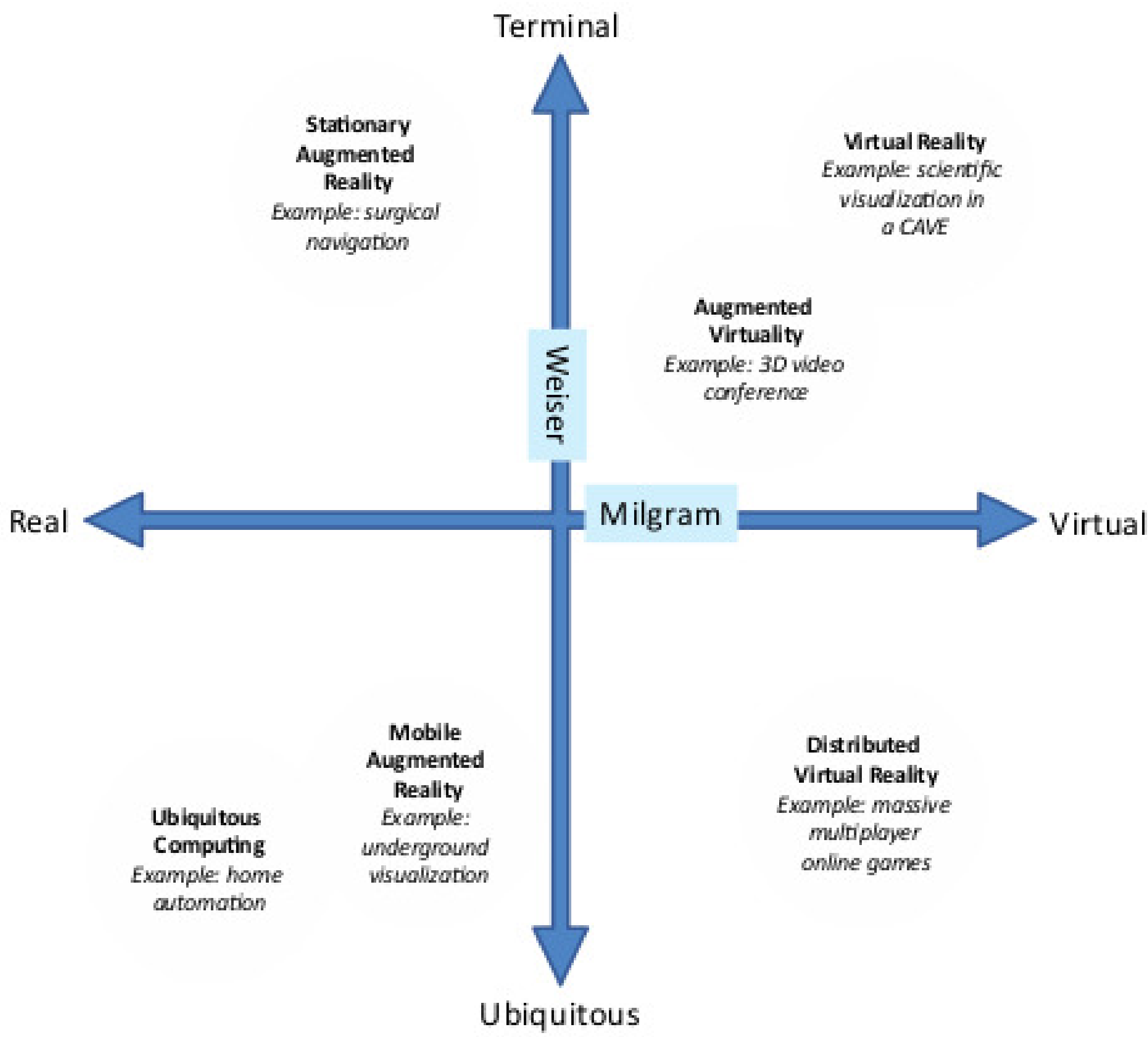

Figure 1.2: AR in the Milgram-Weiser Spectrum (Schmalstieg, 2016).

in terms of the point in time and space where the collaboration is taking place. Figure 1.3 shows examples of different systems that can be found in each quadrant of the classification, which indicate whether the activity is synchronous or asynchronous, remote or co-located.

Complementarily, Kiyokawa et al. (2002) offer two important elements present in a collaborative activity: communication and tasks spaces. In the communication space, people exchange information and can sustain formal and informal conversations. In the task space, the actual cooperative work is done. Those spaces can be unified or separated, and they can be physical, virtual or in a mixed state. Independent of the nature of each individual space, both need to be present for collaboration to succeed.

Based on these classification models, it is possible to define the main field of interest of this research as the use of AR in a mixed task space for face-to-face collaborative learning, which implies synchronous activities in a co-located space. In this setting it should be possible to use the capabilities of AR for visualization and interaction with the physical space to

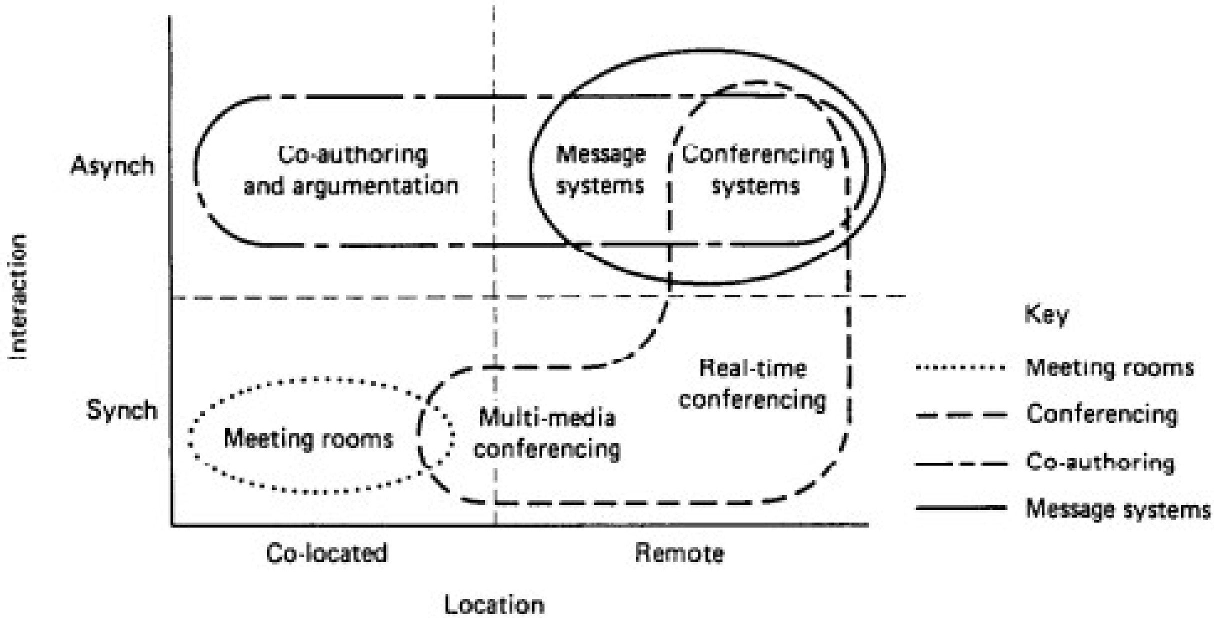

Figure 1.3: Classification of CSCW Systems by Rodden (1991).

augment the collaboration process in terms of:

- Activities related to the communication space, the flow of information and the creation of knowledge in the group.
- The use of the physical space as an asset in the collaborative process.
- Management of the collaborative process itself and the structures that can be created to facilitate effective teamwork and cooperation.
- Guidelines to acquire and refine skills related to teamwork and collaboration.

The main proposal is that the unique characteristics that AR provides in the field of MR are well suited to experiences in co-located collaboration, and that a learning design that is using a collaborative approach can benefit from an AR tool mediating in the communication and tasks spaces of the activity.

The interest in collaborative learning derives from the analysis of the use of AR in education and the tendencies identified in the available literature. It was possible to note that AR is often used as a tool to explore concepts and practices in hands-on approaches that apply previous knowledge in different situations and contexts. Unfortunately, the technology is seldom used in cooperative scenarios, despite the tendency to use mobile platforms for deployment (Gil Parga et al., 2023).

Collaborative learning is a subject rich in discussion. The conversation can focus on the benefits of groupwork in the classroom like promoting different points of view, facilitating complex understanding of a topic and creating the space for an integral development of the student (K. A. Smith et al., 1992; Van Wyk & Haffejee, 2017). On the other hand, the challenges associated with collaborative work are also amply explored (Baloche & Brody, 2017; Buchs et al., 2017; Popov et al., 2012), and there is plenty discussion around issues related to classroom management, inclusivity, effective assessment and good practices.

Many studies have identified and compiled the advantages of using AR and MR as learning tools. For instance, Penn and Ramnarain (2023) offer an analysis of the educational foundations used to design learning interventions. They conclude that most projects converged into constructivist foundations and embodiment theory, opting to create spaces for students to experience and explore the learning subject, with an emphasis on immersion, practice and hands-on interactions. Despite these commonalities, the authors also remarked that most reports do not mention explicitly or do not rationalize the theoretical background of the intervention. They also note that a lot of researchers still hold the idea that introducing a new technology in the classroom is enough to reap its benefits, without caring to build or follow a proper integration plan.

This issue acquires even more relevance now that the mediation of different technologies in the collaboration process has grown due to the Covid-19 pandemic, which forced a quick and often poorly planned implementation of remote learning (Drueke et al., 2021; Nuere et al., 2021). Over time, this prompted the re-design and re-thinking of how to use technology to support the sudden disruption in learning methodologies and approaches (Adi Badiozaman et al., 2020; Fischer et al., 2020). The forced isolation also highlighted the importance of social spaces in the constructions of learning environments, and the need to provide those social interactions even in online spaces (McInnerney & Roberts, 2004; Walas-Trębacz et al., 2025).

The post-pandemic classroom is dealing now with the need to balance the advantages of online spaces with the problems associated with socialization solely mediated by technology, which tends to be asynchronous and impersonal. There is value in the physical space for the learning process (Paechter & Maier, 2010; Petchamé et al., 2021; Thai et al., 2020), and a technology that is able to connect the physical and the digital is worthy of a deeper exploration to identify proper approaches and successful use cases.

The idea is not to oversimplify the presence of digital technologies in the classroom, nor follow a misguided tendency of selling all the benefits of a technology without acknowledging its problems. If the proposal is to identify the effects of a collaborative learning design mediated by AR, it is important to also identify all the problems that derive from this approach, either because of the issues that naturally come from teamworking scenarios in the classroom or the negative behaviours that the excessive use of digital technologies promotes, especially in the specific environment of mobile platforms.

With that context just detailed, it is possible to summarize the goal of this research as: The construction of an AR tool that serves as aid in a collaborative learning setting by helping students to communicate and develop teamworking skills. This can be achieved by means of interactivity, visualization, immersion and augmentation of the physical space.

1.2 Research Structure

The goal of the research can be justified by an abundance of literature showing the elements of AR used in the past to offer distinctive value for learning, and the open opportunities to explore and understand better the development of multi-user AR experiences for collaboration.

The nature of the proposal implies that the research has two main fronts:

- In the *technological* front, the goal is to build a software solution that considers the development challenges of an AR tool in a multi-user and multi-platform environment, and the specific requirements derived from the learning goals proposed in the educational front.
- In the *educational* front, the goal is to identify, describe and develop a case study based on collaborative learning, and to analyse the behaviours that students develop when learning in a collaborative environment mediated by technology, comparing the use of AR with other tools or approaches.

1.2.1 Research Questions

The guiding research questions of this project are aimed at obtaining information of what constitutes an effective collaborative learning design using AR. The answers can be stated in terms of learning goals, interaction design and integration plans with the classroom. The questions also seek information about how AR is creating an effect in the behaviour of the students while using the tool for learning. These behaviours can be related to communication, socialization, planning and engagement with the learning material. The main research question will be stated as follows:

> MQ: How does the behaviour of a group of students changes when using an AR tool designed to aid in the achievement of a learning goal framed in a collaborative learning scenario?

To guide the implementation of the tool through a strong design, a lot of information can be extracted from current literature and from previous projects using similar approaches. Information gathered from real and incremental usage off the system also offers valuable insights in terms of usability, viability and end-user perception. With this in mind, an iterative development process is proposed to facilitate the exploration of ideas regarding interaction design and a proper fit to the learning scenario where the tool is going to be used.

The implementation of the tool alone is insufficient to answer the main question. For this purpose, a case study research methodology is proposed with the objective of identifying and describing a strong context in which collaboration is used for learning and in which the implementation of the AR tool can be contrasted and compared to other approaches. In this context it is also easier to observe more natural behaviours from the students and the observations will show results closer to how the tool could be used in a real deployment.

It is very important to carefully select a case study that not only offers unique circumstances valuable enough to be analysed in detail but also has the proper learning context to explore the themes of interest of this research project. Although the specific learning field of a prospective case study is not that important, it needs certain characteristics, independently if it is developed through a single activity, a section of a course or through the whole course. The case study must incorporate all the following elements:

- A clear learning goal, either as a whole or as a step to achieve a more complex goal. Assessment activities alone are not enough because the research objectives require the observation of collaborative learning, not the development of a single teamwork activity.
- A collaborative approach to the learning activity. Students must be expected to work in groups to achieve the task, or at least, to have access to a space for collaboration in the development of the activity, even if it is individual in nature.

Ideally, the case study also contemplates the process of learning to collaborate, either as its own learning goal or willing to implement such an objective as a secondary goal. This is not essential, but the current proposal borrows a lot of elements from learning designs about developing collaborative skills. If the selected case study already seeks learning objectives in this venue, then the proposed intervention could offer more value.

The selected case study offers an excellent opportunity, not only to experiment with the proposed thematic of collaboration and technology mediated learning, but because it is also a unique and interesting context in terms of learning design and the opportunities for coordinated teamwork, project development and innovative use of technology.

The Industry Project course of the Built Environments program is a complex scenario that provides students with opportunities to develop industry-level skills through a project-based approach. It presents the opportunity to intervene in the whole course and to use the software solution proposed across different stages of development of the course and different learning objectives, an ideal scenario for varied observations and a rich space for experimentation and discussion.

A set of secondary question can be described that will help to focus the data gathering stages of the research, especially those related to the observations and activities in the context of the case study:

> SQ1: What differences in behaviour can be observed between students collaborating using the AR tool and those that are not?
>
> SQ2: What differences the students report in their perception of the activities done in the course between those using the AR tool and those who are not?
>
> SQ3: How does the collaboration and communication process changes between students using the AR tool and those that are not?
>
> SQ4: What differences can be observed in the end results of the course between all the different collaborative approaches observed, especially of those that incorporate the AR tool?

The focus of the observation process should be in identifying behaviours and processes rather than only differences in the end results, that is, the final assessment of the course project. Relying solely on possible improvements to grades or assessments can be a valid but flawed tool to understand the effects of a technology in the classroom (Dalziel, 1998; Katz et al., 2017). Additionally, the selected case study proposes a complex course to analyse, where several and diverse factors influence the work of the students. Taking this into consideration is why a positive or negative result does not tell the full story and why is more important to observe the whole learning process, from planning to development, documenting the behaviours developed to fulfil tasks and reach goals. The objective is to identify if the AR tool proposed is an influence, either positive or negative, and if it provides a more meaningful impact to the learning process as a whole and not only to the end assessments.

Assuming there are actual substantial differences between the students using the AR tool and those that are not, then it is also important to ask:

> SQ5: What elements of a technological intervention using AR create the most impact in changing the behaviours, perceptions or results of the students?
>
> SQ6: What elements of a multi-user AR tool creates the most impact in changing the behaviours, perceptions or results of students collaborating and communicating in a learning scenario?

These last questions are meant to provide an evaluation to the decisions taken in the development of the AR tool, especially the multi-user interactions, and how the tool was integrated to the learning design of the course. It also offers the possibility to highlight elements of the tool that could be polished or reframed in future work.

1.2.2 Scope and Limitations

Using a case study methodology, it is possible to approach the research questions through a qualitative lens, which is better suited for understanding the information gathered about behaviours, relationships developed between students and their subjective perceptions of the course, the activities developed, and the results obtained. All those elements would be hard or contra productive to try and quantify. In this light, it is important to acknowledge the limitations in scope and reach derived by the selected methodologies.

In relation to the software development aspect, it is important to establish the scope of the product as a prototype that explores the ideas of collaboration and multi-user AR. A thorough analysis of the design and architecture of the software was conducted to describe different possible approaches and solutions suited to the context and the needs of the tool, keeping an analytical mindset for all the ramifications and consequences derived from the implementation of a particular idea or design.

In terms of the concrete implementation of the software, the work was undertaken by a sole developer and limited in time and resources related to access to the case study. Most of the design and architectural proposals were tested via mock-ups or wireframe prototypes (Coleman & Goodwin, 2017; Sutipitakwong & Jamsri, 2020) in pseudo-structured test scenarios, aiming primarily for quick feedback and impressions rather than structured data. This allowed constant iteration and exploration of different ideas. The final implementation of the software prioritized showing the most critical use cases and the most relevant interactions with the tool, most of them tailored for specific observation sessions.

This approach had an important influence in the structure of the observation process and the nature of the data gathered, fomenting a more guided use of the tool and a more structured response from the participants rather than the free observation initially intended.

This was an acceptable concession made to facilitate the development process and to give the participants early and consistent access to the tool, albeit in a more incomplete state. This also minimized the disruption to academic work while maximizing the opportunities for meaningful observation in the available time.

There is also an inherent problem in the development of a case study in relation to the applicability of the results, generalization and subjectivity. The Industry Project course was chosen because it fulfilled all the characteristics needed for the research, but convenience aside, the course also provides a unique opportunity to analyse a collaborative learning design in action, with scenarios that could be extrapolated to other similar contexts, but also a unique take on the design of project-based learning and assessment, as well as a complex context for implementation and challenges that had to be tackled, all elements worth of a detailed analysis.

Nonetheless, it is important to identify the limitations of using one sole case study, which can be summarized as issues with the transferability and applicability of the data gathered. Other factors that impact heavily the nature of the data are the overall seniority of the students, which were all close to finish their studies and the lack of a concrete study plan of the course, which focused more on the development of the project as its main teaching instrument, offering just a relatively vague and general set of learning goals.

Chapter 3 will describe in more detail the scope and limitations of the case study, as well as the measures taken to minimize or acknowledge the flaws and restrictions of the chosen framework. For now, these issues can be summarized the main limitations related to focusing the software development into a gradual an iterative progression of a prototype and the validity of the case study in terms of the characteristics needed for the research and the unique value it offers for analysis and discussion.

1.3 Significance and Contributions

The technological environment in which we found ourselves right now evolves rapidly and at large steps. Despite critics of stagnation, the classroom evolves and adapts in response and constantly tries to understand how the available technologies can improve the way we learn (Watters, 2021). It is not always a positive relationship, it is often marred by overselling products and forcing a technological implementation without a plan and a proper consideration of the context and needs of the classroom (Cochrane, 2012; Odunaike et al., 2013). Currently, it is also a field that experiences a lot of social scrutiny as we react to the idea that we have become somewhat addicted to digital content and to the devices used to consume it (Gui & Büchi, 2021; Liu et al., 2022).In a context in which we are banning the use of smartphones in the classrooms and obtaining somewhat positive results because of it (Rahali et al., 2024), why, then, rely on a proposal that fundamentally uses that technology?

As Murray and Perez (2014) stablish, exposure to technology does not equal comprehension. In an age in which almost all aspects of life are mediate by digital activities, it is important to create a positive, critical and informed relationship with the technology that we have to use in our daily lives, and that relationship does not develops naturally (Margaryan et al., 2011; Selwyn, 2009). This is a task that can and should be executed in the classroom at different levels of schooling, in different contexts and at appropriate age and field depths.

The classroom is the ideal scenario to create and nourish this relationship in a structured and positive way, yet this is a factor that is seen as lacking in education institutes, especially in middle and high school (Burton et al., 2015; Jeffrey et al., 2011).

This project is an exploration of that philosophy, and an experimentation with a technological design that uses mobile devices in a healthy approach that promotes the development of skills needed today. Although the capacities of AR as a learning tool are evident, this endeavour needs to follow a due process. It is important to properly understand how to use the technology and what elements of the design and implementation process need to be highlighted and worked on to create that intended positive impact. The technology does offer a big opportunity to enhance and reframe the way we communicate and work together, but it also has proven to impose certain challenges that have to be analysed, understood and dealt with.

This document will describe the proposal, design, implementation, observation and analysis activities done in the development of an AR tool for collaborative learning and the test deployment conducted focused on developing skills related to teamwork and effective communication. The development of this research was carried in two interlocked steps. In one hand is the design and implementation of the software *WorkshopAR*, an AR tool used as a companion in the face-to-face work that was part of the activities proposed for the Industry Project course. Specifically, the software was used during the weekly group sessions of work that the students held for the development of three different assessment milestones in the course-long project.

The development of *WorkshopAR* consisted in solving two main issues, both related to the technological front of the research. First, there was a design challenge that required identifying the most common interactions performed in an AR environment for cooperative work. Different use cases were identified that were unique for this context and that posed an interesting design challenge, like sharing information, sharing and coordinating resources, managing the status of the team and coordinating the development of a task. A user-centred interaction design process was used to outlined solutions for the identified use cases based on important restrictions and particularities of the problem.

This stage was accomplished through a detailed exercise of user-centred design, a process meant to identify ideas and decisions born from the direct interactions of the users with prototypes and mock-ups, a structured process for gathering feedback, integrating the user in the design loop and refining ideas in an iterative process. The results obtained highlighted interesting issues and ideas related to multi-platform interactions, the expectation of users when presented with familiar and unfamiliar controls, complications related to interactions in a 3D space and the integration of the tool in the overall workflow of the students.

The second focus in the development of *WorkshopAR* was the technical challenge of creating a multi-user, shared AR space. Several commercially available frameworks and APIs were explored, such as the AR Foundation suit provided by the Unity game engine and the Lightship API provided by Niantic. It was possible to extract valuable insights about the current state of the development of AR tools, especially in the field of education, and it was possible to describe the lessons learned while integrating the development process with the educational goals of the course. An effort was made to relate the results obtained to those reported in similar studies by identifying common and divergent results and issues, especially those related to finding the technology distracting and the technical development of the tool too detached from the role of the educator.

Parallel to the construction of *WorkshopAR* was the development of the case study and the design of the intervention of the Industry Project course. In the design phase, the course was analysed based on the written material available and on interviews with the teaching staff. This information was used to guide the development of *WorkshopAR* and to identify the instances in the progression of the course in which the intervention could create more impact without disrupting too much the academic activities of the students. It was also an opportunity to detail and analyse the learning design of the course and to understand the context in which the observations were going to take place.

During the observation phase it was possible to obtain detailed observations of the work done by the students with and without the AR tool, about the development of the project through the conception, design and implementation phases, to adapt elements of the tool to better suit the circumstances of the course and to observe the role of the teaching staff in the development of the course and as part of the usage of the tool. This helped to create a rich record to report in the case study and plenty points of analysis and discussion.

The analysis phase used the recorded information from the observation stage plus all the material that could be recollected from the execution of the course. This material included the written communication given to the students via the official channels of the course, the workshops executed during the semester as material to help in the development of the project and the final products created by the students, such as presentations, reports and designs. The observations were also complemented by a set of semi-structured interviews to the students and the teaching staff. The interviews focused on gathering better records of issues related to the personal perceptions about the use of technology and the collaborative activities of the course.

From the observations obtained and the analysis of the results it was possible to formulate some possible and satisfactory answers to the research questions. The data points to interesting behaviours promoted by the intervention, such as students creating a more structured work process when the tool prompted or encourage them to do it. It was also observed that this enforced structure helped the students with the organization of activities, prompting good practices for communication and the organization of ideas and plans.

But the structure also prompted some unintended behaviours, like a tendency to create a hierarchical organization of the group around the student managing the tool, which was not in tune with the objectives of the course. The app also forced the students to be aware of their phones more than they were planning or willing to do, highlighting a current trend among students of regulating their phone consumption. It was also noticed that the tool was most useful for teams struggling with their organization and offered little value for those already with a clear work plan.

Data and observations also provided a lot of insights related to the design of collaborative interactions in virtual 3D environments, to the relationships of students with their phones and how students use different resources, properly or not, to sort out the difficulties of a learning project. It was evident that the social aspect of the collaborative work resonated the most with the students and with the general satisfaction they reported with the activity, giving more importance to the tools created for them to communicate, express ideas and personalize their experience. The case study also opened an unexpected opportunity to analyse the role of the teaching team in the use of the tool, offering a clear path for future work.

1.4 Document Structure

The purpose of this document is to provide an organized recount of the development of this research work, the theoretical background that supported the proposals given and the methodology followed to gather data, analyse it and build a set of answers for the research questions.

This introduction chapter was meant as a summary of all these elements, highlighting the personal motivations behind the chosen research field and the general reasoning that accompanied the design process, the hypothesized and desired results, as well as the methodological pathway taken. Chapter 2 of this document will describe in detail the literature explored that helped in shaping the research questions and that provided the theoretical basis for the educational methods used as anchors for the software implementation.

Chapter 3 details the methodological framework of the research. It briefly explores the research paradigms picked to guide this project and explains the reasoning for choosing them and their relevance to the learning and technical fields this project is based on. The chapter will also offer a complete explanation of the educational front, detailing all the instruments used in the development of the case study. In the technical front, it details the software development methodology followed, especially the reasoning and procedures used for the interaction design process and the methods used to ensure usability, flexibility, expandability and a general user-centred approach.

The description and analysis of the results have been divided in three chapters. The technological front has been described in Chapter 4, where a detailed explanation of the requirement analysis can be found, including use case scenarios and quality objectives. The chapter also details the most relevant elements of the architectural solutions proposed for the tool. Chapter 5 continues with the description of *WorkshopAR*, now focused on the implementation process followed. The chapter describes the design phase to the development cycles followed to build the software, the testing strategies used to ensure validity and the feedback received by the users during testing and the observations in the case study.

The case study itself is described in Chapter 6, explaining the context of the Industry Project course, the teaching objectives guiding its design and the points of interest for the intervention using *WorkshopAR*. The chapter will also provide a narrative description of the observations taken over 20 weeks of development of the course, and a preliminary analysis of the data based on the secondary material extracted, like the written documentation, the work produced by the students and the information gathered from the interviews.

The final analysis is presented in Chapter 7 along with the necessary conclusions gathered from the data and the process described in the previous chapters. The chapter will detail what elements of the development of *WorkshopAR* and the case study support the proposal of AR as a tool to promote positive learning behaviours and effective collaboration, what important technological hurdles were identified and what surprises were found during the development of the project. The chapter will also offer and outline for future work to address the most outstanding issues derived from the research limitations and the problems encountered during the development of the project.

Chapter 2

Literature Review

This project is centred around the concept of CL: the process of acquiring or creating knowledge as part of a group, and the dynamics that emerge when a team needs to manage participation, the overall success of an activity and the learning objectives that need to be fulfilled (Laal & Ghodsi, 2012). It is also possible to analyse the concept as learning to collaborate, to acquire the skills needed to be an effective team member.

The purpose of this literature review is to find the benefits of adopting a CL approach in a learning design and the challenges that any educator needs to consider when implementing a strong focus on collaboration. It is also important to understand why collaboration and teamwork are such important skills to develop, why is it so valued in most professional environments and why is there a general feeling that educative institutions are failing in teaching those skills.

This research will approach the field of CL using the lens of TEL: the use of technologies to support processes or management of teaching and learning (Passey, 2019). Understandably, this is still too wide of a subject, so, a particular focus is set on AR technologies. The objective will be to understand how AR can be used in the classroom, and the particularities that it offers in terms of benefits, challenges, common implementations and relevant success stories.

AR can be a valuable collaborative learning tool based on the unique abilities it possesses. Specifically, the technology excels at incorporating strong visuals into the surroundings of the user(s), which provides an opportunity to design enticing experiences for both learning and collaboration. For this purpose, it is important to identify what collaborative experiences have been created in the past with AR, what learning goals align with the technology and what challenges in design and development can be expected.

This chapter explores these questions and themes across three sections. The first section tackles the topic of collaboration, how it is defined and measured, different implementations in the classroom, perceived benefits, common challenges and lessons learned from previous projects. The second section presents a systematic literature review aimed at identifying the use of AR as a tool for education, how the technology has evolved, which elements of the technology have derived in success stories, and which have become challenges. The final section will combine both subjects and explore the literature for previous experiences using collaborative AR, both in education and in other fields.

2.1 Collaborative Learning

2.1.1 Collaboration as a Transversal Skill

The subject of CL is wide and can be tackled through different angles. For instance, we can talk about collaboration as an isolated term, and relate it to similar concepts like teamwork, cooperation and intrapersonal skills. In need of a definition, the approach taken by Andriessen and Baker (2020) is very complete. The authors state that "(…) collaboration means developing, in an equal and mutually respectful way, a shared view of a situation". Is a simple definition that implies two important things. First, collaborators find themselves in equal terms, there is a sense of equality and mutual correspondence. Authoritative or hierarchical relations hardly produce collaborative settings. Second, collaboration goes beyond cooperation, beyond working together to fulfil a goal. Collaboration implies building, through time, a shared view. The group has a shared set of goals, purpose and values. The classical definition of collaboration proposed by Roschelle and Teasley (1995) also complements this idea by stating that cooperation involves the division of labour among participants, of individual responsibility, while collaboration is a coordinated effort to solve the problem. Echoing this distinction between collaboration and similar ideas like cooperation or teamwork, Salmons (2019) states that collaboration is a process that needs to constantly be managed, while teamwork is a more *natural* or *improvised* way of working together when needed.

Andriessen and Baker (2020) later expand all these ideas by stating how collaboration is built through time in three different dimensions. In a personal sense, we collaborate because we gain something from it, or in the worst-case scenario, because we need to do it. Our goals align with those of others, creating a second inter-personal dimension, in which we also care about the relationship we create with others and the social advantages we obtain from the collaboration. Finally, with strong inter-personal interactions, we create a group dynamic, in which decisions are taken beyond the individual level and align with the necessities of the group. This evolution can be fractal, which means that after a strong group has been created, it can be developed in a fourth dimension, an inter-group relationship in which several tight-knitted groups collaborate in a shared purpose within a bigger system.

Now is possible to build an initial definition to be use as conceptual backbone. Collaboration will be understood as: the process in which a group of people build a set of values, shared views and coordinated efforts to fulfil a purpose or reach a goal. This definition considers the most important aspects highlighted previously by other authors: collaboration goes beyond just coordinating tasks because is strongly linked to building relationships and communities, therefore, it is a very social skill. Collaboration is not a single action, is a process that needs time and effort to be constructed. Finally, collaboration is aimed, it needs a purpose or a goal. More important, that goal needs to be shared among the collaborators instead of being imposed. Collaboration can only be developed when all the members of the group find themselves as equals and share the perceived value or the necessity of working towards that aim.

It is possible to add more elements to the definition, creating a wider and stronger framework, if we analyse the concept of collaboration from other perspectives and in different contexts. For instance, collaboration can be viewed as a skill to be developed within the context of professional development and employability.

As stated by Crant (2000), it is common for a workspace to seek in a professional a proper balance between theory and knowledge, that is, the information gathered through practice and experience. This position can be read as employers seeking people that have built a career rather than only have a profession. For employees, it means that they must build that career identity: the right set of skills, experience and personal traits that show value beyond a profession. Additionally, employers seek candidates that can properly response to a variable environment, either due to high pressure or a fast-moving market. Employees must then demonstrate personal adaptability, a willingness to explore, and life-long learning values. Finally, employers seek for the ability to properly construct and use the surrounding social context, the employee needs to build social capital by cultivating abilities in cooperation, networking and teamwork.

This framework shows the importance on adding value to the professional profile, and highlights skills related to adaptability and collaboration as key value pointers. The framework also relates easily to other popular concepts like social intelligence, personal development and problem-solving attitudes, all part of what has been known as transversal skills. The UNESCO defines transversal skills as those that can be used in a wide variety of situations and work settings (UNESCO, 2014) and are valuable for their general applicability and transferability. These skills can be organized in five broad categories:

- Critical and innovative thinking.
- Interpersonal skills.
- Intrapersonal skills.
- Global citizenship.
- Media and information literacy.

To understand better the type of abilities that the previous categories encompass, we can extract some ideas from diverse sources in the literature, a set of studies focused in understanding the work market in different contextual variations of geography, professional field, years of experience and even nature of the job-hunting process (Abir et al., 2024; Boahin & Hofman, 2013; Garst et al., 2019; Huynh et al., 2024; Llorens et al., 2017; Nadarajah, 2021; Ng et al., 2021; Smaldone et al., 2022; Zhang et al., 2024). This sample of literature shows the type of skills most demanded by employers around the world. Although skills that are considered *hard*, or better described as field-specific are understandably diverse, there is almost a complete coordination related to *soft*, transversal skills. These skills are common to all contexts and independent of the region or work field. Figure 2.1 shows a visual distribution of the most common transversal skills sough after in potential candidates, extracted from the studies mentioned above.

Despite being a small sample, the data shows a clear emphasis on relational skills, and it seems to portray the added value of hiring and developing people with high communication abilities, teamwork, good social interactions, and flexibility (Al Asefer & Zainal Abidin, 2021).

Another important sentiment that can be extracted from all the literature mentioned is a perceived discordance between the skills and abilities focused on by high educational

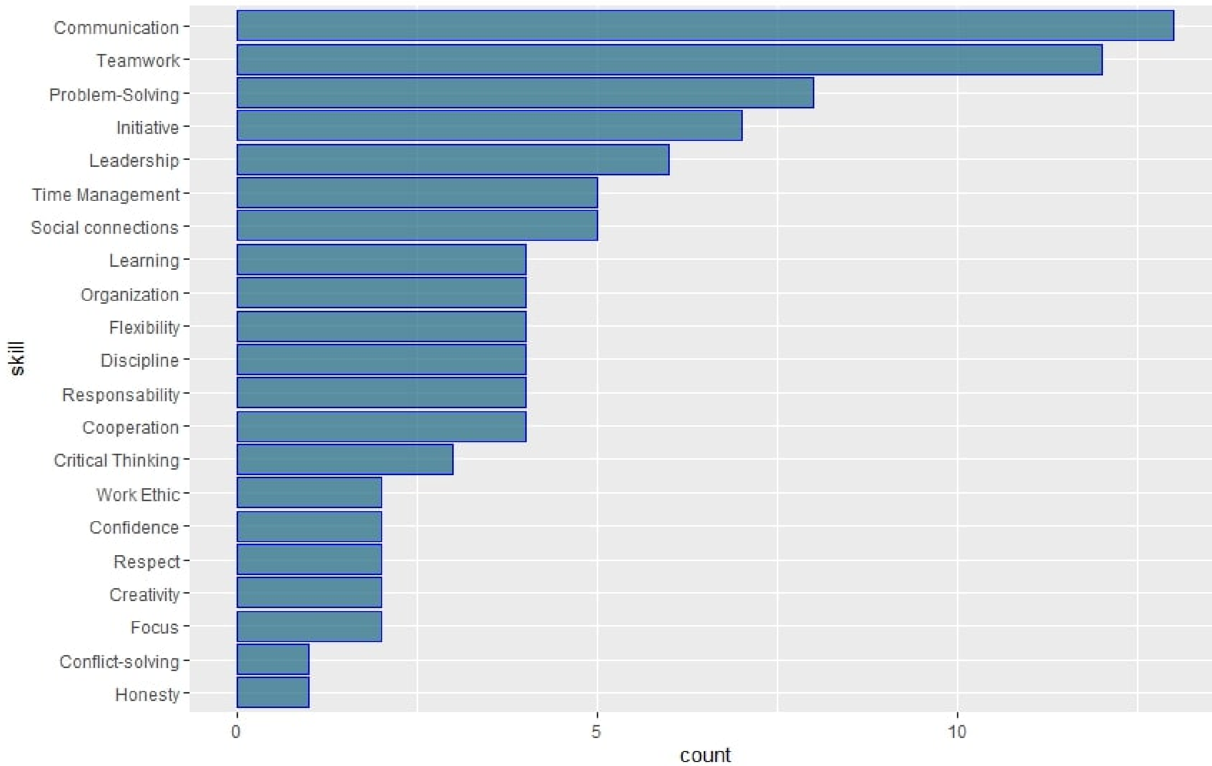

Figure 2.1: Most Common Soft Skills Identified

organisations and those sought after by companies and employees, as identified so far in this section (Damoah et al., 2021; Ng et al., 2021).

Tulgan (2023) identifies specifically three soft skills missing in young workers: professionalism, critical thinking and teamwork. Tulgan argues that this gap is closely related to a generational and technological shift where there is a lower need to develop communication skills because we have offloaded most of our communicational needs into electronical devices (mails, meetings, conversation, social interactions), forgoing the opportunity to develop such skills in person.

Tulgan keeps explaining that in recent years *good old-fashioned* teamwork became an issue for young people, who have developed a different set of values that clash with those of previous generations. Where older generations built a long career as part of a company or institution, newer generations have little to no loyalty to a single employer, and they are more likely to search for better deals. Young professionals find little value in adapting to the ways and stablished institutions of a group when they know very well, they won't be there for too long.

Secondary and tertiary education programs are often seen as the main responsible agents in the development of skills deemed crucial for the general professional life (Kechagias, 2011) and is in this light that modern curricula in almost all disciplines now have a higher emphasis on building those transferable skills (Earl et al., 2021). This is not without problems, and there is a noticeable consensus that teaching such skills is not a trivial task (McCale, 2008). Several factors influence the ability of a curriculum to provide such learning pathways. Most common problems are related to how little educators are prepared to incorporate those skills in their classrooms (Kemp & Seagraves, 1995), the costs of adapting the curriculum, training the personnel and changing old mindsets to effectively provide the space to create such learning (Chadha, 2006).

Considering these extra contextual elements, it is possible to modify the previous definition of collaboration as follows: Collaboration is the skill to work as part of a group by identifying, building and acting upon a set of values, shared views and coordinated efforts to fulfil a purpose or reach a goal. This modified definition not only states a set of characteristics of what the process of collaborating implies, it frames it as a skill to be executed by each member of a group, and not as something that happens naturally when people gather. It also acknowledges the need to learn and develop such skills.

The theoretical framework around collaborative learning can be expanded by exploring the definition of a second concept, that of Communities of Practice, with the intention of later linking this idea with the general definition of collaboration built so far and the approach proposed by this project to use it as an educational tool.

2.1.2 Communities of Practice

A term proposed by Lave and Wenger (1991), a Community of Practice (CoP) can be defined as a set of mechanisms of situated learning used by communities to transmit knowledge internally, especially to newer members. These communities are formed around a focused domain and were initially used to describe the learning process used in different trades like craftmanship and health practices. The meaning of the term has been broadening with time and can be applicable to any group that gathers towards a common goal over an extended period, that is, enough time for several generations of members to come and go.

Strongly linked to CoP is the idea of legitimate peripheral participation. This can be defined as the process in which newcomers learn mostly through observation and gradual participation with their senior peers (McDonald & Cater-Steel, 2017). CoP are used to give some structure to the process of informal learning, especially the type of learning found outside of academy, and which is more linked to professional practice, personal development and the relationships forged in the day-to-day activities with peers and a community in general.

A healthy CoP is one that is generated naturally due to the interaction of a community around a specific domain. It is different from a task force or team of people gathered around a goal or a problem in the sense that the general objective of a CoP is more undefined in terms of time, is focused more on the practice rather than the goal or product itself and is constituted by a social community rather than by just a collaborating team of people. These are the three fundamental elements that define a CoP: a domain, a community and a stablished practice, as described in Figure 2.2.

The domain is the knowledge base in which the community chooses to work because it offers value or just interest in general. New members come voluntarily and interact with elder members gaining experience in the practice, with the expectation of eventually gaining mastery.

The community is a group of people that can be driven by complex social dynamics. The community will create norms, stratifications and behaviours over time. The community is

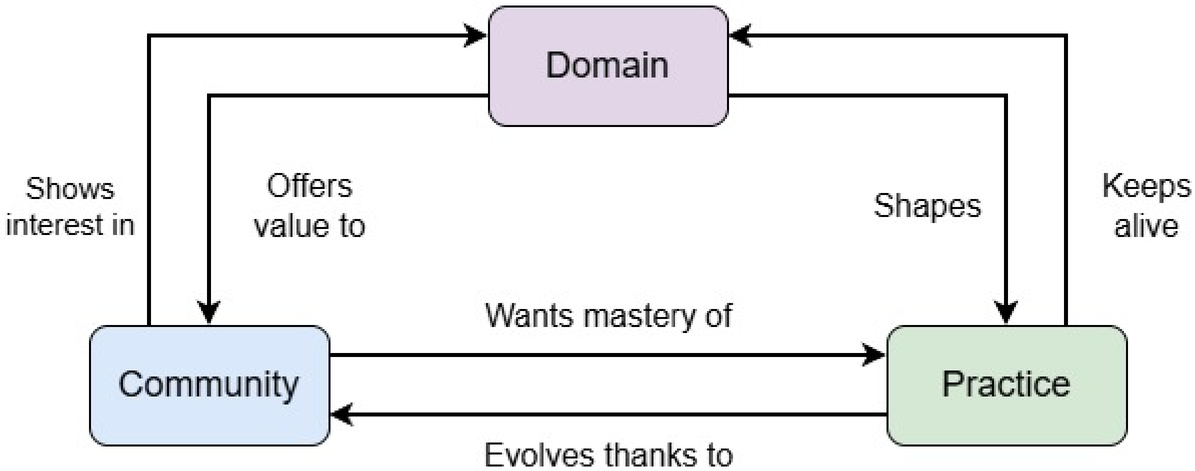

Figure 2.2: Elements of a Community of Practice

expected to be in a constant flux of members and to evolve with time, in terms of social interactions and the relationship it has with the domain.

The practice is an integrated set of activities and knowledge executed and transmitted by the community to create mastery over them. Some CoP focus more on nurturing members into the mastery of the practice, while other focus more on the transmission and preservation of the knowledge generated, as noted by Klein et al. (2005).

CoP have shown to be the cornerstones in the development of informal and social learning, leadership development and organizational change (C. Anderson & McCune, 2013). Often, high indexes of performance and employee satisfaction are present when a CoP has been identified and nurtured within the informal structure of an organization (Blackmore et al., 2010). Also, they have proven to be valuable integration, communication, and identity reinforcement tools in communities with a high degree of disparity or diversity among members (Hildreth & Kimble, 2004).

A particular emphasis on the idea of nurturing a CoP is important. A community can perish due to lack of proper nourishment or caused by artificial intervention. Previously it was noted that it is not possible to force the creation of a CoP, new members must voluntarily engage in legitimate ways with the practice. But even when a community develops and grows in members, and its value is acknowledged and sough after, forced practices that run in contradiction with the spirit of a CoP can end up killing it. Authors like Ardichvili et al. (2003) and Brown and Duguid (1991) have analysed the lifecycles of successful and failed CoP inside the organizational practice of many and varied industries, from insurance brokers to high education institutes. Of all the key factors identified in the failure or disappearance of a CoP, it is useful to highlight the following ones:

- Not allowing enough time for members of the community to form relationships and sustain them. This can be due to frequent administrative reorganization or the prevalence of temporary or part-time staff.
- A culture of tight management, where every aspect of the practice is highly standardized

and supervised, and there is no possibility for a natural development of the practice due to the relationships formed in the community.

- A culture of highly individualized work, either due to the very nature of the practice or because of other factors, like a very competitive environment. This inhibits collaboration and discourages collective engagement.

- Practices with extreme time or performance pressures, which leaves little time to develop a collective or communicative environment.

- Spatially fragmented work, so that there is no opportunity to establish a relationship, or because there is not a common, unsupervised space for the community to gather.

- Communication and activities are heavily mediated by technology or even other people. This makes interactions (arguably) less immediate and intense, therefore, lacking importance.

So far, all the elements that have been illustrated paint the concept of a CoP as a natural-occurring social structure formed around the need of preserving, communicating and nourishing knowledge within a well stablished community. It is a manifestation of informal education that thrives and offers the best results when acknowledged at an organizational level but not tampered with (Brown & Duguid, 1991), and when given enough resources to grow and evolve (Klein & Connell, 2008).

Given the importance of naturality and informality within the definition of a CoP, it is possible to see how these fundamental elements can be at odds with the expectations and common practices found in more traditional and institutionalized education, where learning is more structured, participation is hardly voluntary and collaborative processes are more task-focused rather than long-term approaches to the practice. These issues will be acknowledged in the following section.

2.1.3 Communities of Practice and Education

A common image of what could be considered *collaborative learning* in a classroom setting can be of a group of students working together to fulfil an assignment, say, a presentation on an assigned topic. Although it is possible to define this group of students as a community that came together with the goal of learning something, it fails several other criteria needed to be called a CoP.

In first instance, the example scenario breaks the idea of *natural occurrence*. The group was not formed by the voluntary interest of the students to engage with the domain, but as a necessity to complete a particular task. Also, it would not be uncommon for the composition of the group to be a forced assignment, either by chance or by the designation of the teacher, which would not constitute a community formed by the natural connection of its members.

The community itself also lacks perpetuity because its existence is linked only to the fulfilment of the task. Although the students can benefit from the work done alongside their peers, the knowledge and mastery acquired benefit only the immediate members of the group, they will not be transmitted to other members, nor is it expected for the group to grow and evolve.

Despite all these short comings, there are advantages in using the CoP approach in this scenario. For example, it can be argued that the activity does offer a genuine peripheral participation with the practice, allowing the students to engage with elements of the domain and work towards mastery. This engagement is achieved more through structured lessons rather than by informal observation of the actual practice, but it still offers the general idea of a gradual and guided approach.

To properly identify the community formed around this practice more elements need to be added to the analysis. In that way, the natural structures that are normally present in a CoP will get clearer. By including the teacher in the community, it is possible to identify expected elements like eldership and meaningful learning-focused forms of communication. By considering all the groups formed by all the students in the classroom, not only the initial single group, it is easier to see the opportunities for more meaningful relationships to be formed between peers, for more variations on the ways they approach the task or for more opportunities for that practice to evolve. The same results can be obtained by extending the timeframe in which we are analysing this activity, and considering not only the current cohort of students, but also all previous and future groups that participated and will participate in the activity. It is possible to keep adding more elements to the scenario, like all the teachers that have overseen this course and all the courses that are part of the same department, for instance. This exercise exemplifies that, although there are fundamental problems in the definition of a CoP in the scenario of institutionalized education, a lot of the core elements can be found with interesting variations.

A study group could be a better example of using a CoP as a useful framework of analysis in this context. In these circumstances, it is common for the group to form naturally, as a voluntary activity surging from the necessity of mastering a subject (for an upcoming test, i.e.) or because the students have identified that having the support of other peers helps them achieve better academical results. In any case, the core purpose of the group is learning and practice, rather than the fulfilment of tasks. The relationships between the participants extend beyond organizational roles and the focus lays on the transmission and collaborative generation of knowledge and expertise.

Cremers et al. (2008) analyse the formation of CoP in higher education in a similar way to the one proposed before. They argue that knowledge sharing and knowledge creation require a different set of skills than those needed for groupwork. A CoP becomes an ideal tool to promote those skills and to prepare students for the type of informal learning processes they will find later in life. Yet, forcing students to create a CoP or creating a lesson around the use of a CoP goes in contradiction to the spirit of the term, which is meant to be voluntary and informal. Giving continuation to the initial example, a study group work the best when the students themselves want to be part of it, and it is because they recognize that they need the help of their peers or because the collaborative activity is beneficial, not necessarily because it is part of a lesson proposed by the teacher, or even worse, because is a mandatory part of an assessment.

It is then necessary to come back to the idea that a CoP must be cultivated rather than created (Wenger et al., 2002). When designing a lesson, a course or a curriculum, several steps can be taken to promote the natural formation of CoP. Klein and Connell (2008) propose a series of principles used to nourish CoP that can be followed to create proper spaces for them to form naturally. For example, students should be allowed to propose and develop their own activities, with possibilities to grown on their own and to have active channels of communications with external sources, like teaching staff . It is also important for students to have access to physical spaces to develop such activities, encouraging both public and private work. The teaching staff should also be aware that they are stakeholders in this scenario, being an integral part of the *value* that the students create by engaging with the CoP and the natural *rhythm* the creation of this CoP follow related to the activities proposed by teachers, like due dates and exam weeks.

Cremers et al. (2008) go a step beyond and propose that students should actively learn to be part of a CoP, arguing that it is no different to learning the proper skills to be a good team member. Speaking in terms of learning outcomes, they stablish that a student should be able to:

- Reflect on the state of the practice and the necessities for future development and evolution.
- Be able to communicate and make explicit the results of their engagement with the practice.
- Be able to use the resources of the community in their engagement of the practice.
- Construct, organize and create new knowledge in collaboration with the community.

Rather than asking students to create a real CoP or forcing their participation in one, there is more value in giving the students the tools to understand the dynamics found in a CoP and the proper way of participating and creating value in one. Keay and O'Mahony (2016) echoes this idea by arguing that the significance of a CoP lies in the process rather than in the product. The process of interacting and collaborating with the community has an inherent value, even if a CoP is not properly constructed, and that is what a student needs to learn and be exposed to.

It has been noted by several authors that the dynamics found in a CoP can also emerge and be fostered in online environments, such as Riverin and Stacey (2008) in the perspective of the students and Kirschner and Lai (2007) in the perspective of the educators. These online CoP can emerge and be effective provided that similar circumstances allow the natural growth of the community, like an open participation and clear opportunities for mentorship. CoP for online environments have been found to boost participation and engagement in students, especially young ones (Thomas, 2005), which reinforced the idea that the creation of a community is crucial for a successful remote learning experience (Archambault et al., 2022; Swan, 2002), which circles back to the importance and usefulness of collaboration as a learning tool.

In summary, a Community of Practice is an analytical framework better used to understand processes of informal learning strongly linked to the social dynamics of its members (Wenger, 1998). These characteristics often find themselves at odds with the more formal and structured forms of learning in educational institutions but still offer valuable insights and working tools when analysing the activities done by students outside the structure of a classroom, and more related to their practice of being students.

As discussed in Section 2.1.1, collaboration is as skill that needs to be developed, not only because it is a well sough-after *soft skill*, but also because it is an integral component of professional development, community integration and life-long learning skills. As students, we become part of a particular Community of Practice that works around the development of useful skills for learning and academical success. The development of these skills is, at risk of proposing a circular definition, a collaborative process, one that is hard and unproductive to create in a forced or artificial manner, but than can be promoted and grown given the proper resources and the correct nourishment.

2.1.4 Learning Teamwork and Collaboration

A lot of literature can be found that try to create a model of how individuals come together as a group to achieve a goal, identifying all the factors (internal, external, personal, emotional or circumstantial) that contribute to the success and failure of a group. For example, Bion (2003) offers a good explanation of the Tavistock approach, a systematic view of the psychological dynamics present in the conformation and life cycle of a group. This model identifies important structures that appear inside a group like authority figures, responsibilities, boundaries (both internal and external), crisis responses and organizational structures. Participants form and engage with these elements even if they do not properly identify and name them.

A group can also be analysed in terms of the behaviours adopted by its members. Bales (1970) categorizes the possible interactions performed by team members into task-oriented and emotional. The latter category can be charged with a positive, neutral or negative tone. These behaviours can be expressed through every form of communication that the group uses, including non-verbal cues like body posture or voice tone.

Following the behavioural line of thought, E. S. Knowles (1982) identifies some factors that strongly influence the unique identity that a group develops through time. Factors like time constraints, physical environments, group size, composition, overall goals, patterns of participation, communication tools and the set of norms, procedures and structures that emerged within the group.

All these elements create a valuable toolbox of concepts that are useful to systematically describe the work, behaviours, relationships and experiences of a group. They create an essential common language that describes problems within a group, propose solutions, set goals or creates awareness of the elements that are expected to emerge or the issues that must be addressed when working with others. These elements shouldn't be the focus in the process of teaching collaboration but do offer a solid structure in the design of a solution.

A better approach to the learning process is to understand why we collaborate, not only in terms of the employability necessity that has been exposed previously, but also in terms of finding value in teamworking, even if the approach was not voluntary or when collaboration becomes a necessity rather than a goal. When framing our group relationships through the idea of creating value for the group and extracting value for ourselves, we can focus on identifying all the aspects that develop an effective collaboration, those that create more value (DAWSON, 2017; Wagner et al., 2010). When identified, these factors can be replicated and taught in controlled and varied scenarios so that students become aware of their value, build their own experiences of success and failure in a safe space and gain proficiency in identifying and replicating those same elements by themselves.

Hibbert and Huxham (2005) use the terms collaborative advantage and collaborative inertia. We navigate towards collaboration because we find a tangible advantage in the process (for example, the task cannot be accomplished by an individual). But even with a concrete advantage, no interaction can be initiated if the initial collaborative inertia is not surmounted. This inertia is defined as all the barriers that impede collaboration, things like:

- A personal unwillingness to seek input or to learn from others.
- Hoarding expertise, defined as an unwillingness to help others, share knowledge or skills.
- Inability, due to lack of skills or resources, to transfer knowledge.
- An impulse to compete.

The role of competition, especially in academic settings, has an interesting antagonistic nature to collaboration. A dose of competitive elements in an academical setting creates interesting results in terms of providing motivation and building a sense of belonging (Bergin, 2016). Yet, it has also been analysed that focusing too much on the competitive aspects of schooling can derive in undermining the intrinsic value of learning (Tauer & Harackiewicz, 2004) or creating an environment in which the competition is more important than the learning outcomes (Kowalski & Christensen, 2019). This pervasive sense of competition in schools is one of the main issues behind the generalize reticence of students to participate in collaborative activities, which can also derive in lack of motivation, loss of focus and general disinterest in the goal of the collaborative activity itself (Driscoll, 2025; Kloepper, 2023).

One of the first steps to implement a framework for learning how to collaborate is to identify which of these barriers are at play and provide tools to deal with them and not just hope for people to continuously stumble into these problems and finding solutions by chance or by trial and error. Salmons (2019) provides an extensive analysis of what collaboration is and how to create an environment that teaches and promotes it. One first effort is to acknowledge and identify different types of collaboration, defined by the overall achievements gained and the value obtained by participating in the collaboration. Salmons identifies these four degrees of collaboration:

- *Co-creation:* Collaborative generation of new knowledge, processes or products.
- *Acquisition:* A team effort to acquire or develop new skills or knowledge that none of the participants had prior to the collaboration.
- *Transfer:* Sharing expertise or knowledge between the members of a group to create a wider and more complex pool of knowledge or skills.
- *Exchange:* Sharing physical and human resources to fulfil a goal that could not be achieved individually.

Salmons also offers a complementary approach to the idea of value generation explored in other literature. She stablishes that before any form of valuable collaboration can be generated, it is important to reinforce two cohesive elements of any group trying to work together: trust and communication. These building blocks are the main forces used to create meaningful connections inside the group, which in the future leads to the creation and perception of value.

Trust refers to feeling comfortable while working towards and sharing goals with other people. *Trust* can also be seen as a form of social capital (Addison & Teixeira, 2020) that determines how willing is someone to cooperate with others and accept relationships and situations of dependence (Zand, 1972), risk, uncertainty (Frederiksen, 2014) and hope for mutual benefit (Kramer & Tyler, 1996). On the other hand, a lack of trust can be evidenced trough feelings of dread and exhaustion within the group, forcing collaboration only when strictly necessary. Vangen and Huxham (2012) also identify the ability to create a safe environment for all the members of the group as an exercise of trust, where the social risks associated with dealing with other people are negated or mitigated, so it is possible to concentrate in creating results consistently. A group that can create trust within is more resilient to external issues, to errors and to conflicts. It also shows a better capability for self-reflection and faster course correction (Shinya et al., 2016). A high level of trust is also correlated to safe environments for innovation and risk-taking (Bichard, 2005).

The second building block proposed by Salmons is to develop proper channels, skills and habits of communication. The author describes two types of communications in a collaborative setting:

- *Dialogue:* Defined as formal and structural communication best used for planning and coordination. It is also the best form for stablishing rules and initiating collaboration itself in new groups.
- *Conversation:* Defined as informal and loose communication used to create relationships. It is also the best tool to build and convey trust in the group.

Proper and structured dialogue is also the main venue for students to learn and practice collaboration. Essential aspects of teamwork can be evidenced and assessed through activities guided by dialogue that students engage with. The structuredness of dialogue also helps in creating activities and resources that show collaborative skills that students can develop and provide evidence of, such as:

- Proper use of interpersonal skills.
- Understanding of the task or project.
- Work design.
- Task allocation.
- Quality assurance.
- Coordination.
- Communication of progress.
- Accountability.

- Conflict resolution.
- Decision making.

With these elements is possible to create a roadmap that guides a lesson, a project or an assessment in which collaboration is an important learning goal.

Through dialogue it is also possible to create a third communication skill fundamental for a successful collaboration, *review*. This is defined as a process of self-reflection focused on the work or progress done by the group. A review process can also be structured, facilitating teaching and assessment activities, and can be mapped, as shown previously with the main indicators of dialogue, to key aspects like:

- Ask and give support.
- Analyse and challenge other's contributions.
- Analysis and incorporate feedback.
- Encourage peers.
- Monitoring peers.

Other authors like Keay (2006) proposes similar criteria for proper collaboration in terms of trust, respect and effective communication. Additionally. example implementations like the one showed by Lingard and Barkataki (2012) highlight some other success factors like developing a sense of self-organization, inclusivity and responsibility. Marasi (2019) proposed a series of activities that highlight important attitudes and approaches for healthy teamwork like listening before responding, being affirmative of other's ideas and being mindful of nonverbal communication. Davis and Ulseth (2013) and Asrori and Tjalla (2020) iterated over these concepts and proposed different frameworks, activities and observations that identified other important ideas like accountability, feedback, ownership, empathy and long-life learning.

Table 2.1 serves as a general summary of the concepts and ideas explored throughout this section and shows a map that links fundamental criteria of effective collaboration with activities and behaviours that acts as helpful evidence of those fundamental elements in the work and interactions of a group of students. This map provides a structured tool that will be used as an initial standpoint for all the educational and technological developments of this research project that have collaboration as the main learning goal. They are also a useful tool to frame the analysis of the work done by student in the development of the case study and to describe the learning design proposed by the Industry Project course.

2.1.5 Conclusion: Approach to Collaborative Learning

At this point this literature review has explored some of the tools needed to properly characterize the benefits and challenges around collaborative learning. From the definition built in section 2.1.1, collaboration is a skill developed by students to achieve a common goal through a shared set of values, views and efforts. The common goal here is learning.

Table 2.1: Evidence Factors and Assessment of Collaborative Work

Criteria	Evidence	Examples in Literature
A shared common goal	Work Design	Fisher (2014) and Kernaghan & Cooke (1990)
	Goal and task ownership	Kropp & Dodd (2024) and Sohmen (2015)
	Co-creation and acquisition of knowledge	Drejer & Jørgensen (2005) and Salmons (2019)
Willingness to collaborate	Inclusion of all members	Keay (2006) and Zubiri-Esnaola et al. (2020)
	Affirmation of all ideas	Marasi (2019)
	Group bonding and empathy	Blanco et al. (2017) and Widayati et al. (2022)
	Share knowledge and resources	Nissen et al. (2014)
Proper management	Task allocation	Hirshfield & Chachra (2015)
	Self-Organization of the group	Keay (2006)
	Communication patterns	Amponsah (2003) and Jahng et al. (2010)
	Flexible structure	Gross (1997) and Jacobs (2015)

Section 2.1.2 introduces the concept of a Community of Practice and all the positive implications of using such framework in terms of understanding the motivations that drive the people partaking in collaborative work. In a CoP there is a feedback loop between participants and the constantly evolving field of work in which they are immersed. A proper and healthy CoP has an antagonistic view of the type of work normally found in a classroom, with little agency from the students and a practice limited in time and scope to the fulfilment of a task.

Nonetheless, exposing students to the general dynamics of a CoP has value, especially considering both the ubiquity and the transparency of such dynamics in the workspace. It is also valuable to refocus the approaches used for a CoP in the classroom not as a fancy synonym to teamwork or classroom group work, but as a tool to analyse how students in general approach their learning tasks. This can be achieved by identifying a whole classroom (or cohort, or course, or department, etc.) as the community. The practice is not a task or a project, but the process of learning in the specific domain. In the proper environment, students develop tools to manage their learning and the relationships that naturally grow between peers and teachers.

Finally, in section 2.1.4, the information found on current literature was used to establish a set of criteria and evidence-base assessments that can guide a pedagogical design based on collaboration as a learning tool or that focuses on collaboration as the learning goal.

Collaborative learning can become a useful tool for the classroom based on the approach taken to encourage and manage teamwork. While trying to collaborate in the fulfilment of a task, students learn how to summarize and share information, co-construct ideas through dialog and debate, and improve retention and comprehension through the simple action of raising questions and answering them (Andrews & Rapp, 2015). Activities requiring teamwork also help in the development of critical thinking abilities (Zandvakili et al., 2019) and prepare students for a realistic work environment (Chowdhury et al., 2002).

Collaborative activities help students create a better understanding of the task at hand. Salmons (2019) uses the concept of *workflow*, where students must discuss, design and implement a concrete plan that help them achieve their objectives. Even if this plan involves a simple coordination of consecutive or parallel tasks, or a more complex synergetic form of work, the process of coming with ideas and solutions to tackle the needs of the task creates a stronger involvement with the team and the project. Similarly, coordinating work efforts develops proficiency in communication, especially in developing literacy (and digital literacy) around tools for remote communication, file sharing, document cocreation and asynchronous work coordination (Lo, 2009; Murphy & Cifuentes, 2001).

Collaborative activities are also often associated with creativity and transformative learning by providing the proper conditions to bring ideas together, compare solutions and expose different viewpoints (Peppler & Solomou, 2011; Turnbull et al., 2010). Rubbing ideas together offers extra value to the generation of knowledge by also providing the circumstances to make connections around that knowledge and challenging preconceptions or stereotypes (César & Santos, 2006; Eden & Ayeni, 2024).

But working in a collaborative setting is not without its challenges and hurdles. Just the act of being in a group is not enough to boost the educative process of the students. If the proper tools are not provided and major barriers are not acknowledged, the situation can flip around, and teamwork can become more of a liaison than a boost.

One of the most common issues is power imbalance, a natural occurrence whenever there is a group of people gathered towards a common goal (Essabbar, 2015). Elements related to gender, social status or economic issues, among other things, are always factors to consider when using teamwork as the main driver of a lesson. When a strong imbalance is present in a group, it can be evidenced is several behaviours and outcomes that could hinder the learning process of the students. Different authors have identified some of these patterns in different contexts, like inconsistent exposition to the learning material, coarse relationships forming between students, lack of agency in work distribution (Griffin et al., 2015), stress, downgrading performance and lack of response to feedback (Van Der Vegt et al., 2010).

Educators need to develop the necessary foresight to acknowledge these difficulties and provide the students the tools to solve them. In general, it has been shown that self-reflection, an effective mediation and clear communication methods work the best when dealing with issues of imbalance. A closely related analysis in search of solutions for this situation is related to academic imbalance, especially when it affects the educator and different expectations in terms of performance and achievements formed for each group member. A lot of studies have worked on strategies to identify these discrepancies and how to work around them, proposing tools like peer-assessment (Rezaei, 2018) participative rotation, close time management (Davies, 2009) and smart grouping (Akhrif et al., 2019; Q. Wang, 2009). The important note is to be aware that issues with academic imbalance are common, and that there are several tools to manage them and even take advantage of them.

Management chaos is also always associated with groupwork, not only from the point of view of the students but also from the educators. It is expected that students will have problems with managing elements associated with groupwork, like distribution of work, problem solving and self-reflection. It is also expected that the students will find the educational material needed to learn how to cope with those management issues in the educators or the course itself. Nonetheless, it is surprising how often it is expected that students collaborate without giving them the tools to learn how to do it. Bruffee (1973) provides an interesting record of his experiences implementing a collaborative approach with his students, and a lot of information can be extracted from this case study in terms of the work needed to set-up and manage the tools, environment and mindset that allows a proper collaboration. This extra management time is noted to be present both for the students and for the educator.

Jaques and Salmon (2007) also identify other behaviours that arise when students don't feel comfortable working in groups. Poor participation and general distress can be associated with cultural differences and expectations around how to behave in a group, or, as Biggs et al. (2022) also noted, around personalities that inhibit themselves easily when in a *formal* or *high stakes* form of participation. Other issues reported by students themselves are related to frustration around low performance groups, previous bad experiences or having to deal with personalities that are too outgoing, and which completely dominate the whole group in detriment of other students.

The myriads of advantages present in building a learning experience around collaboration can be attained when the proper tools and the proper environment is provided to the students. It is also important to acknowledge and provide pathways to deal with inevitable issues around conflicts and overhead management time. With all this material is possible to propose a guiding conceptual framework for the educational and technological design that is going to be undertaken in this research. This guideline consists of four principles:

1. Collaboration is a skill that students need to learn and develop. This learning offers value to the employability profile of the student, to the development of lifelong transversal skills and as a useful tool to facilitate more learning.

2. Students develop a form of CoP when learning together, with emerging relationships with their peers, their environment and their learning goals. This communities can be nurtured, which in reciprocity generates a rich, heavily knitted and unique development, both as learners and professionals.

3. It is possible to aim for the ideal characteristics that help the development of collaborative attitudes and use them to design and develop learning tools that take advantage of a collaborative learning approach.

4. The advantages inherent to collaborative learning can be minimized or negated if proper attention is not given to the equally inherent problems that immediately arise. Any learning tool needs to be designed around these issues and provide resources for both the students and the educators to propose solutions fitted to their context and particularities.

This set of guidelines is meant to inform the design and development of learning objects, tools or experiences. Therefore, it is now necessary to analyse all the ideas so far exposed about collaborative learning in the context of technology-enhanced learning, and particularly around augmented reality as a tool for education. With this approach, we will have all the theoretical foundations that inform this thesis, and this is the objective of the next section.

2.2 Augmented Reality in Technology Enhanced Learning

Throughout chapter 1, it was briefly explained how the interest in using AR as a tool for education is born from an inherent interest in the technology and a perceived value that the characteristics of AR could offer unique solutions to educational problems related to motivation, engagement and communication. It is now necessary to explore the current use of the technology in the education sector to identify advantages, disadvantages, common issues and, in general, the current state of the technology as related to the field of education.

2.2.1 Context

The idea of technology-mediated education has often been linked with that of collaborative learning. Lehtinen et al. (1999) highlight key ideas about the communication process developed in learning scenarios and how we use technology to facilitate and manage such activities. Many of the main concepts explored by the authors echo those already presented in the previous section . Ideas like the importance of nurturing a community, selecting proper tools for communication, structured process of review and mediation are explained in similar detail as before, as well as other common issues identified, like excessive competitiveness and factors for social friction.

Dillenbourg and Fischer (2007) propose some recommendations to incentivise effective collaboration in students, like physical spaces that facilitate interaction instead of individualisation, a common but flexible structure for collaboration that is transmitted to the students and techniques that the educator can use to orchestrate the technology used in classroom. It is key to acknowledge that the process needs constant moderation and motivation to create consistent results. These examples reinforce the idea that collaboration needs an extra effort to transform the work performed by teams into knowledge and learning. Coordination between the technology mediating the interactions and the expertise of the educator building the experience is crucial to achieve that goal.

Complementary, Kirkwood and Price (2014) offer a systematic review which aim is understanding what is being enhanced when talking about TEL. When categorising the data, the authors found that interventions often sought changes in the current state of an educational material, and that the chosen technology could offer positive quantitative or qualitative changes, either in terms of markings, assessments results, motivation or deeper understanding of the teaching material.

The literature in the realm of TEL is rich and it is possible to identify some form of evolution in the topics analysed when following the jump from teaching machines (Watters, 2021) to computer-aided learning, to online and mobile learning. Among all the information that can be found it is easy to get lost or confused in all the definitions and terminologies proposed. A good example is the term Virtual Learning Environment (VLE), which will often be used to describe the approach of this research. It tends to be used interchangeably with another concept, Learning Management System (LMS), which refer to the tools used in the administration of educational content and other administrative functions common of educative institutes, which is not the aim of this research. The term VLE is often abused a little to try and give more value to a system or set of tools more akin to an LMS.

To stablish more precise use of the term VLE, we can use the characteristics given by Warburton (2009) like immersion, a sense of presence, visualisation, contextualisation of content and simulation of impossible or dangerous experiences. It is interesting that the author also emphasises these elements:

- Tools for affective, empathic and motivational interactions.
- Intelligent interactions between individuals, communities and objects.
- Opportunities for individual and collective identity and role-play.

The emphasis on the social aspects of the interaction with any form of educative technology is crucial when analysing the enhancement provided by the tool in question. That emphasis can be summarized in three important elements: Visualisation of content, sense of presence and immersion, and a sense of collaboration or participation with other people.

With a a more precise definition of a VLE at hand and a set of tools to draw comparisons between technologies, in terms of visualisation, immersion, interaction and collaboration. The next question would be, why focus on AR? What makes AR an interesting technology for education? What makes it special or different enough from other technologies in the same spectrum, like VR?

AR saw a surge of popularity between 2011 to 2013 (Akçayır & Akçayır, 2017; Fidan & Tuncel, 2019) due to many factors like a wider adoption of suitable technologies for deployment. Successful implementation of applications in diverse scenarios has also been documented and reported (Javornik, 2016), and access to developer-focused tools like ARCore and ARToolkit prompted and facilitated development efforts. In general, and very similar to the case of VR some years prior, successful applications and explorative research was boosted thanks to new toolkits and technologies that facilitated learning and development.

In contrast to VR, the resources needed to develop and deploy an AR solution are less complex, and generally cheaper (Ivanova, 2018). This relative ease of development makes AR a popular technology to try out without the risk of committing too many resources into one single solution.

In terms of comparing the usage of AR with other similar technologies, different studies have shown numerous perceived advantages in the use of AR at different levels of education, and they are mostly consistent between studies: enhanced engagement in the learning activity, high levels of immersion and cost efficiency in terms of deployment and usage (Challenor & Ma, 2019; Chen et al., 2017; Saidin et al., 2015).

Common problems found are also consistent throughout literature and they are mostly related to usability issues: technology instability, steep learning curves and time-consuming usage. The term *cognitive overload* is also commonly found through the literature, normally associated to the unnecessary stress derived from the complications of using the technology in addition to the demands already present in the learning activity.

It is noticeable that most reported results are focused on validating the engagement of a particular development rather than its educational role in the learning activity and the educational objectives associated with the project in the first place. To broad the analysis of why AR is a valuable technology for education, a complementary question can be asked: What educational goals are being fulfilled with the use of AR technologies?

The idea behind this question is to identify actual pros and cons of AR technologies through the lens of educational achievements rather than usability or feasibility, as is common in the current literature. To facilitate an answer, four supporting questions can be proposed:

- Are there other advantages or positive outcomes beyond a spike in the student's engagement?
- Are there circumstances in which AR technologies work better than other technologies?
- What are common practices and mistakes emerging from the reviewed literature?
- How do advantages, mistakes and general results change when the field of study or the educational level changes?

Given that AR is the main focus of this research, it was important to propose a deeper review of current literature in order to give substantial answers to the questions just proposed, and to position the technological development of this research in a proper educational basis that both uses well established educational practices and explores avenues, solutions and ideas missing or seldom touched by current literature. Sections 2.2.2 through 2.2.4 will then discuss in detail the systematic literature review performed for this purpose.

2.2.2 Systematic Literature Review Methodology

One common way to analyse learning objectives in any given scenario is by using Bloom's Taxonomy (Bloom et al., 1964). This classification proposes 6 categories in which a learning activity can be positioned depending on the cognitive complexity it requires from students. From easiest to more demanding, these categories are: Remember, Understand, Apply, Analyse, Evaluate and Create. Each category is meant to be seen as actions the students must take to achieve any educational goal.

The taxonomy is a flexible tool in education, useful for both design and evaluation processes. It can be used to understand how a course or lesson is trying to achieve a goal, and if it is doing it with the correct approach (L. W. Anderson & Krathwohl, 2001). It can be seen as the initial blueprint for the design process of any educational material (Healy et al., 2011) and it has also been used as the basis for establishing standards and policies (Guskey, 2001).

Nonetheless, it can be argued that Bloom's taxonomy offers only a one-dimensional approach to education, relegating knowledge of any subject as the simplest aspect of cognition. Krathwohl (2002) proposes a revised taxonomy in which knowledge is a second dimension that gives context to all the other cognitive categories, considering all the ways in which different domains at different levels of education approach learning. This knowledge dimension proposes four categories: Factual, Conceptual, Procedural, and Metacognitive. These categories are

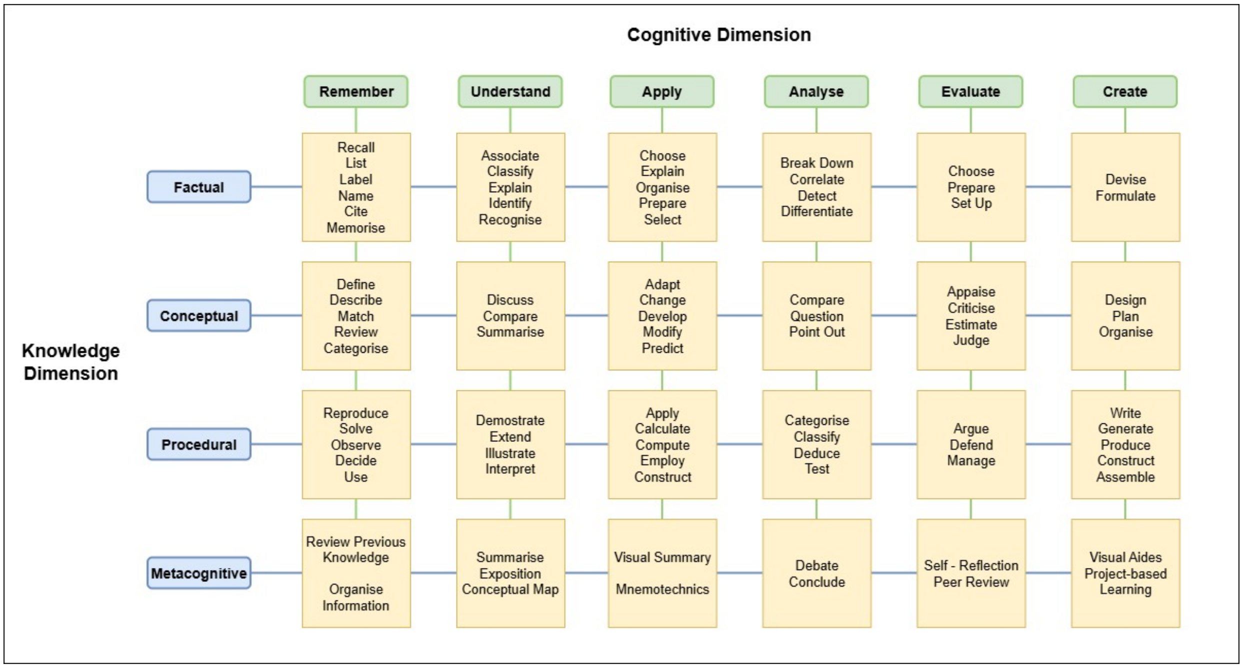

Figure 2.3: Bloom's Revised Taxonomy with Example Verbs

not meant to be hierarchical like in the cognitive dimension, just a representation of the type of knowledge that is acquired. In conjunction, the knowledge and cognitive dimensions can give an educational goal a more pinpoint description of what it is trying to achieve, and a more substantial tool for comparisons.

Figure 2.3 shows a visual representation of the taxonomy as the intersection of the complexity of a task with the type of knowledge that is asking the students to acquire. The figure shows some examples of action verbs that can be used to describe each intersection. These examples were constructed using as reference the works of Newton et al. (2020), Radmehr and Drake (2017), and Stanny (2016), which perform an amazing job at compiling, synthetising and organising reported lists in the literature of common action verbs used at each level.

This visualisation will be used as a reference to analyse the academical material found in the web to determine the pedagogical goals of a particular project. The Bloom's Revised Taxonomy (BRT) framework is valuable because it can offer insights about the complexity of a goal, the nature of its educational setting, and a concise and easy description of the goal itself to use during the analysis of the information gathered.

A SLR methodology is used due to its main premise of being replicable (Siddaway et al., 2019), to serve as a tool to encourage discussion around the type, quality and quantity of the sources used for the review, and to enable the comparison of results given different search, filtering, or codification parameters. This SLR process consisted of four distinct phases: selection, filtering, codification, and a final analysis.

Selection Phase

This phase focused on finding relevant work in AR tools for education at different learning levels and fields as possible. The main selection criteria in this phase were the year of publication and the technology used. The timeframe selected was from 2017 to 2021 due to the amount of work already done in the 2011 to 2016 period (Altinpulluk, 2019; Arifin et al., 2018; Chen et al., 2017). This timeframe also helped to focus the analysis on recent projects with a more contemporary understanding of AR technologies and access to modern development tools and products.

The technology had to be reported as AR in the title, abstract or key words list. The term had not to be exclusive or opposed to VR or MR, this with the purpose of not excluding works that could add valuable information for the analysis prior to a more detailed reading. It is also very common to use the terms AR, VR, and MR interchangeably and are common umbrellas for more specific technologies that could be used to define projects of interest for the SLR.

Google Scholar was the selected search engine using the following queries:

- (augmented reality OR mixed reality OR AR OR MR) AND (education OR training OR learning) AND (technology enhanced OR technology assisted)
- (augmented reality OR mixed reality) AND (education OR training OR application) AND (review OR evaluation OR case study)
- (augmented reality OR mixed reality) AND (industry OR job) AND (training OR instruction OR practice)

Due to the nature of the Google Scholar engine, more than 10000 results were found and could not be realistically managed in a reasonable timeframe. Nonetheless, Google Scholar was maintained as a search engine due to the diversity provided by the tool in comparison to other engines and aggregators. Only the first 10 pages of indexation were used to limit the initial selection of papers. At this point, sources were excluded if any of the following elements were identified in the abstract or summary:

- A merely technical approach to MR technologies with no identifiable educational component.
- A systematic review or a collection of case studies that would make it difficult or impossible to access information about each individual project.
- A duplicate of previously surveyed articles.

Filtering Phase

The amount of literature to be reviewed was reduced to truly relevant material with enough reported information for any analysis to be possible. A first set of criteria was used to exclude projects that:

- Use VR rather than AR or a form of MR that did not fit well into the AR definition, as proposed by Milgram et al. (1995).
- The AR implementation assisted or facilitated a task rather than to educate or train, i.e., surgery planning instead of anatomy education or procedure education.
- Show a general discussion of AR as a tool in education or other context, and not a particular application or case study that could be coded and analysed.

It was possible to identify these elements by quickly surveying sections like the abstract, the introduction or the methodological description. A second set of criteria was used to determine if the project in general provided enough information to process it in the coding phase:

- The article had a clear enough explanation of the tool and its pedagogical purpose.
- The article had a description or discussion of the outcomes of the project or experiment.

For these elements, it was necessary to make a deeper reading of the results, discussion and conclusions sections, and because of that, it was done in parallel with the codification phase.

Codification phase

In this phase, two researchers extracted the relevant information from the selected literature and classified it into two categories:

- *Analytical information:* Year of publication, target demographics, target field, particular AR technology used, success of the project (either reported or inferred) and specific experiences during the development or deployment of the tool. Special attention was given to identifying, if possible, reported positive and negatives outcomes with the tool or its usage.
- *Pedagogical information:* The concrete pedagogical goals of the project as enunciated by the text itself or inferred by the researchers. Educational goals were coded in the format *action verb* + *statement* as specified by Fuchs and Deno (1982).

Determining whether a project was successful or not was very subjective and there are no general criteria to give a pragmatic evaluation of this type. It is also debateable that publishing a report of an unsuccessful project is not a common practice either (Cook & Therrien, 2017; Dawson & Dawson, 2018). A couple of criteria were used to determine if a project could not be considered a success:

- A considerable part of the objectives or hypothesis for the project were not met or proven wrong.
- The discussion, results, conclusions, or any similar section heavily focused on problems and negative outcomes without mentioning significant positive or successful ones.

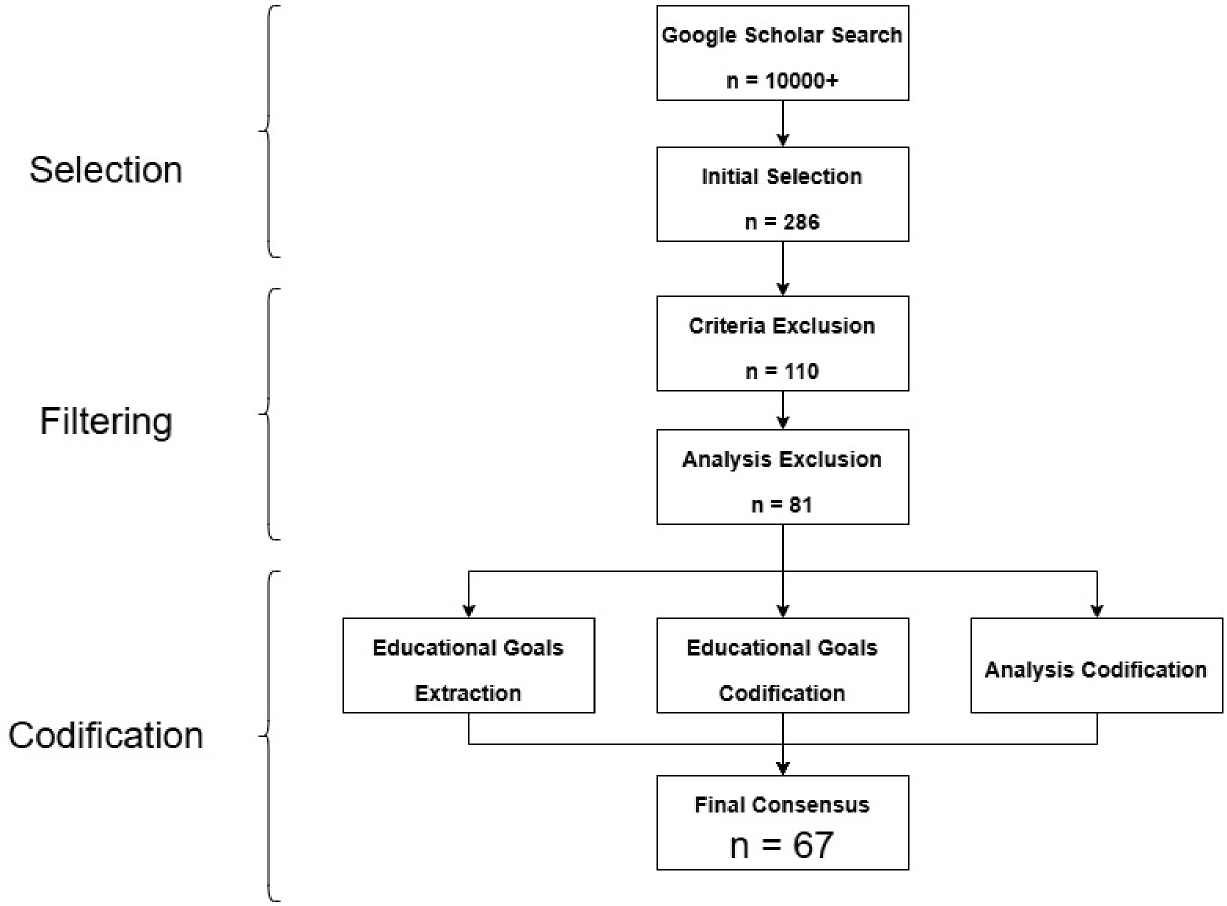

Figure 2.4: SLR Process

- It was not possible to build or test the intended project, or the design had to be significantly altered, especially for technical reasons.

The last stage of codification consisted in positioning each pedagogical goal into the BRT given its perceived cognitive and knowledge complexity. In all stages of the pedagogical codification, each researcher gave its own analysis and classification for every text individually, then reached the final codification by consensus. The final codification for every text surveyed can be consulted in Appendix A.

Figure 2.4 shows the general flow of the process, and the number of documents filtered in each phase until the final count used for the analysis. The process follows the standard recommendations given by the PRISMA statement (Page et al., 2021).

2.2.3 Analysis Phase

In the final analysis, content and thematical analysis was used to extract useful insights of the behaviour and tendencies of the surveyed data, as well as some statistical descriptions used to find categorical distributions, modes and tendencies. Data was organised to identify the general distribution of the surveyed projects among different demographics, educational levels and educational fields. More qualitative data was also organized and categorized such

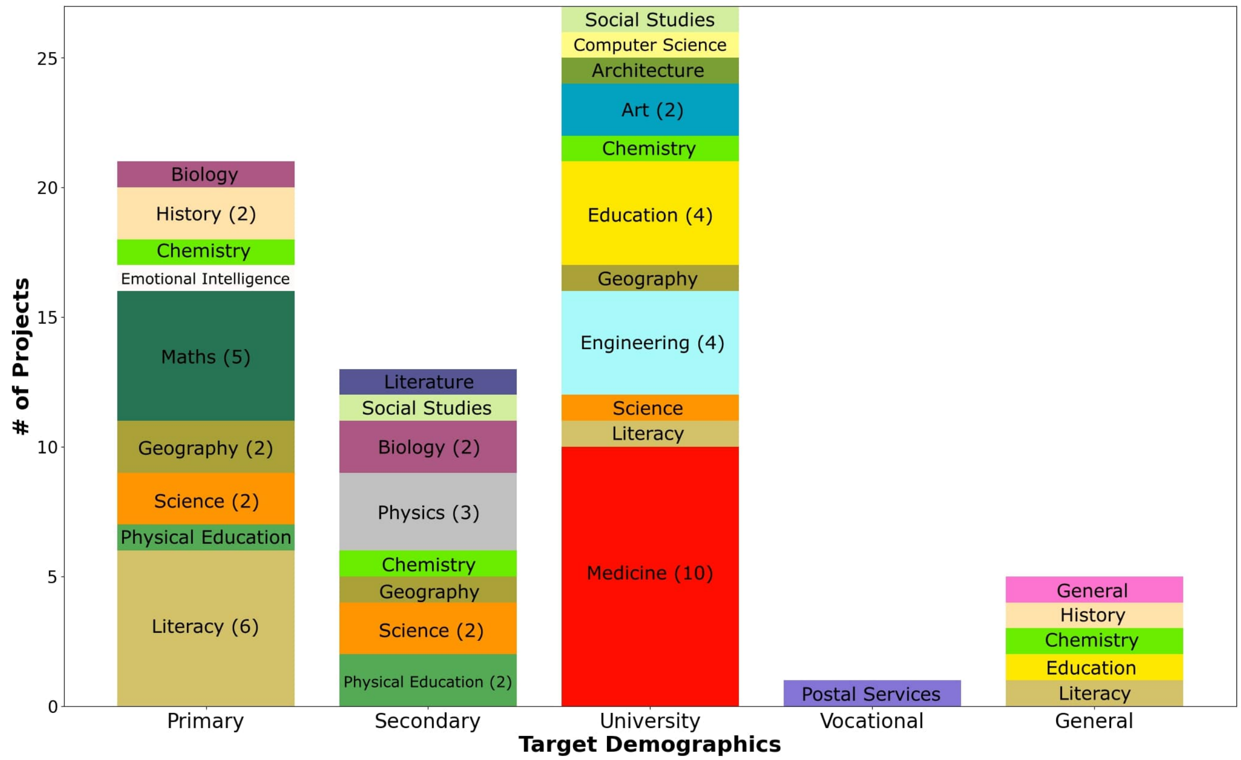

Figure 2.5: Target Demographics vs Target Field

as the experiences reported by the researchers and the users, common trends and positive and negative outcomes. Finally, there was an important focus on identify the pedagogical goals targeted by each project, their distribution and significant trends.

Demographics, educational levels and target fields

Figure 2.5 shows the distribution of projects between the target educational level of the students and the field in which the project is providing the educational material. Demographics were divided into *primary school*, *secondary school*, *university*, and *vocational training*. Target fields where coded as general as possible to avoid too much granularity in the data.

University-level education comprises most of the surveyed population and has the most variety of target subjects. Yet, K12 education in general surpasses higher education if the division between *primary* and *secondary* school is ignored. The division was kept because, as the figure shows, there is little overlap of learning fields between target demographics. This suggests that the focus fields for each educational level are well identified and most worked on. Math, science, and language are the predominant focus for K12 projects, with more complex applications of mathematics, like physics, or new subjects like chemistry being more prevalent in secondary education. University projects either focus on field specialisations (medicine in general, surgery training specifically, for example) or in common learning abilities such as critical thinking or problem solving. Little was found in vocational training using AR.

A fifth category emerged from the data and was labelled as *General*. It refers to projects aimed at creating a tool or experience focused on a particular field but not for a particular demographic, instead meant to be accessible by anyone or to present a broad approach to the field with a loose or no focused learning goal at all. One project was classified with a *General* field and *General* demographics, a metacognitive tool meant to help in the process of studying itself with issues like concentration and motivation.

An effort to create tools for learning languages is notable, used mostly as a tool for heavily ideographic languages, like Mandarin, or as an aiding tool to boost retention or even creative skills. Other projects were related to areas that could not be classified by the BRT, like physical education or emotional intelligence. Those projects were kept in the survey because it was possible to code all other data not related to the classification in the taxonomy and could provide some interesting insights.

Fields inside the STEM spectrum are a major focus for any educational research or development, often seen as a priority for a rounded education, career success or community development (Madden et al., 2020). Fields outside STEM, on the other hand, are often seen as underrepresented in research and development endeavours and not given the same importance or focus as STEM fields. In Figure 2.6, fields in red are part of the STEM classification, 52% of the identified fields. This shows almost a half and half distribution between STEM and non-STEM subjects. In terms of volume of projects per field, STEM fields have a bigger volume, taking almost 60% of all the fields identified. Non-STEM fields are present, but in a lesser quantity. Education, or learning to educate (in any field, including STEM) is the biggest representation of non-STEM fields, followed by tools focused on history, general literacy, and alphabetization.

Positive and Negative outcomes

A list of positive and negative outcomes was coded for each surveyed project based on what was reported in the findings, results, or discussion sections. Outcomes were then coded using a theme codebook to identify recurrent ideas and problems. Figure 2.7 shows the recurrences of all the coded themes for positive outcomes. Figure 2.8 shows the recurrence for negative ones. A summarized label was given for each theme, and it was assigned to a broader category to help readability. A full description of each category can be consulted in Appendix A.

Tendencies identified in other studies can also be seen here for both positive and negative outcomes: Enhanced performance and motivation are the most common results, while technology issues in general are the most common reported problems. Aside of these expected results, more surprising findings were also identified. Providing an autonomous educational experience was a positive outcome for different projects. Control over the pace of the learning process, access to a tailored experience or providing alternative tools for different types of learning were common positive feedback. In contrast, positive results associated with socialisation or collaboration were fewer, showing a high tendency to create individual experiences.

Regarding negative outcomes, the most recurring theme is *None Reported*. There is a general tendency to not publish or report any kind of negative results, either to not undermine the general validity of the research or because it is not considered important information to be reported. Is more common for projects derived from engineering or computer science to be more thorough in describing or commenting on the design and implementation process of the tool, as well as reporting problems or negative results, while projects that are closer

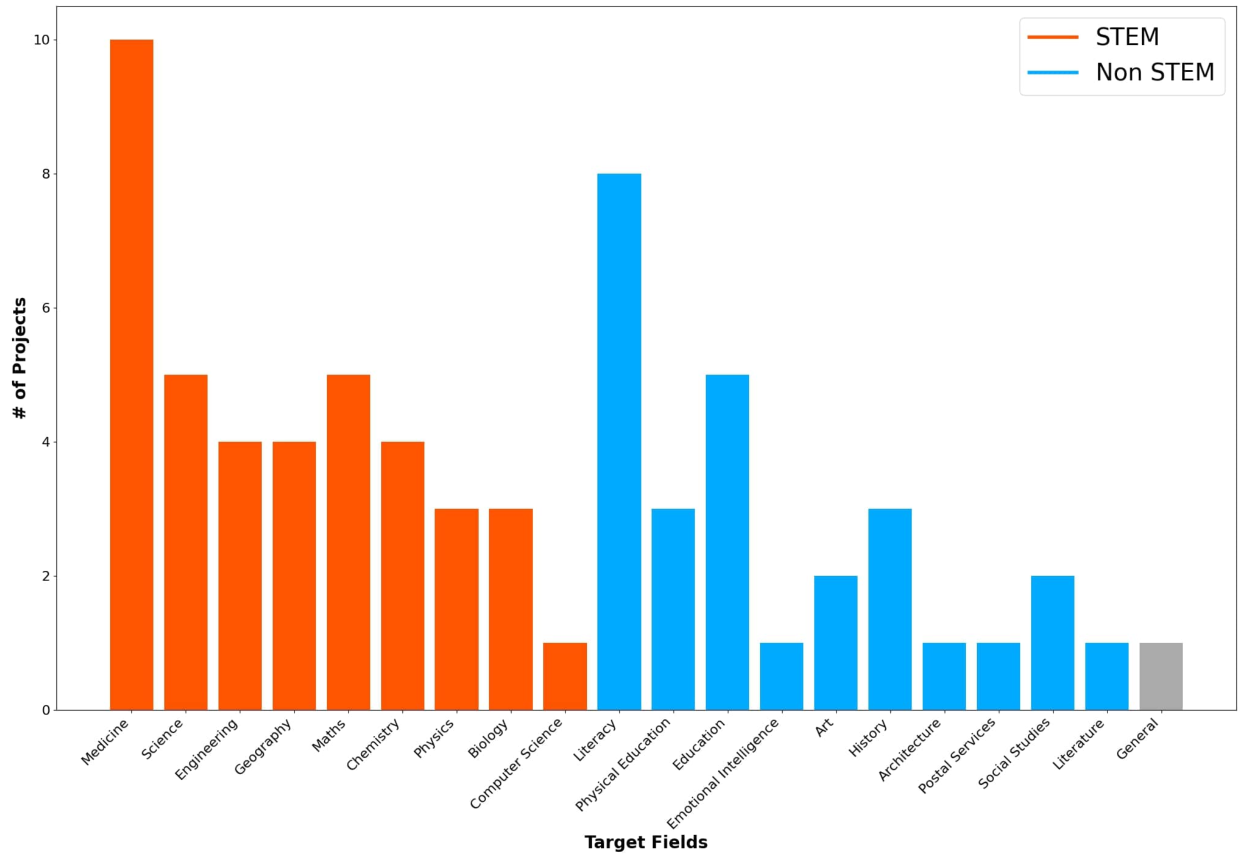

Figure 2.6: STEM vs non-STEM Fields Distribution

aligned to education research or humanities tend to focus more on the results and discussion derived from the experimental design.

Other negative outcomes are related to unintended behaviours caused by using the AR tool. Most popular undesired behaviours reported are constant distractions, hinderance of other skills or excessive time consumption. Another reported outcome is the users perceived lack of quality or quantity of content, or researchers having problems to create content for the tool.

An interesting fact that can be inferred from Figure 2.7 is that most projects guided the validation of their proposed solution by comparing performance gain through the usage of AR. This conclusion is derived from all the positive outcomes related to increased performance, and their absence in Figure 2.8. User validation is another common comparison tool present in discussions and conclusions, commonly in the form of user's perception and acceptance of the tool.

The reported negative outcomes coded were paired with the success classification given to each project by the researchers. Figure 2.9 shows this cross-reference for projects coded as successful, while Figure 2.10 does the same for projects that were coded as unsuccessful.

It can be noted that the density of negatives experiences reported by projects categorised as successful is lower than for projects that were not classified as such. Also, projects categorised as successful are more likely to report no negative outcome at all (the *None*

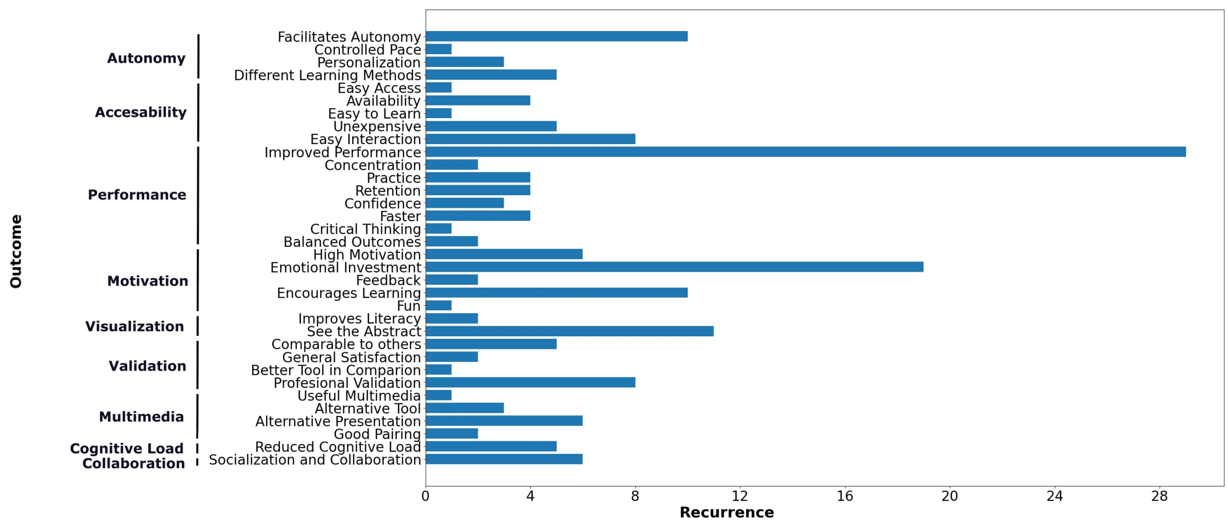

Figure 2.7: Categorisation of Positive Outcomes

Reported category), while projects not categorized as successful where more prone to report technical issues. It could be argued that when technical issues where present, it was harder for users to engage with the tools, resulting in worse perceptions or worse performance results. In contrast, when an immersive, fun, or simply good experience was provided by the tool, technical issues could be glossed over with more ease by users and researchers alike.

Distribution of Learning Goals

Information about the pedagogical goals of each project was extracted and classified using the BRT framework. The cognitive axis is hierarchical by design, and each project was classified with only the highest category analysed for it. For the knowledge axis, several of the possible categories could apply to the same goal and thus were coded like that. Figure 2.11 shows the distribution of points in the BRT matrix for each project. If the project belonged in more than one knowledge category, each pair formed with the corresponding value in the cognitive axis was plotted separately.

The biggest cluster of projects can be seen in the *Understand* and *Apply* cognitive dimensions. *Factual* and *Procedural* knowledge are the most paired with (37% of the total pairings). Most goals tackled by AR educational projects can be found then in the mid to low cognitive complexity, with a preference in applying *Procedural* knowledge. Examples of this can be training in a practice-oriented process, like a surgical procedure, or learning to perform a task, like reading or arithmetical methods. Understanding factual knowledge is the second most common cluster, in which the visualization advantages of AR technologies can help to comprehend concepts difficult to visualise or to get an example of in real life. Concepts in physics or biological processes are common examples.

Several projects were classified in higher cognitive complexity levels like *Analyse* and *Evaluate* throughout all knowledge dimensions, and some efforts were noted to create apps focused on creativity tasks. Another interesting cluster is the *Metacognitive* knowledge dimension, with several projects at all cognitive levels trying to use AR more as a complementary

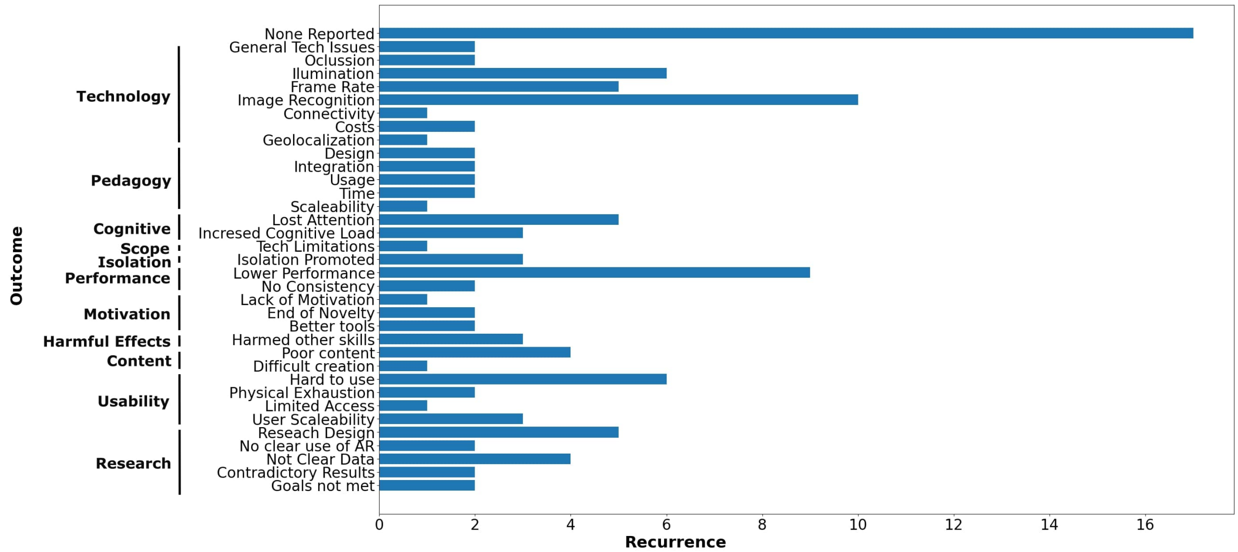

Figure 2.8: Categorisation of Negative Outcomes

tool rather than the main transmission tool of the pedagogical design. This tendency is of notable interest due to metacognitive tools being in line to modern approaches to education like distance learning, long-term learning, learning how to learn and other similar topics. The only gap in the distribution is in the *Metacognitive-Remember* zone, which could be interpreted as mnemonic tools for boosting memory or retention.

Figure 2.11 also cross-references the cognitive-knowledge data with the target demographics of each project. Primary and secondary-level projects coincide with lower cognitive complexity, while university-level projects tend to be in higher complexity levels, though all demographics levels can be found at all cognitive levels in general. University-level projects are more clustered in the *Apply* dimension while K12 levels are more clustered in the *Understand* dimension, which correspond to the common approach each level of education has. The same tendency holds when analysing the knowledge axis: K12 levels are more prone to *Factual*-type knowledge while university level projects are more common in *Procedural* knowledge. The previously identified tendency of projects being in the STEM spectrum can also be seen as a reason for a bigger cluster in the *Apply-Procedural* quadrant in comparison to the lower presence of projects in the *Analysis* or *Evaluation* areas. Figure 12 shows the same distribution of Figure 11 but cross-referenced with a more generalized categorization of the target fields.

STEM related projects are noticeable clustered at the *Apply* section of the distribution. Non-STEM projects are more noticeable clustered in the *Understand-Factual* quadrant and the *Analyse*, *Evaluate* or *Create* regions. In general, the *Remember* region could be considered of little interest or impact for most subjects, hence the lack of points, while higher level cognitive activities get less projects due to the complexity of creating pedagogical designs or tools for those goals. It could be argued that, for those same reasons, no project in the STEM spectrum was found along the *Create* dimension.

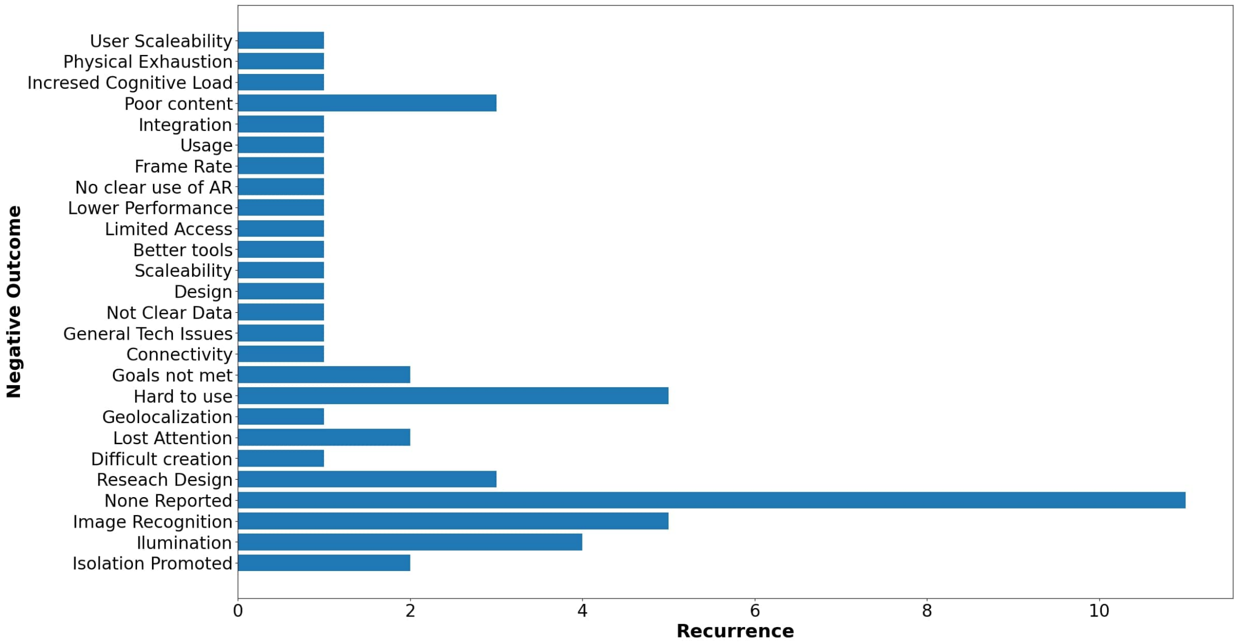

Figure 2.9: Negative Outcomes of Successful Projects

2.2.4 Discussion

The data shows that most projects either focus on *Understand* goals, that is, goals aimed at learning a concept and how to use it, or *Apply* goals, those focused on using obtained knowledge in concrete situations. Primary and secondary level education leaned more towards *Understand* goals while universities and vocational training preferred *Apply*-style ones. When used in a professional context, the prevalent use of AR seems to be as an aiding tool rather than as a learning tool. VR technologies are the main option when training is needed in the workspace, which was outside the scope of this review.

Metacognitive tools were present at all levels, showing an increasing interest in learning-how-to-learn approaches. More complex cognitive goals like *Analyse*, *Evaluate* and *Create* were present in smaller numbers, showing an interest in tackling this kind of endeavour but being deterred by both design and technological difficulties. A small effort to create experiences for the public and with a more product-focus in mind is also noted in a couple of fields.

The visualisation advantages provided by AR technologies are commonly considered as one of its biggest strengths, especially suited for complex concepts, abstract information or elements difficult or impossible to come by. The augmentation element of AR technologies was used often to complement in situ experiences, like museums or laboratories, but it was not as widely used, opting more for a simpler *scan and show* approach. Geolocalisation-based augmentation was used in a couple of cases with a variable degree of success and with several issues associated with the technology. Image recognition approaches are significantly less used and were normally associated with custom-build solutions in need of more development or research.

The primary goal for most projects was to achieve a better academical performance by

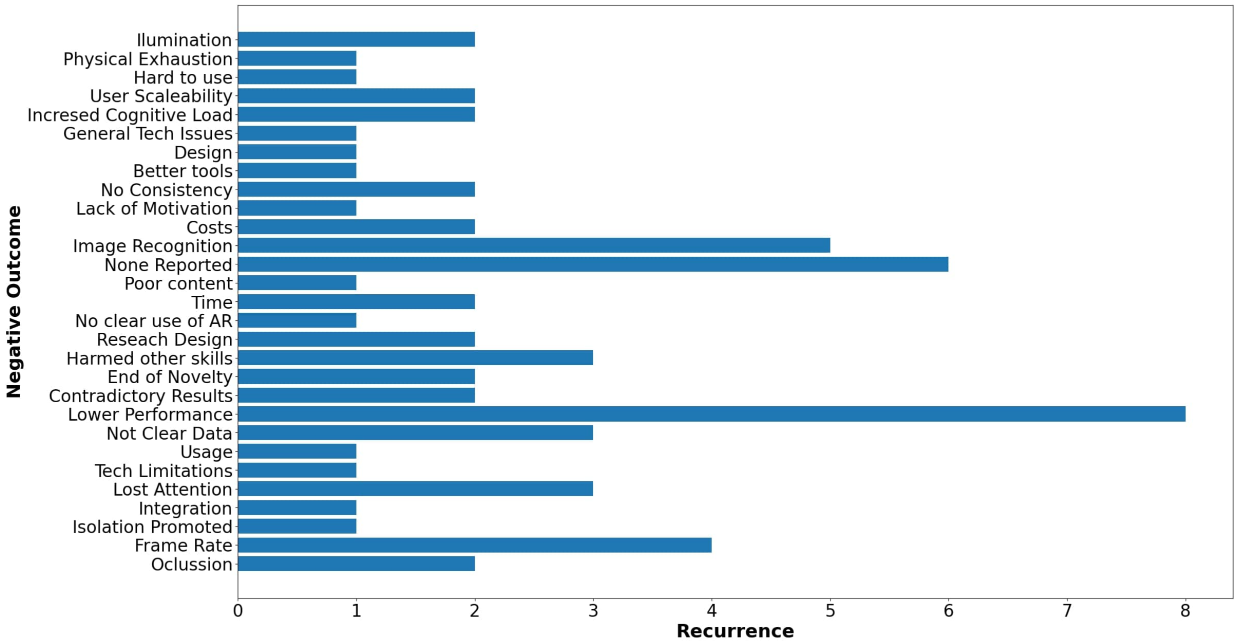

Figure 2.10: Negative Outcomes of Unsuccessful Projects

using AR technologies over other tools or approaches. AR was used to enhance the general motivation and disposition of the students towards learning and to aid the comprehension process through multimedia visualisation, interactivity and assisted practice. Improvements in academical performance were often small but present. AR technologies were commonly seen as more accessible and usable due to the prevalent use of mobile devices as the base platform for deployment and the plug-and-play nature of it.

Although collaboration was a factor in some projects, the general tendency was to create individual tools that helped students to follow an autonomous process. Opportunities for self-paced learning and the ability to have a unique or tailored experience were commonly praised outcomes. This aligns to similar results found in other studies like in Dey et al. (2018) in which the tendency for AR experiences, not only in education, is for individual interactions, with a marginal presence of a collaborative design. It is also clear that AR has been used to explore a diverse number of fields in science, technology and humanities, but is better suited for goals in need of a novel or interactive visualisation or a real-time display of information. Gamification designs that require quick access to information, movement or the visualisation of fictional or no-easy-to-access material are also a good fit for AR technologies. In these scenarios, collaboration is more common but mediated more by the activity rather than by the AR tool.

As for negative outcomes, the most common unintended result is to become a distraction from the educational goals or to impose an extra cognitive load for students, signalling a poor integration between the technological and the pedagogical design. Lack of improvement on academical performance, lack of interest, distractions and lack of content are common reported issues that can be traced to not specifying a clear goal or a clear path to achieve the learning objectives with the use of AR.

Fewer non-STEM subjects were found in the surveyed data, but when present, were often

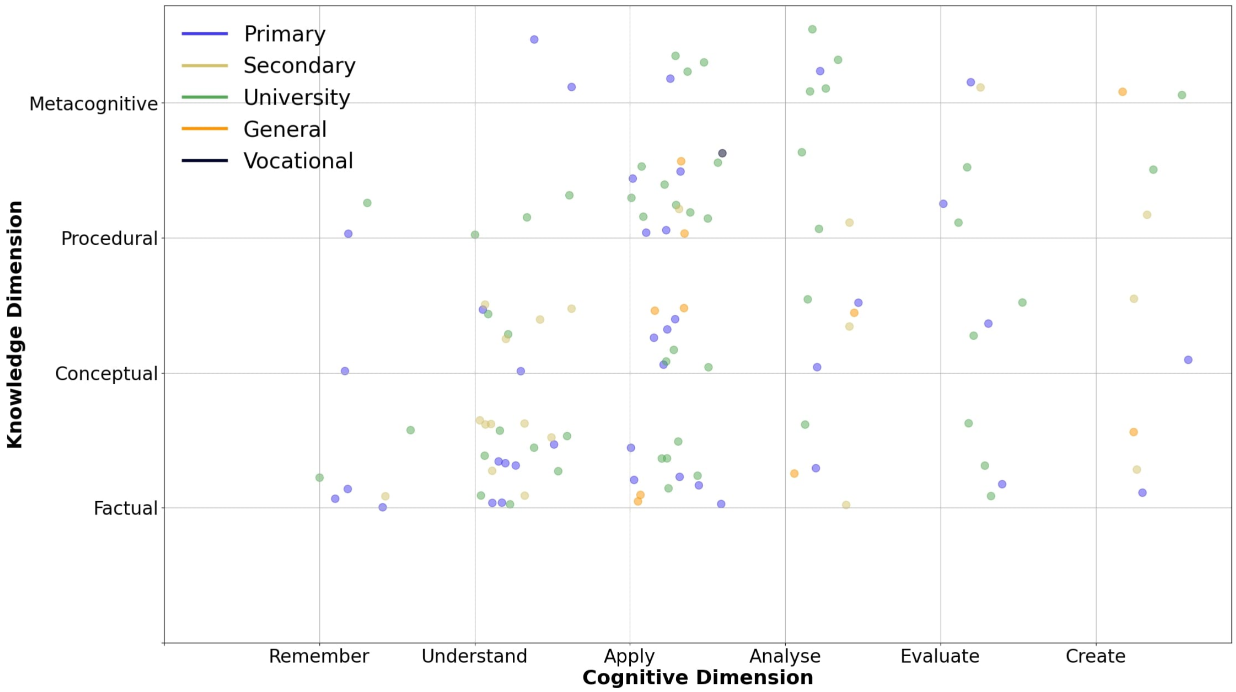

Figure 2.11: Distribution of Educative Goals per Target Demographics

associated with high level cognitive goals. A higher number of projects in subjects like arts or humanities is needed to better understand the advantages and correct uses of AR technologies in these subjects. STEM related subjects, on the other hand, could experiment with more demanding and varied goals, especially in the *Create* dimension.

Validation is mostly done through performance assessment and comparison tests. While offering a good analysis based on data, it also limits the type of discussion to have around the tool, the type of data that can be analysed, and the types of results expected and that can be considered valuable or a success. Understanding not only the statistical gain of the students, but also their experiences, the behaviours observed, positives and negatives, and how the tool relates to the general pedagogical design are also valuable sources of data to consider when analysing the impact and possible benefits of a tool.

Three main ideas can be stated as final conclusions to guide the future development of this research project. First, the identified advantages of AR technologies fit better with *Understanding* or *Applying* abilities, but many examples like Hsu et al. (2019) and I.-J. Lee (2021) show the value of proposing more complex goals or testing the technologies in unconventional ways. There is an opportunity to understand how the technology behaves outside of its most trending usage.

Second, AR technologies are being used mostly as a visualisation tool, with motivation and accessibility as the main driver for the pedagogical design. Improvements in academical performance are achieved using a multimedia approach and appealing to as many different learning styles as possible. The augmentation element associated to AR has not been used as extensively in comparison. This can be explained due to the technical challenges associated with the image recognition process, and how issues in this area affect the overall stability of

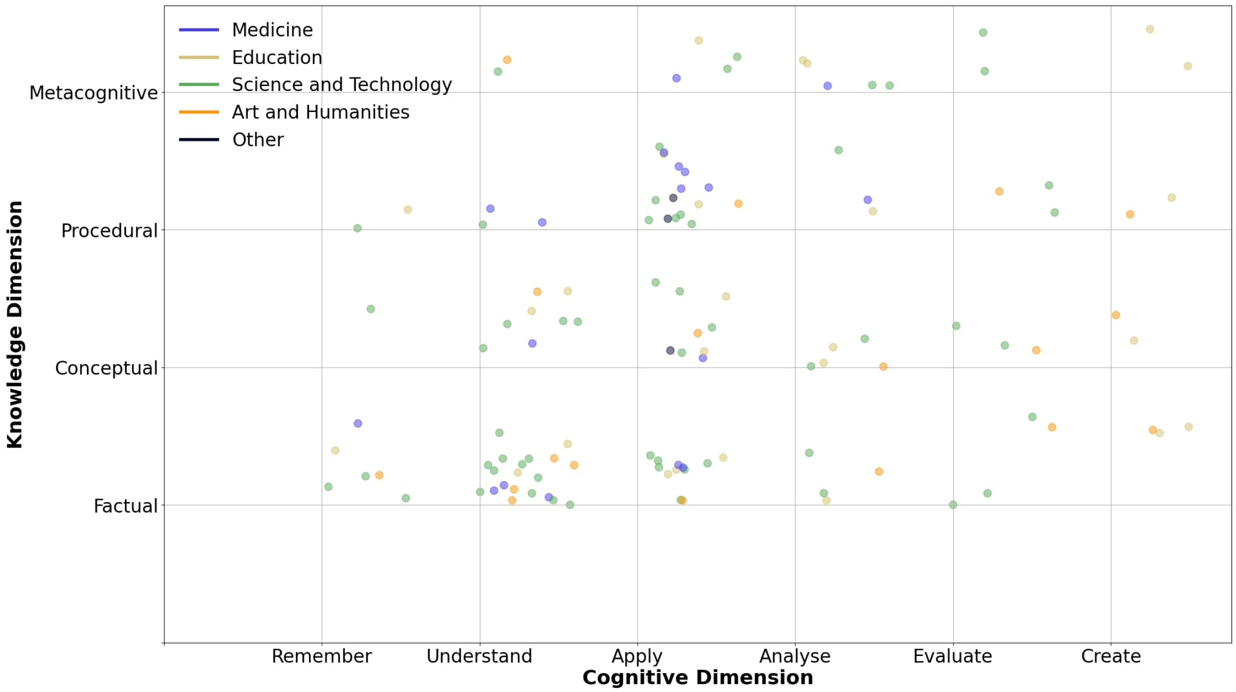

Figure 2.12: Distribution of Educative Goals per Target Subject

the tool and its perceived quality by the users. It is probable that the newest generation of consumer-ready tools will provide an easier way for designers to propose experiences more focused on augmentation rather than only visualisation. Some very interesting results can be achieved when augmentation is the focus of the experience, like in Lim and Lim (2020) or Schaper et al. (2018).

Finally, mobile devices are the most used platforms for development. This facilitates accessibility to the technology and 1-to-1 device usage in any classroom. Yet, most projects found are designed as individual experiences, prioritising student autonomy and promoting a self-pace approach, which under-uses the connectivity advantages provided by mobile platforms. Few examples like Cheng et al. (2019) and Kim et al. (2017) are a multi-user or collaborative design.

2.3 Collaborative AR

From the many insights obtained from the SLR, it is possible to highlight a potential hole in the literature related to collaborative AR experiences in education. A significant majority of the projects surveyed prioritized individual interactions, and those few using multi-user approaches perfectly highlighted the interesting possibilities that the technologies offer around collaboration, teamwork and project-based approaches to learning.

We can expand the information obtained through the SLR and focus some more on the concept of collaborative AR in general, the research done around the topic and the most notable developments found in the literature. If we analyse the topic outside the field of education and consider how AR is being used as an aide in different tasks, we can find multiple

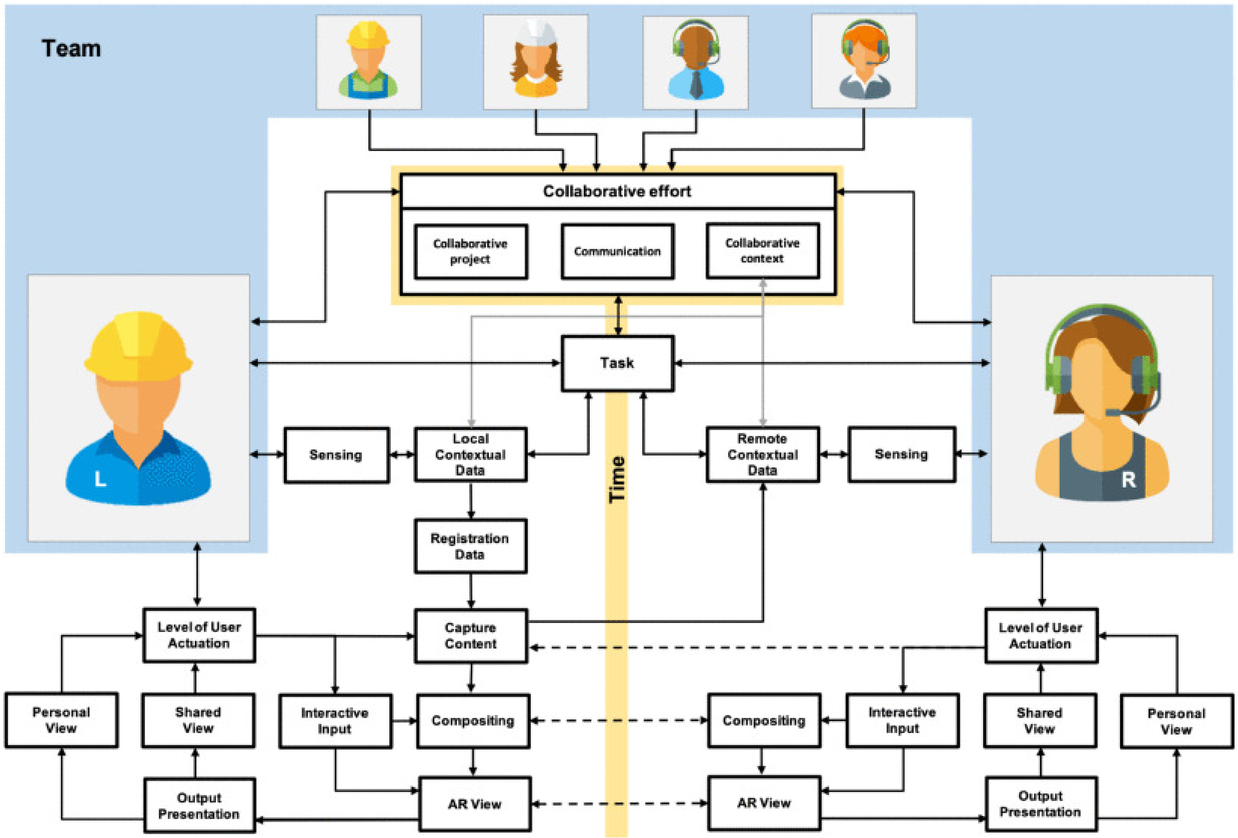

Figure 2.13: Conceptual Model for Collaborative AR as proposed by Marques et al. (2022)

tools that have been tested in diverse scenarios and have been proven to offer benefits and advantages in terms of performance, communication and workload management.

A first approach, before getting into the realm of analysing specific tools, is to understand the domain itself, identify the most important concepts in the core of a collaborative tool and their relationship with AR. Marques et al. (2022) approached the domain of collaborative AR with that same perspective and offered a detailed mapping that can help in categorising and describing any project that could fit in this field.

Figure 2.13 shows the most important components in a collaborative AR tool that interacts with the user and within each other to provide a multi-user environment. Of the key components identified, three main categories can be identified:

- *Context awareness:* Gathering data from the context, either the real space or the actions of other participants.
- *User interaction:* Sensing the actions of each user, each point of view and each flow of information.
- *Composition:* Using data from the previous categories to create the AR view of each user.

These components do not differ much from a generic conceptual model of any AR or MR interaction. The main distinction lies in coordinating all of them through time and different users. The biggest addition is the category *Collaborative Effort*, composed by the components project, shared context, communication and coordinated effort. These components resonate with the fundamental characteristics off an effective collaboration exposed in section 2.1.4. It also serves as the connective gear between users and holds the bulk of interactivity with the app.

It can also be appreciated that the model does not forces a particular architectural design, save for the general communication pathways proposed between components. Another noticeable decision is to coordinate all the users through the *Task* component, giving the collaborative effort a project-oriented nature and a more hands-on approach to the type of applications that could be designed using this model, which can be seen as a first stepstone for a more detailed proposal in the future. It is also useful as a shared language to understand, communicate and analyse the decisions and particularities of a specific implementation, such as this research project.

In terms of extracting interesting implementation cases from specific applied fields, *Collaborative Design* is a domain in which AR has been integrated into work processes a lot. Different projects experiment with tasks requiring communication of ideas, co-creation of products and balancing the workload of the team members. X. Wang and Dunston (2011) for example, proposed a comparative study that showed performance improvements when using a design pipeline that integrated a MR component. They focused on offering both a face-to-face collaboration process as well as a tangible interaction design using image markers.

Fata Morgana (Klinker et al., 2002) is a similar proposal in terms of implementation, but focuses more on the presentation of info rather than in the interactions with models. The collaboration aspect lies more outside of the tool; its main purpose is to show information while being aware of the context and giving the user freedom of mobility. The Magic Meeting (Regenbrecht et al., 2002) offers a similar concept by providing a physical setup to interact with the markers and the digital models. Again, the emphasis is more on the communication process, where the tool offers the means of interaction with the digital construct, but the main objective is what is happening between the users.

Another prominent field for the use of AR is medicine. The technology has been tested several times in diverse areas like procedure aide, data visualisation and training. When exploring collaborative instances of AR tools, we can find some modern projects focused on helping in decision-making process based on data and consensus, such as the examples shown in Friedl-Knirsch et al. (2024) and Yasojima et al. (2012). But the most prominent application is the use of the technology in procedures using telepresence, tele-assistance or tele-mentoring. AR has been used as an enhanced remote collaboration method, allowing for communication enriched with visual data and real-time interactions. Wisotzky et al. (2019) is one of the most radical examples in exploring the boundaries of the technology. The authors propose a framework for full digital reconstruction of a surgical procedure to enable remote collaboration through AR, highlighting the challenges related to image recognition and network stability. Other projects aim at helping visualise and guide complex procedures and research goals that relate more to feasibility and safety (Alismail et al., 2019; Jun et al., 2023). Other common examples, like Augello et al. (2023), Pooryousef et al. (2024), and Yu et al. (2023) focus more on the display of information, remote or collocated, using a more common marker or feature recognition approach, and providing aid in brief processes, evaluation or mentoring.

Mentoring or aiding students, specially aiding in practice scenarios for surgical procedures, is a major research venue in medicine, evidenced by most of the literature found supporting these types of projects. Just to name a few outstanding examples: Gasques et al. (2021), C. Y. Huang et al. (2018), Moro et al. (2021), and Schott et al. (2024) all show interest in providing students with tools to understand and practice learning material through visualisation and interaction, while also using a collaborative learning approach.

Other fields also focus more on using AR for communicating ideas rather than direct interaction with the immediate context. The augmentation is focused on giving form to abstract concepts or superposing non-existent elements into existent ones. The collaboration aspects focus on sharing the visualisation or interactions between users, or giving presence to remote users in the collaborative task. Some examples of note can be the project proposed by Garbett et al. (2021) in the field of construction, Shen et al. (2010) in collaborative product design, Ahn et al. (2019) in architectural design and Sanabria and Arámburo-Lizárraga (2017) in arts and creative processes.

Telepresence and remote collaboration appear to be a common goal, but it is also possible to find examples that prioritise face-to-face collaboration in diverse activities. The Collaborative Web (Billinghurst & Kato, 1999) explores the concept of collaborative browsing, sharing information and points of view. The work of Billinghurst and associates in general provides a fundamental view of the challenges and possibilities related to collaborative work in AR and focuses on identifying key design issues around interaction and the technological framework that can be used to propose a solution. Another fundamental example is The Studierstube (Schmalstieg et al., 2002), which shows the necessity of an interaction design process of tools like this. Operations in a 3D space that are shared with other people are too distant from more common desktop or mobile interactions, thus requiring extra work to properly build an interaction proposal.

Modern and early incursions with the technology offer a valuable list of considerations for common scenarios like shared interactions, communication between users and co-located activities. From the reviewed material it can be stablished that several issues need to be addressed in the design stages of any AR solution of this nature:

- A library of interactions with the physical and digital environment that is intuitive, usable, easy to learn and that accommodates the hardware being used. These interactions must consider the 3D nature of the technology.

- A protocol for communication and interaction between users. Co-located experiences should capitalise on the ability of the users to be close together, but solutions must consider issues related to coordination, change in points of view, distribution of tasks and general connectivity issues.

- A clear integration with the overall purpose of the tool. This is important when the tool is meant to be a complement to bigger processes, like and assembly line, maintenance or training. The tool should considerer the time and effort required for set up, tracking, interaction, changes of focus and priorities, establishing communications, among others issues. Good examples of projects dealing with these types of issues can be found in W. Huang et al. (2013) and Tang et al. (2003).

This small sample of projects show an interest in providing the users tools to discuss and communicate ideas, using the capabilities of AR to give a visual anchor to the concepts being exposed and to interact with them. An emphasis on offering tangible interfaces shows the importance of providing a sense of *physicality* to the digital information used in the task. It also provides an interesting analysis of different approaches to interaction design. AR solutions often need to deal with issues related to interactivity and interface problems that are unique to projects delving in 3D spaces or that want to use the user's natural body movements as a mean for interaction.

One final step in the analysis of the available literature is to situate collaborative AR in the field of education. The SLR in section 2.2 analysed analysed AR in general as a tool for education and found an important focus on single user experiences. Those projects providing collaborative designs were few and showed us a research field that offered interesting possibilities and was currently underexplored. Upadhyay et al. (2024) is a more recent SLR that not only explores AR technologies in education but also focuses on experiences using collaborative design and collaborative learning.

The review identified the importance of mobile technology for AR development and collaboration practices, being the most common deployment environment and the most common technique used for implementing collaboration. Even more notable in comparison to the results found in the SLR of this research is the educational approach used by several of the reports reviewed. Projects like Radu and Schneider (2023), Schiffeler et al. (2019), Vassigh et al. (2020), and Xefteris et al. (2019) not only explore diverse learning objectives, but they also explore AR technologies beyond engagement or performance gains, evaluating aspects like communication behaviours, social interactions, induced behaviours (positive and negative), attitudes towards the learning materials and changes in learning processes.

Overall, reported benefits align to concepts already discussed in previous sections and fortified the position of this research to focus on visualisation, interactivity and mobile communications. When analysing challenges identified in the reviewed material, different authors highlight a couple of interesting ideas. Primarily, there seems to be strong trade-off between offering a recognisable environment through mobile technologies that users can easily interact with and the difficulties the students and teachers found in managing new systems and new process. It is important to consider that any technological intervention in the classroom implies a high degree of effort in integrating it to the normal flow of the learning experience. It requires effort for the technology to become and enabling factor and not an inhibitor or a distraction.

There is also an interesting side effect when assessing the impact of the technology in learning outcomes. Section 2.1.4 briefly mentioned that collaboration is difficult to assess. This difficulty can be exacerbated when paired to other, sometimes contradictory learning outcomes, and when the assessment does not consider the raw effects of collaboration in the results, positive and negative. The same can be said about the noise in the students' performance created by the addition of an intervening or mediating technology. In conjunction, a collaborative approach that uses a mediating technology (not only AR) poses a significant challenge for evaluation, performance assessment and other similar tasks. This can be evidenced by the diverse and sometimes contradicting reports found in the review by Upadhyay et al. (2024) and the one performed for this project in relation to what works and what not.

2.4 Final Discussion

This chapter explored the theoretical framework that supports this research project and examined a varied selection of relevant literature related to the most important issues and research trends in the field of collaborative learning, TEL and AR.

This analysis helped in defining and understanding collaboration as a skill that must be built over time with proper nurture and resources. Collaborative approaches to learning can both aim at developing the necessary skills for effective collaboration and at using those skills, such as teamwork, communication and resource management, to boost individual performance, create opportunities for emergent learning and offer educators tools for better classroom management.

It is especially relevant in the exploration of different TEL scenarios to understand the diversity of configurations and tools that can be used and the short-, medium- and long-term consequences for learning, skill development and teaching. Any technology offers benefits and detriments to all these dimensions, and as researchers and educators, it is part of our responsibilities to understand them and manage them.

AR, as any other technology, has proven to offer unique advantages for education, capable of creating strong visuals, immersion and interactivity. It excels at integrating with the user's context, at providing means for exploration, at data-driven scenarios and, central for this proposal, at facilitating shared experiences. On the core of this research project is the idea that, in conjunction with mobile platforms, AR provides important advantages to the design of collaborative learning experiences by using the physical context of face-to-face collaboration and by innovating and recontextualizing the design of tools for communication.

But it is also necessary to highlight the challenges and issues posed by the chosen technology. It is especially interesting to explore solutions for problems related to connectivity, the diverse ecosystem of AR devices, the negative effects in the development of social skills and the challenges students face when adapting complex process and requirements into their learning projects. It is clear that the technology can be both an enabler and an inhibitor. The goal is to identify the proper approaches to boost the advantages while minimising or mitigating the problems.

In the next chapter, the objective will be to use the information gathered in this chapter to properly structure the research and development goals that will guide this project. The current state of the art helped at defining a useful conceptual model about collaboration, the development of collaborative skills, collaborative learning and collaborative AR development. With this information at hand, now it is possible to explain the formal methodological approach for this research and define the proper instruments that will help in the constructions of and answer to the questions formulated by the motivational background of the research and the gaps in knowledge found in the literature.

Chapter 3

Methodology

The following chapter will detail the methodological decisions taken for the development of this research, as well as a quick outline of the theoretical framework that supports the ideas and procedures followed, the analysis strategy and the constraints and limits identified in the methodology proposed.

The first section will focus on the research strategy, the philosophical background used to support the decisions taken, the approach selected for the research design, and the limits and constraints imposed by those decisions. The second section will explain the most important elements of the theoretical framework that supports this research, considering that this project uses both, an educational research lens and a software development.

In the educational field, this research uses as foundation the views proposed by social constructionism of how knowledge is built and the effect of social interactions and cultural context in the educative process. The collaborative learning approach uses those fundamentals to propose methodologies for learning design, assessment and analysis.

Finally, the technical approach will detail the process used for the design, development and evaluation of *WorkshopAR*, emphasising the design philosophy followed to involve as much as possible the input of the user. This approach is used to better integrate the learning design (of this project and of the target course) into the development of the software. This can be evidenced in the design exercises proposed for the interaction with the functionalities, the overall user experience and the discovery of functional requirements.

3.1 Research Strategy

This project is qualitative and exploratory in nature. Qualitative because at the core of the research questions, there is the idea of description. The goal is to describe behaviours and characterise the technology selected as it is used in the classroom. It is exploratory because, rather than trying to corroborate or to formulate a theory, the goal is to gather enough data to understand the characteristics of the scenario being studied. This will allow to formulate a way forward to deeper analysis of the issues and questions that stood out through the review of relevant literature and previous experiences in the field.

Critical Realism (CR) was identified as a good fit for the philosophical approach of the research. Observation will be at the core of the enquiry process and the data gathered will be subjective in nature, linked to very specific points of view: the observations of the researcher, the perceptions of students and teaching staff. CR proposes an approach to research that takes into consideration the imperfections inherent to understanding reality through such subjective perceptions, finding value in identifying and gathering different views of the same issue (Bryman, 2003; Hastings, 2021).

Since very specific points of view are going to be the main source of analysis, it is important to understand and accept that any data and observation will be flawed and that all results proposed, especially those that try to identify the underlaying mechanisms in the social phenomena observed, will be in need of interpretation and contextualisation to assess validity or applicability beyond the case study here exposed (Easton, 2010).

Section 2.2.4 explores the tendency found in literature on justifying any educational development through improvements in average grades or similar gains, to give the research validity using a very positivistic framework of analysis, while neglecting or obscuring the human and social core of education. CR offers a midpoint between positivism and social constructionism by acknowledging the social dynamics used to build knowledge and learning processes, the ontological framework used to model and interpret qualitative data as well as linking all these ideas with a more empirical understanding of the mechanism working underneath what is observed (Parra et al., 2021).

Both Carter and New (2004) and Scott (2005) highlight that one of the most important tools in CR research is to identify agency and structure, that is, to understand the phenomenon being observed by identifying who is doing things, what are the social connections at play and the reasons behind the actions being observed. This approach requires the presence of both empirical, hard data that can be analysed, as well as the necessity for a more interpretative and qualitative approach to those elements that cannot be described with just numbers.

CR is also a framework well suited to answer *how* and *what* questions. Saxena (2021) explains that the Empirical-Actual-Real model used in CR (see Figure 1) can be used to explain ideas about causation, correlation and the underlying structures driving the empirical events we are experiencing. The research questions proposed for this research both align to this view or were tuned to align with this approach. The main objective is to describe what is happening in the classroom around the mechanisms of collaboration and the technological tools created as an engine of those mechanisms (the what, the empirical), and to create an analytical and interpretative framework to explain those behaviours (the how, the actual). The research process will potentially identify a link between what was observed with the ideas, designs and tools created.

Given the emphasis on observational qualitative data, the best research type that fits the methodology so far described is an inductive approach. The main question (MQ) of this research is aimed at understanding how AR promotes observed behaviours, while questions SQ1 to SQ6 help describing the data with the goal of linking the empirical data with the actual mechanisms in play throughout the educational scenario. The assumption here is that those mechanisms in fact exist, and that they are being built or promoted using the technology that is part of the educational design of the activity.

It is important to clarify that the explorative nature of the data will not be totally blind. It will be informed by a series of expectations and hypothesis about how AR has been used in the past for educative scenarios and the theoretical background found in the literature about collaborative learning, project-based educational design and TEL experiences, all which were

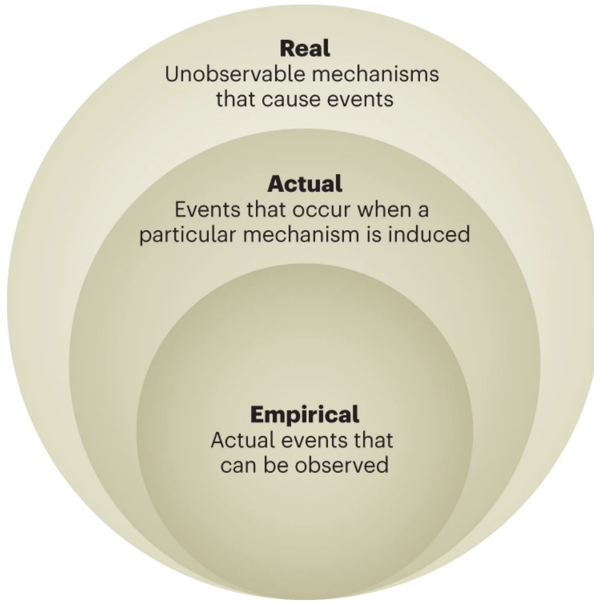

Figure 3.1: The Relationship Between the Real, the Actual and the Empirical (Blackie, 2023)

explored in Chapter 2. This background provides the biggest foundational ideas that will guide the analysis process of the data:

- AR can be used in education as an effective tool for communication and it works by exploring the advantages inherent to the technology in the form of visualisation, interaction and connection with others.
- The core of collaborative learning lies in using social connections as the main driver of the educational design, and the main behavioural changes that could be observed will be present in those points where AR is part of the social interactions taking place.

The analysis process will naturally try to find indications of those ideas in the data, mainly because it has been apparent in the results found in previous works. It is also important to understand that the main research drive of this project is to identify all those elements that do not align with those ideas (if any), try to link them to underlying mechanisms and refine those initial ideas based on what has been observed, and create a new or distinct starting point for future research.

Following the research strategy chosen, it was decided that a case study would fit best with the philosophy and research types adopted. This strategy offers a promising approach to manage the kind of observations proposed, as the research is more interested in behaviours and personal perceptions and the way they change thanks to the intervention designed. A detailed case study is well suited to create the scenario where these changes can be observed, even if it is difficult to identify and understand all the variables and mechanisms operating behind the observed scenario.

Case studies produce the type of context-related knowledge that is most valuable in learning scenarios. They are also a very useful tool for contexts in which human behaviour needs to be observed and in which a community is an integral part of the context (Hamilton & Corbett-Whittier, 2013). Case studies are ideal tools to identify problems, outliers and out of the norm observations. Moreover, thematics and problems characterised as rich or complex are perfect subjects for case studies, which have a good set of tools to capture, organise and analyse extensive narratives, which value lies more on its density and richness rather than the factual data that can be extracted out of it (Grauer, 2012; Seale, 2007).

Cober et al. (2015) and Tay et al. (2017) represent example projects of educational case studies centred around a TEL intervention. They demonstrate and set a precedent of many of the foundations used in this research around technology in the classroom: a design process that involves the users and a feedback loop that adapts to the circumstances observed. These examples also highlight some unique benefits of a case study approach like involving the community as an active part of the research, enriching the discussion of the ideas generated by using different points of view and facilitating a two-way avenue of the benefits of the research, often allowing the researcher to actively and directly make an impact on the community well beyond the research scope.

Finally, and given the prevalence of qualitative data in the research strategy selected, the analysis technique chosen for the project will focus on content and thematic analysis. These approaches offer good flexibility to analyse the data obtained from different points of view that are subjective and anecdotal in nature. They help in synthesising large quantities of textual data, organise them and extract relevant points of interest to start building results and conclusions (Kiger & Varpio, 2020; Neuendorf, 2018). Specifically, the thematical analysis of the research will focus mostly on identifying themes related to:

- The use of technology to manage team activities in the classroom.
- The prevalence and importance of technological tools while creating channels of communication among the groups.
- The interaction between the different tools used by the students.
- Instances of reflection and analysis of the activities in the classroom thanks to the use of the tool proposed for the intervention.

Although these themes are the focus of the analysis, it will be very important to track and identify any other relevant content that emerges during the observation phase, especially if it is related to the collaboration process or the use of technology in the classroom. Additionally, these same analysis strategies can be used in the evaluation process of the software development, offering a structured method to gather and analyse the qualitative data gathered about the usage of the tool and the perceptions of the users while using it during the instances of controlled test and during the intervention in the classroom.

An important effort is also proposed to offer some triangulation to the data obtained from the case study. On one hand, the software development process requires a substantial data analysis coming from the reported results of similar past projects and from the exercise in user-centred design that will be followed during the initial stages of development of the tool. This process is also iterative in nature, and the feedback collected and analysed during the testing stages will inform not only the development of the tool, but also the observation process.

A second approach for triangulation is the use of two extra data instruments to complement the observation process of the case study. One is focused on unstructured interviews with the students with the objective of gathering data about their collaborative process, their approach to teamwork during the course and the tools they used to make work possible or easier. The second instrument is a similar set of unstructured interviews focused on the teaching staff. These interviews will gather information about the influence of teamwork and collaboration in the educational design of the course, the criteria used by the staff to assess a successful teamwork and their personal views on how this project-oriented activities could be used to prepare the students for their professional lives.

3.2 Research Constraints and Limits

The biggest constraint on the timeline proposed for the development of this research is the access to the Industry Project course. Any cohort of the course runs through two semesters, and it was important to follow the development of the project from the first contact the students have with the group to the final presentation of the project. The observation window to achieve this objective is one academic year, which can be translated to 24 academic weeks along two semesters of 12 weeks in which observation of the development of the course is possible.

This situation forces the project to adapt a cross-sectional approach, which tries to describe in as much detail as possible the effects observed of the technological intervention for one cohort of the Industry Project course. The main objective is to create an internal comparison between those groups participating in the course as they would normally do versus those using the AR tool. As with most cross sectional studies, this approach offers a quick way of gathering data and the observations cover multiple variables of interest that can be analysed later, at the cost of creating a situation in which the data will be biased to the particularities of this sample, making it harder to propose any generalisation or causality (X. Wang & Cheng, 2020).

Sampling inside the cohort of the course will be guided by convenience and the disposition of the students to participate in the research, since it is important to prioritise their academical work and minimise any disruptions to the classroom. Observations will be organised in singular work sessions that the students will be asked to participate in. Each work session will be guided by the students, setting the goals of the session based on any requirements they have during that instance of development of their project. The observation will try to focus on different aspects of the course that had not be observed previously. The researcher will also try to show the students a different aspect of the AR tool one functionality at a time, facilitating their involvement in the observation and helping with the iterative development of the software. If possible, the observation will try to gather different work sessions focused on:

- The initial contact and formation of the group.
- Understanding the project and gathering information about it.

- The initial design proposal and the flow of information between the different roles in the group.
- The creation of material for the project.
- Evaluation and correction of the material created based on the feedback of the teaching staff and their peers.
- A final evaluation of the work done by the group at the end of the project.

Ideally, the observations can gather instances of these activities for groups using the AR tool and other groups working as they normally would. The observations themselves will gather data of the student's work using mainly audio and video recordings, as well as the personal notes of the researcher. Data will be initially organised based on the week it was taken (to determine the overall development stage of the course's project), the composition of the group observed and the functionality of the AR tool that was prioritised.

There are clear limits and constraints inherent to the research methodology chosen that are important to acknowledge and that will affect the results obtained:

- Case studies are not well suited for generalisation and extrapolation (GERRING, 2004; Wiersma, 2000). The results obtained through this project will be linked mainly to the case study selected and its unique characteristics. An important effort was done to select a case study that uses all the elements of collaborative learning needed for this research, as well as presenting a unique an interesting scenario which can provide insights usable in future research.
- The time constraints of the project cause the data points to reflect only one cohort of the course. Though it was possible to follow the whole development of the project, comparison among other cohorts would minimise the noise generated from difficulties and circumstances present in this instance of the course that could have affected in a negative way the observations obtained.
- The prototyping nature of the technological intervention also causes a lot of noise generated from the friction with learning and using the tool. This often is evidenced by observations that highlight the most comments and opinions of the tool around these disruptions and not the educational design.
- The predominance of qualitative data generated from this study needs to be interpreted and contextualised. In the same way, the focus on behaviour and underlying mechanism of social activities makes it impossible to create answers and observations that are not subjective in nature and marked by the biases of the researcher. An important effort is done to separate the observations from the analysis. The interpretation of the researcher will be well documented, but hopefully this research will also offer the opportunity for future work to recontextualise and propose different views to the data obtained and documented here.

Considering these constraints, it is important to follow the documentation of this project as an exercise in experimenting with the application of a unique technology in the classroom, basing the evaluation method of the proposal on its observed effects in the learning process rather than in the assessment and end results. This is done in hope of identifying good and bad influences that could be linked to the use of the technology proposed. In the end, the analysis of the results documented in this thesis can be used as a starting point to develop similar experiences in the future in different contexts, either trying to emulate the same positive results or working on the negative ones, inciting more discussions and pushing forward the understanding we have about the use of AR technologies in the classroom.

3.3 Theoretical Framework

So far it has been explained that the approach to this research is through the themes and situations found in the classroom using two complementary fronts. This is first and foremost an exercise in TEL, with the firm idea in mind that it is impossible and neglectful to ignore the use of technology in the classroom. The first premise that supports the proposal is that mobile technologies can be used in a positive and constructive way to promote learning and good behaviours, when used correctly and with the proper guidance. The educational approach that guides the project focuses on the tools and theories that view education mainly as a social phenomenon and as an ongoing construction based on our environment and experiences, as it will be explained in the following sections.

The technological development followed for this project also acknowledges the challenges found in the implementation of educational technology. The focus, besides a successful implementation, is on executing and documenting the design process followed to create ideas and promote discussion about how to integrate the educational and software design processes, how to align functional and learning goals and how to incorporate the students and the teaching staff into the process.

3.3.1 Educational Approach

The theoretical framework behind the educational proposal is in CL used as a teaching tool to promote the communal construction of knowledge in the classroom, to play with the strengths and flaws of both students and educators as well as facilitating the development of transversal skills such as teamwork and communication.

Collaborative Learning Structure

CL is seen as a philosophy of education that prioritises communal exploration over uniformity (Eskiyurt & Özkan, 2024). It assumes that learning is an active process that depends on rich context, diversity and is inherently social (B. Smith & MacGregor, 1993). In a collaborative design students benefit from a joint effort to solve problems, a diversity of skills put together for a common purpose and a transmission, active and passive, of knowledge and skills between members of the group (Buchs & Butera, 2015; Laal & Ghodsi, 2012).

CL also goes beyond grouping students for an activity, it is an active educational design that seeks to integrate the learning goals of an activity into the dynamics created in the group.

Laal and Ghodsi (2012) remarks the importance of creating positive interdependence between the members of a group, where the success of one individual is linked to the success of all the others. This can only be achieved if the learning design allows for the group activity to create considerable and meaningful interactions, mechanisms for personal responsibility, development of social skills, and spaces for the group to self-evaluate (Johnson & Johnson, 1990). CL can be distinguished from similar concepts like peer learning, tutoring and cooperative learning by the way it frames the relationship between students. It proposes an equal relationship between peers that develops high levels of mutuality, that is, interactions that must be executed in both ways of the relationship (O'Donnell & Hmelo-Silver, 2013).

The education proposal of this research is founded on these characteristics. It is possible to summarise the goals of the TEL intervention with three objectives:

1. Use the visual tools provided by AR to reinforce ideas of interdependence and mutuality by highlighting important structures of the collaborative work such as roles, tasks and communal evaluation of progress.

2. Leverage the communication features of a mobile deployment to provide extra channels for meaningful socialisation.

3. Explore the concept of *augmented collaboration* using visuals and interactions to give students a different way to approach elements of teamwork, team management and resource management.

Paired with the requirements of the course, the learning activities that *WorkshopAR* is meant to support are analytical and creative in nature, aimed at evaluating the collaborative work of the students at a high cognitive level. If the design purpose of the tool is to provide a technological approach to develop the skills needed for an effective collaboration, the observation stage of the project will evaluate if the students that are engaging with the tool are using it in a way that helped them with these learning objectives, either:

- To better organise or show the analysis done to the project through an understanding of the context of the problem, the use of their previous knowledge to propose a solution and communicating their analysis to their peers or the clients.

- As a tool that helps the creative activities needed for the proposed solution by pooling resources, help the discussion of ideas or approaches, help to organise tasks or keeping track of progress.

Observations of this type of activities, positive or negative, will be used to construct a satisfactory answer for the research questions, and will show that the tool is indeed highlighting the aspects of collaborative work that are meant to be reinforced: organised and meaningful interactions plus a structured way to manage the interdependent activities taking place in the group.

Available literature has also identified major obstacles that have to be acknowledged when proposing a CL design. Section 2.1.4 highlighted the most relevant literature found around this discussion. Other relevant work like the one present by Hansen (2006) and Pfaff and

Huddleston (2003) also offer an important analysis of the challenges found in collaborative settings that where critical in the design of *WorkshopAR*. Of the ideas mentioned, it was considered important an interesting to emphasise:

- The imbalances in dynamics, power and assessment that are often found in teamwork and how the students negotiate and work with them.
- The hurdles in communication that often are the focus of friction and failure in a team.

Finally, this is also an exercise in collaboration mediated by technology. There is ample discussion on the phenomena of online collaboration (Hathorn & Ingram, 2002; Rodchua, 2017), but the proposal of *WorkshopAR* is to identify how this trend of asynchronous and discontinued communication is used outside the web and in the physical classroom, with the main intention of identifying, if possible, positive or interesting behaviours that can be nourished to create a more positive relationship with technology inside the classroom, akin to the proposals made by authors like Dyer et al. (2015) or the analysis of available options and approaches from Olson and Olson (2012).

Social Constructionism

If CL is the central pillar of the educational proposal of this project, then social constructionism is at the core of the educational philosophy found in collaborative learning. One of the main premises of social constructionism is that the cognitive development of people matures thanks to the guidance and collaboration with others (Rogoff, 1990). It is impossible to learn in complete isolation.

In this model of learning, communication is a key ingredient since language itself is the main driver for the construction of knowledge and the core medium for its transmission (Renzl, 2007). The immediate context, the cultural background, cultural artifacts, the relationships that had been build and the sense (or lack of) pertinence to a group are all fundamental factors in the process of creating knowledge, and they are all inherently social (Burr & Dick, 2017; Edley, 2001). It is no surprise then that CL echoes all these elements and bases most of its methodological approaches in social constructionism premises:

- Focus on learning and working in small groups to facilitate the creation of meaningful relationships.
- A prevalence of discussions and debates as the main tool for the creation and transmission of knowledge.
- Use of tools that support communication.
- Guidance and mentoring as the primarily teaching intervention.

It is important to clarify that this research takes a stance in social constructionism and not in constructivism. The main reasoning follows the definitions used by Rob and Rob (2018). This research is focused more on the social aspects that can be observed in the case study, it is more interested in the behaviours of the group and the way communal knowledge is built (which aligns better to the approach of constructionism) rather than the cognitive process of the individual interacting in the group (a more constructivists view). In the end both terms are rather interchangeable and complementary, and both views agree in the fundamentals, but it is an important distinction to establish and to keep consistent through the research.

Another important distinction is between weak and strong constructionism. The main difference lies in accepting the existence (or not) of *brute facts*, absolute and fundamental realities (weak constructionism), or rather considering that reality is completely created by social convention and interaction (strong constructionism) (Lawson, 2002; Travers, 2004). In its core, the distinctions are not fundamental for the proposal of this thesis, but a strong constructionist view aligns better with the analytical methods proposed in the methodological design, since all observations are of social behaviours, and need to be filtered and understood through the social context, cultural and linguistic frameworks of the case study. This also supports the view that any observation and analysis done in the case study cannot be objective or completely unbiased. The context around the classroom and the background of the participants and the researcher, among many other factors, are important determinants of the results of this research.

Andragogy and Educational Design

The last fundamental guideline in the educational approach of this project is andragogy and the design principles in education for adults and life-long learners (M. S. Knowles et al., 2014). This document has purposely tried to avoid the term pedagogy because it recognises critical elements that characterise an adult learner and that are integral to the educational design proposed for the project:

- **Adult Learners thrive better with self-direction:** The proposal can offer the flexibility needed for the students to explore the material at their own pace and with personal objectives that the learning process could be inciting.
- **Adult learners have a bigger set of past experiences:** There is a bigger pool of knowledge and skills that can be leverage for the educational design.
- **The classroom has a more heterogeneous background:** The classroom benefits and uses the different backgrounds and abilities of the students to boost the collaborative aspect of the learning goals and uses the different roles that the students can support as a backbone for the course project.

An additional consideration is the approach to technology proposed by this project, which makes more sense, is safer and offers more value in a context with adult learners (Jimoyiannis & Gravani, 2011; Shopova, 2014). Since the proposal focuses on identifying the benefits of communication and collaboration mediated by technology, it is important to balance the approach with the evidence that also shows benefits by withdrawing access to mobile and smart technologies in young learners (Cheriti, 2025). There is value and a necessity to create sufficient and actionable digital literacy, but not at the cost of other aspects of development.

Within this framework, as well as based on the ideas mentioned in previous sections, it is possible to summarise the educational design behind the development of in three main ideas:

- A project based on CL, aimed at using collaboration to reinforce the work done in the course and apply it in a complex context, but also as an exercise in learning the essential tools of collaboration and teamwork to apply them in the professional space.

- An educational philosophy guided by social constructionism and highlighting the importance of transferable knowledge, learning communities and learning spaces.

- A teaching approach based on the necessities and opportunities found in adult learners, benefiting from a deeper and more diverse pool of knowledge and experiences that can be applied in new contexts and transferred to the community.

The following educational design is proposed based on these ideas. To define the learning objectives proposed by *WorkshopAR* it is important to distinguish between the objectives proposed by the Industry Project course and those that *WorkshopAR* is proposing or reinforcing. The development of the case study will be explained in more detail in Chapter **??**, but the general idea behind the learning objectives of the course can be explained as an exercise in the analysis, proposal and evaluation of an architectural design for a construction project. Within this context, the teaching intervention proposed for this project is based around the following learning objectives:

> LO1: Plans and follows a structure for a work session that prioritises the accomplishment of a clear goal and identifies the roles followed by all members of the group and the resources needed during the session.
>
> LO2: Collaborates to accomplish a well-defined task that requires the coordination of resources, discussion among members of the group and a process for making decisions.
>
> LO3: Communicates and explains ideas or opinions about a decision to be taken and collaborates with the team to reach a communal decision.
>
> LO4: Evaluates the feedback received by peers or clients to determine what actions must be taken to correct or expand the work done.

These goals prioritise a simple structure in the collaborative process, starting by analysing how the students plan a task, how they organise their work and how they incorporate feedback into their project, closely following the structure proposed for the project itself in the Industry Project course. Additionally, the goals are also separated from the medium and instructional techniques selected for the project. The use of AR, interaction and visualisation is meant to be tested against these goals, which means that observations and comparisons with the other groups not using *WorkshopAR* (and using other types of approaches to their collaborative work) are possible.

The main learning activity associated to the educational proposal of *WorkshopAR* are the work sessions of the students. Within a work session, it is expected that a group focuses on a set of tasks to fulfil during the session and uses the tool to organise their work and support any discussion or debate. This is a very open approach to the activity, meant mostly to facilitate the observation process during the execution of the case study, and to disrupt the work of the students as little as possible. Some structure can be given to the students to guide their work and emphasise better one or more of the learning objectives. Three types of activities can be suggested that are in line with the learning goals and can be useful, especially if the group finds itself a little directionless or without a clear goal in mind for the session:

- Identify long term goals and divide them in smaller achievable tasks. Propose and discuss these tasks, considering priorities, resources, dependencies and the responsibilities of each role.
- Discuss a problem that is impeding progress in the work right now. Propose solutions and discuss how to implement them.
- Identify individual tasks that each role needs to fulfil, either alone or with the help of others. Distribute those tasks and periodically report your progress to the group.

As with the learning objectives, these activities run along the work that must be done with the course without enforcing approaches too specific or that cannot be adapted to the situation of a group. It is worth mentioning that these activities also facilitate observation of how the tool is helping to give structure to those activities by allowing flexibility to use it (or not) in different ways and with different objectives in mind.

The assessment process of this educational design is minimal, and it is concerned mostly on comparing the assessment done for the course between the different observations throughout the case study. Hopefully, it is possible to observe that there is a positive change in results in the groups using *WorkshopAR*, but this research is more concerned in identifying changes and differences during the development process. Nonetheless, some criteria can be established to guide what can be considered a positive or negative impact.

Positive observations can be characterised by constant communication between the team members, coherency between the different aspects of the proposal and a clear visualisation of the decisions taken for the project and the reasoning behind them. In contrast, poor performance can be characterised by a disjointed work without a common or clear objective, noticeable lack of communication between the group and a failure in meeting deadlines or show consistent progress in their work.

3.3.2 Technical Approach

Yadav et al. (2015) establishes that an agile development process with an iterative or cyclical approach is better suited for rapidly changing environments and for the exploration of ideas. The nature of *WorkshopAR* requires a development process that can quickly be adapted and refocused as the testing and observation goes on, when the context of the course is better understood and when feedback from the students is gathered and analysed.

As explained in the introductory section of this chapter, the exploratory nature of the process does not imply that the process must advance completely in the blind, it is guided by several premises that focus the development into achieving the most important aspects of the research goals. First, the development focuses on visual and interaction design, multiuser interaction and communication features. Additionally, despite the initial open nature of the case study, a set of common functionalities can be used to represent the core ideas of CL that can be enhanced through AR. Independent of the particularities of the case study, *WorkshopAR* is conceived and designed to provide solutions for:

- Visualising the interdependence of the group by making the composition of the team clear, the structure implemented and the work being done at a given time. This to offer guidance and promote discussion as suggested by Salas et al. (2015).
- Keeping track of the visual resources used in the work session (ideas, models, discussions, experiments).
- Create a shared experience for visualisation, interaction and communication.

In a similar way, the conceptualisation of the tool was first tackled through a focused design process with the aim of understanding interactions in 3D and how to effectively use the inputs and tools often available in AR technologies. This focused design identified ideas to create an interaction experience that highlighted the advantages of AR for users that in general terms are not familiar with the technology.

User-Centred Design

The first step in the development of *WorkshopAR* is a user-centred design aimed at the interaction aspects of the experience with the hardware and with the AR tools built to accomplish the learning goals established. The main objective is to identify expected interactions with AR technologies and formulate a proposal for the overall experience based on the aggregated feedback gathered from the users.

Chammas et al. (2015) establishes that a user-centred design is characterised, among several other things, by:

- Being project-oriented, tailored to specifics needs and driven by data.
- Interacting and engaging with the user. It is an exercise in communication.
- A constant process of analysis and refinement.

A user-centred design can move in a wide spectrum of implementations depending on how much involvement the user has in the process and the steps in which input has been gathered and implemented. Marti and Bannon (2009) highlight the need of considering the context of the project to determine how much access the user can and should have in the development process, the points in the project life-cycle in which the input of the user is more valuable and the importance of identifying and working with all the different users that a product may target.

The development of this project benefits from a user-centred approach because it is an exploratory context. This type of design process is better suited for scenarios in which the user has a personal or strong connection to the product or when tailoring a solution to a particular population. User input is also more valuable when used to guide research, explore

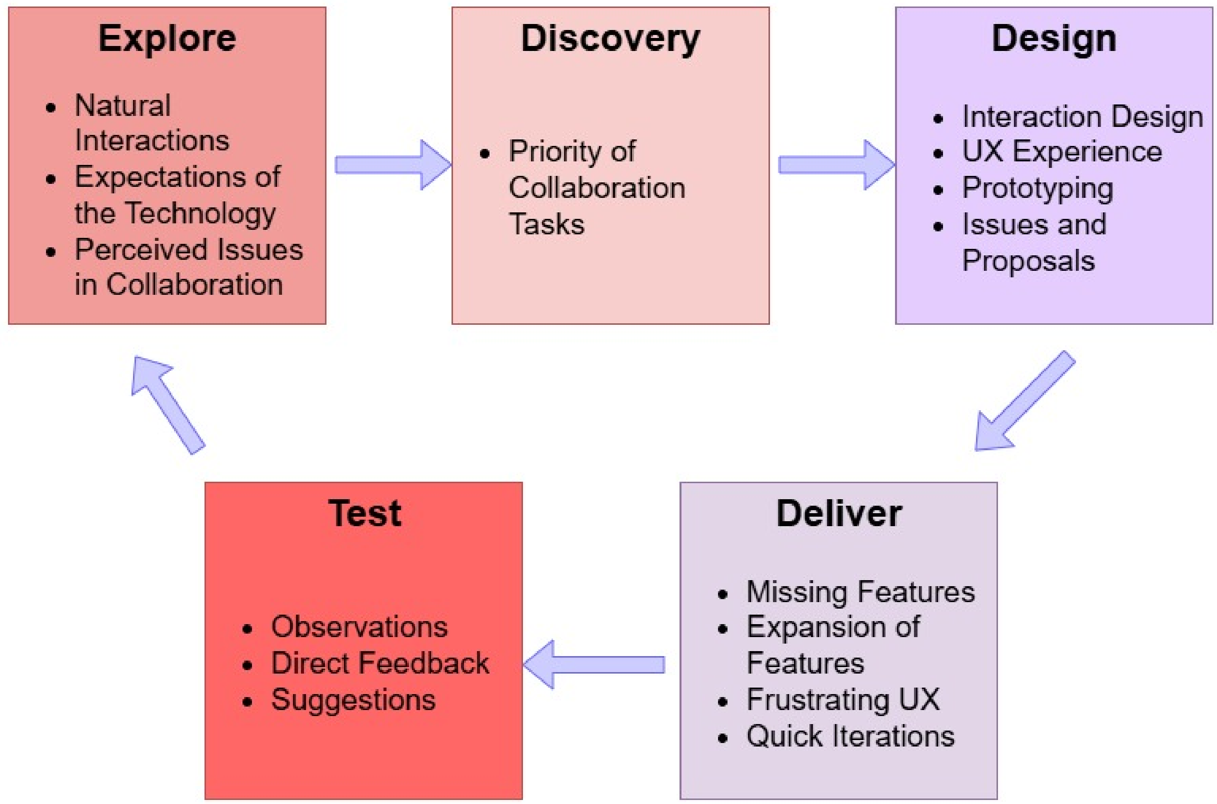

Figure 3.2: User Input in the User-Centred Design Process

design pathways or identifying issues with an idea or a prototype (Goodman-Deane et al., 2010; Tosi, 2020). Figure 3.2 shows the involvement of the users incorporated through the different stages of the research, but more significantly in the software development process of *WorkshopAR*, using as a basis the stages proposed by Pelegrin (2024).

The biggest risk that needs to be acknowledged when implementing a user-centred design is to mismanage the involvement of the user in the process. As Wilson (2014) exemplifies, this situation can take the form of:

- A lack of expertise guiding the project, which results in only community-driven ideas than can turn out amateurish in nature or not considering the whole extent and context of the project.
- A disorganised input that lacks cohesion and a clear goal.
- Stumbled progress, usually due to late decision making or relaying on the necessity to please many disjointed voices.

The clear goal of creating a shared space for collaborative work and the learning goals for the general software development and the industry project course help in mitigating these risks. Other guidelines have been established to keep the interaction design focused and to help the analysis and incorporation of feedback in the different stages of the process:

- The development of *WorkshopAR* is mobile-first. Several aspects of multiplatform interactions will be discussed and tested, but the priority in design will be given to the technology and input methods found primarily in mobile environments, leveraging the handholds and methods present and expected in the medium.

- The main design space will be in 3D interactions and the use of AR to solve problems like the selection of objects, positioning and image identification. Aesthetical decisions and design are important but not the focus of the research.

- AR is a new interaction space for many users, but plenty of sources have shown and tested different designs with positive results. The design guidelines for AR interactions developed by Google DevWorks (ARCore, 2025) are a good example of the available solutions in the current market than can be leveraged and expanded upon without the necessity of reinventing the wheel.

This philosophy behind the user-centred design was also used as part of the iterative process for the software development, especially in support of the analysis and testing steps, as will be described in the next section.

Software Engineering Process

The process proposed for this development is based on the stages and activities recommended by Salo and Abrahamsson (2007) to conduct and agile development process that is constantly checking for improvement and for effective use of resources. The research context in which *WorkshopAR* is developed, and the exploratory nature of the objectives behind the software proposal require for the approach of the development process to be flexible and agile.

The process will be used in the context of a development that tries to deliver functionality fast and in increasing complexity. This behaviour is a response to the necessity of tentatively explore solutions for the educational and technological challenges of the project, to change course, pivot a proposal and be able to incorporate the input of the user participation as explored in the previous section.

As Elbanna and Sarker (2016) explains, agile developments are useful for getting unrestrictive progress in rapidly changing environments, but they run the risk of aimlessly keep producing bloated functionalities and lack a clear direction. For the prototyping nature of *WorkshopAR*, this is an acceptable use scenario, and the same guidelines established to focus the user-centred design process can be used to guide the software development. Additional milestones can also be proposed to evaluate progress and to set a baseline of what can be considered a finished version of *WorkshopAR* within the context of the project.

Milestone 1: A multiuser experience with shared visuals and coordinated interactions.

Milestone 2: Visualisation of relevant digital constructs for the learning activities and varied interactions with them in a 3D space.

Milestone 3: Tools for communication and team management.

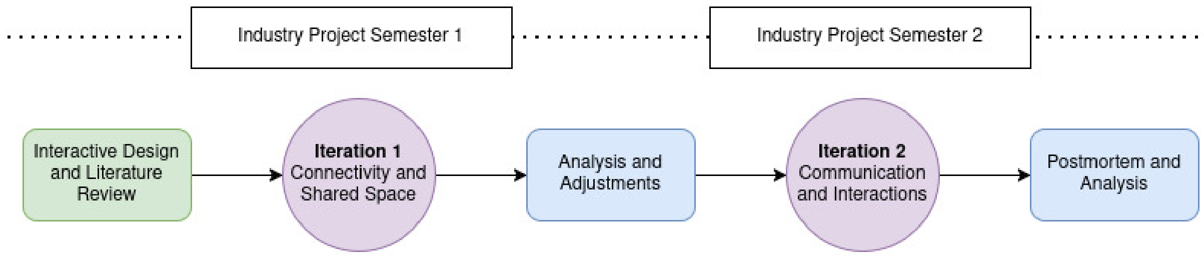

Figure 3.3: Top Structure of the Development Process

These milestones are general enough that allow movement and flexibility around the objectives to achieve, but they also offer specific criteria to determine if an idea or proposal is outside the development scope.

Time constrains are also an effective tool to avoid an iterative process to spiral indefinitely (Kula et al., 2022; Waldmann, 2011). The nature of the research itself establishes a hard time constraint for the development that needs to consider access to the case study and the work needed for the research outside the software development. Figure 3.3 shows the top-level structure of the development process that contemplates the research and evaluation steps done before and after the development, as well as the need to fit the process with the schedule of the case study.

For the timeframe at hand, it was considered possible to divide the development process into two iterations aimed at two complementary objectives. The first iteration could build upon the previous literature review and early design stages and increasingly offer different functionalities to the users throughout iteration 2. This model also takes advantage of the flow of the case study that proposes a similar structure and is also iterative in nature.

Each iteration establishes clear requirements and runs through the process several times, gathering user feedback to determine if it can derive in a meaningful change or needs to be tackled in the next iteration. Figure 3.4 shows in more detail these dynamics in a single iteration.

3.4 Conclusions

This chapter has identified and described the major decisions taken for the methodological framework of this research. The strategy proposed is based on a quantitative approach that focuses on exploring the topic of CL using AR through two complementary perspectives:

- An learning design that establishes clear objectives for the development of the TEL intervention. These objectives are meant to develop fundamental skills for effective teamwork and communication and are based in strong educational approaches of social constructionism and good teaching practices for adult learners.

- An exploratory software development aimed at proposing solutions to challenges of interaction design for shared AR experiences. It also needs to propose effective mechanisms to properly integrate itself with the learning design of the target course. This

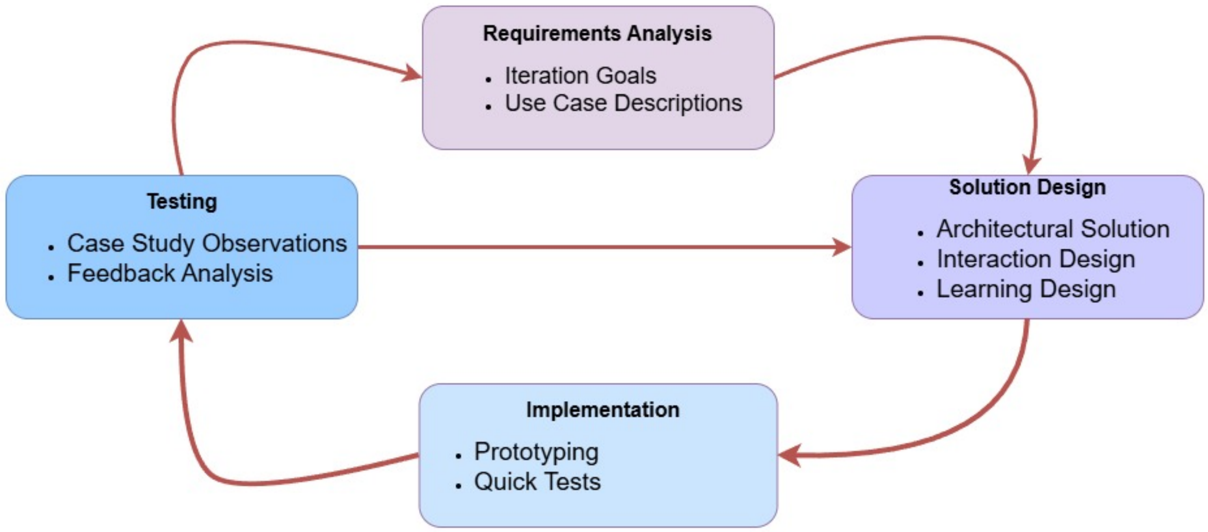

Figure 3.4: Activity Flow of an Iteration

software is developed through an iterative approach that is centred on user feedback, quick adjustments and fast delivery.

The theoretical background explained supports the main objective of this research of integrating a strong learning design in the implementation of a mobile technology intervention in the classroom.

The next step is to describe the results of applying the experiences and ideas gathered through the literature and in the methodological guidelines, first in the design and development that supported the technological approach of the intervention, an then in its use during the interaction with the Industry Project course.

Chapter 4

WorkshopAR - Architectural Design

The following chapter will discuss the architectural design of *WorkshopAR*, the educational software developed to explore the research topic of this thesis and to gather data related to the research questions formulated. This and the following chapters are dedicated to the technical identity of this research and will discuss the development process followed, the technical analysis done to accommodate the educational context of the development, the technical issues found through the design and implementation stages, the results obtained and the path forward for future implementations.

This chapter will focus on the description and analysis of the conception and design stages of WorkshopAR, a prototype implementation and proof of concept of an AR technology for education. Specifically, the intention of this chapter is to discuss these main points:

- The architectural design of the software, the challenges that were present in the context of the research questions and the different solutions proposed to tackle them.
- The technology integrated into the proposal and used for the development process.
- The changes in approach and solutions that had to be made in response to the learning design context, the challenges posed by the selected technology and the data gathered during the testing stages.

As stated in Chapter 1, the educational pillar of the research has never been far from its software development counterpart. This is especially important in the conception-design-implementation cycle used for the development of *WorkshopAR*. It was essential to stablish the proper motivation behind the software development in terms of educational research, its broader position in the field of TEL and its more particular purpose in the context of the case study selected for this project.

Chapter 1 also offered a detailed explanation of the research questions that motivate the construction of a specialised software. To summarise that information, this thesis aims to identify what elements, unique to AR technologies, can be used to enhance the classroom experience by offering benefits that are different or comparable to other technologies. Through an extensive review of the literature, which was detailed in section 2.3, four main factors that distinguish AR from other technologies emerged:

- Visualising and interacting with abstract or difficult to come by concepts and objects.

- The ability to keep and incorporate the immediate context as part of the experience.
- A mobile-first deployment philosophy that eases development to an extent and proves to be more accessible to the end user.
- Easy integration with other tools in the mobile environment like cameras, movement sensors, GPS and ubiquitous connectivity.

The literature review showed that there is an increasing tendency to develop AR applications that leverage the capabilities of a mobile hardware. Access to diverse sensors detect the context and environment of the user and diverse image recognition tools are used to blend in it digital information that creates an experience that is unique to AR. Nonetheless, most educational apps are single-user experiences, which undermines the equally powerful connectivity tools provided by the mobile ecosystem.

Multi-user AR experiences (educational or not) excel the most at co-located, in-person activities. GPS-based augmentation is also used to create strong experiences in shared spaces, building connectivity on top of a different framework in the mobile network infrastructure and prioritising the sense of presence in a physical space rather than an abstract connection with other users.

With this information it is possible to establish one of the main motivations for the development of *WorkshopAR*: to create an AR application made for a multi-user interaction that explores the technology's capabilities for digital visualisation, context awareness, seamless connectivity and a strong link to the physical space shared by the users.

The literature review was also used to provide an analysis of the TEL field. It was possible to identify two problems that proved to be the most relevant for this project.

First, there is a clear trade-off when adapting any technology in the classroom between the benefits obtained and the issues that are related to the implementation, usage and management of the technology. In this relationship between technological development and learning design, it is important to consider all the instances in which one aspect informs the other, and it is even more important to be aware of all the issues that arise when there is a mismatch between both approaches.

Available literature also shows a lot of instances in which a technological development was poorly received. Many studies pointed out the dangers and inefficiencies of a particular technology in learning scenarios, and it is possible to identify a push towards reconsidering the role of information technologies in the classroom (Bećirović, 2023; Boonmoh et al., 2021; Ifinedo & Kankaanranta, 2021).

The second area of motivation for WorkshopAR is to explore the synergy between the developer and the educator. Hopefully, the documentation of the design and development process followed for *WorkshopAR* provides valuable insight of the challenges of creating educative technology, to the point of becoming useful for future projects in similar scenarios. A good step forward from this research would be to polish the positive experiences and to discuss the negatives in future work.

Section 4.1 of this chapter will provide a general description of the context in which WorkshopAR was developed and the broad requirements that were identified. Section 4.2 will show a more detailed description of the use cases and scenarios designed for the tool and their relation to the overall research goals of this project. Section 4.3 is dedicated to the architectural design of the software, focused on the high-level decisions taken to accommodate the learning goals present in the development context and in the research objectives. Finally, section **??** offers a broad summary of the results obtained and the conclusions reached from the analysis process made throughout the chapter.

4.1 Project Description

From exploring the literature, it was possible to extract the idea that AR experiences build around the connectivity tools available in the mobile-based deployment environment showed very positive results and interesting pathways for future research , but a lot of developments were not taking advantage of these tools. With those findings in mind it seems fair to propose a software development using AR interactions in a multi-user environment as an avenue to gather data and build a case study that explores the use of technology that supports collaboration as a teaching tool.

By setting the development in the proper scenario, it is possible to analyse how students use AR technologies as an aid for the collaborative aspects of their projects. It would be of interest to identify what problems are the students solving or finding, how the technology shapes the structure of their work, the relationships between peers or the development of their learning process. By comparing these observations with the behaviours of other students in the same scenario, it would be possible to identify key behavioural changes promoted by the implemented tool, opening the possibilities for further analysis and pathways for future development.

In general, the development of WorkshopAR aims to fulfil three main objectives, which are independent of the learning goals proposed by the case study context. These objectives will be labelled as Observation Objectives:

> OO1. Visualising my team. Sections 2.1.1 and 2.1.4 of the literature review exposed the challenges of teaching collaboration skills, and described the process in terms of the time, guidance, and exposure to proper tools needed to succeed. WorkshopAR is meant to use AR to graphically convey the information needed to manage a group, for establishing a discussion or for organising a conversation. The idea is to offer tools that convey a good grasp of who the group members are and what they are doing.
>
> OO2. Visualising decisions. *WorkshopAR* should provide the ability to show digital representations of the learning material and allow the user to hold a discussion around that. The discussion can derive in solidifying knowledge or building learning, or could help in managing decisions, debates with peers or giving a visual representation to the workflow followed by the group. It should be possible to determine from the observations if interacting as a group with a digital representation of the learning material promotes any change on how the group is learning, how the material itself is being used, or if the augmentation helps managing the learning activity or not.
>
> OO3. Augmenting the workspace. *WorkshopAR* was initially envisioned as a tool for presential collaboration with a physical shared space where the students

are working. The tool should be able to use that space as an anchor for the visualisations and use the capabilities of AR to give extra meaning to that physical context, either by identifying and using elements of the environment in the augmentation or by adding digital elements anchored to the space that can help the user to create such meaning. In terms of research goals, observations should be able to identify if the ability to augment, visualise and customise digitally the physical environment of a group helps with aspects related to team identity, productivity or team management.

So far, these objectives are devoid of context. Their aim is to support the research questions proposed, especially SQ1 to SQ3. The objectives describe activities common in any collaborative situation, and the insights found in the case study can provide answers to questions related to the value of AR in education. The main hypothesis does state that AR offer advantages using clever visualisations and quick interactions with the context, and *WorkshopAR* is proposed to offer such scenarios.

4.1.1 Context: Built Environments Industry Project Course

Industry Project (ENBU 79X/89X in the university's enrolment system) is the final course offered for a variety of programs under the Built Environments department at Auckland University of Technology (AUT). This department includes the undergraduate programs of Architecture, Construction and Architectural Engineering, Civil Construction and Spatial Design.

ENBU 79X/89X is described as a multi-disciplinary course focused on providing an industry project experience that bridges the academical studies of the program and the professional career of the students. In educational terms, this is a project-based course, providing the students a year-long case study that they must work out as a team to provide a solution design, a construction plan, and an estimation of costs. The course does not provide a conventional curriculum or study plan, as it is mostly devoid of content. Instead, the course acts more as a final opportunity to assess the skills and knowledge gained in previous courses, akin to a research project or thesis. Nonetheless, the course has very clear learning goals:

- Propose, organise, design, execute and reflect on a multi-disciplinary project.
- Apply existing knowledge and skills.
- Demonstrate initiative and independence in decision making scenarios.
- Deliver comprehensive client and progress reports.

The course has an overall focus on giving students opportunities to develop crucial soft skills related to teamwork, communication, planning and management. There is also a substantial emphasis on developing the relationship between client and team in terms of communication, progress and stakeholders. All course assessments move around the idea of reporting progress and decisions to the client. All elements of the project involve a two-way communication to understand views and requirements about relevant policies, possible designs and bits of information needed for the proposals.

The course also promotes a strong multi-disciplinary approach. Four different roles are represented from the different bachelors that are part of the Built Environments department: Architectural Engineer (AE), Construction Engineer (CE), Construction Manager (CM), and Quantity Surveyor (QS). Each role represents an important aspect of the project, and every team is composed of at least one representative of each role. Every role has a particular set of goals inside the project and works towards specific learning outcomes. The project is built around constant communication within the group, distribution of tasks, and effective collaboration to succeed.

In summary, ENBU 79X/89X is a course that focuses on developing crucial soft skills on students in the context of project development and management, emulating in different ways common processes and scenarios found on the construction industry. In general, the learning goals can be enunciated as:

1. Develop, document and communicate the design, plan and cost estimation of a construction project, considering the expectations of the client and the limitations exposed by the context, relevant policies and local constraints.

2. Participate in and contribute to the project as part of a team, considering the expectations of the role as well as the processes and tools agreed upon by the group.

3. Create and use the proper tools to communicate decisions to the client, report progress, receive feedback and ask for information.

The course provided other set of particularities and complexities that made it a very interesting case study to explore in detail. All that information can be found in Chapter **??**. For this section, this is enough context to position the development of *WorkshopAR*. The course provided an ideal scenario to test goals associated with teamwork and CL, as well as to provide the opportunity to have evidence of this process in all the stages of development of both a collaborative project and a course, not only in an individual assessment or lesson.

With this context at hand, we can position *WorkshopAR* as an educational technology development meant to offer support to the students of the Industry Project course their coursework and the management of their team. It is possible then to elaborate on the educational and development goals of *WorkshopAR* in relation to the research questions of this thesis, which is the objective of the following sections.

4.1.2 Learning Goals

The expected learning outcomes for the Industry Project course highlight teamwork and communication, and as such, the intervention designed around *WorkshopAR* should focus on identifying the tools that AR technologies can provide to facilitate or boost that collaborative process. The aim should be in facilitating group management, communication among members and aiding in the production of knowledge and materials for the project.

The guiding prompt of *WorkshopAR* is to provide students with a digital environment that helps in visualising the activities performed by the team and to highlight possible tools at their disposal for proper team management, discussion of ideas and effective decision making.

By aligning these goals with those of the course, *WorkshopAR* can provide a tangible feel of abstract concepts used in team management like participation, ideas and task allocation. It can also reinforce in the students the need to develop a strategy to deal with issues related to groupwork. It can also prompt ideas about self-reflection, evaluation of performance and the development of each individual role inside the group.

The main goal can also be aligned to research questions SQ1 through SQ3. The case study around ENBU79X/89x provides an ideal observation environment to compare different approaches to collaboration, not only with those using AR or not, but also behaviours related to the use of different types of technologies and to the circumstances fostered by the design of the course.

Since the case study is complex and strongly based on observations, three scenarios have been selected to focus the development of *WorkshopAR* and to simplify and direct observations, comparisons and analysis, making the data gathering process manageable for the timeframe of the research.

These scenarios represent the general framework used to provide context to more specific and goal driven use cases in the future, which will be exposed in Section 4.3. The scenarios are born from the learning goals of the course but not necessarily from the particularities of its development. They represent common instances of collaborative work and common roadblocks in the development of effective teamwork (see Section 2.1.4). A deeper analysis of the implementation of the tool is described in Chapter **??**, and a description of the changes and adaptations needed to properly incorporate *WorkshopAR* as part of the observed cohort of ENBU 79X/89X can be found in Chapter **??**.

Scenario 1: Team Management

The first scenario is about tools that can be used to manage the composition and dynamics of a group and to help in answering questions like: Who is my team? What are their roles? What is my role? (Bannister et al., 2014; Delgado Piña et al., 2008).

The proposal is to use visual cues and digital constructs to represent information about team members and their work, offer the means to interact with such information and to share it with others. This *digital profile* can help students to start a conversation around the structure of the group and the evolution of the dynamics of its members. The same reasoning can be applied to give a visual anchor to the tasks allocated and the roles assigned.

A specific manifestation of this scenario is the first contact the members of the group have. *WorkshopAR* can help in that first process of identifying who is who, to create the proper environment dialog and to establish the first rules of communication. It can also be extended to scenarios such as a new member trying to integrate into the group or as a management tool to experiment and consolidate the group's structure.

WorkshopAR can also provide a digital presence for the team itself and represent information about the work being done and who is doing it. This digital presence can be defined in terms of the physical space where the group is meeting. This will serve as the positional anchor for all the digital constructs that can be used as part of the team's activity. A digital room can be created for the team members to enter or connect which would signify the disposition to start working. The room will be constrained or superimposed to the physical space where the group is gathered and can be composed by the digital constructs the team is using to work and all the physical elements the application could recognise.

This is a good moment to explain the first design constraint taken for this project. *WorkshopAR* will provide support only to face-to-face collaboration. Online or asynchronous collaboration won't be considered.

Sections 2.1 and 2.3 of the literature review exposed the importance of network technologies in the analysis of CL and the expansive field of online collaboration. The opportunities of exploring the connectivity capabilities of mobile devices as part of AR application was also part of the discussion, and a big focus of this development is to explore such possibilities and to leverage the ubiquitous connectivity offered by mobile devices to power seamless connectivity between users.

It is also true that this research wants to understand the unique capabilities provided by AR technologies in terms of augmentation of the physical space, a scenario that adapts easily to face-to-face interaction and connectivity with users in proximity. This does not mean that such elements cannot be together with online interactions, but they do add another degree of complexity to both the research and the development of the software, risking the possibility of taking the spotlight from the main objective of understanding CL and the influence of AR in the process.

To conclude, Scenario 1 proposes a solution in which the team and each of the team members have a digital profile that can be interacted with. The information in the profile can be used as the main anchor for the software to provide the visualisation and augmentation tools that aid in the process of team management, communication, roles, task allocation and team structure.

A visual representation of all these concepts can be seen in Figure 4.1, where the room represent both the physical space used by the group to work and the current instance or session the group is participating in. The room will hold all the digital objects used during the session as well as framing all the other activities the team can perform.

Scenario 2: Work Session Management

For this scenario, we can define a work session as an activity the group decides to execute for a given time with a set of goals in mind. It can also be described as the group members participating in a lesson or a learning activity. The session would be composed by participants, the space the session is taking place in, and the goals proposed.

It is envisioned that *WorkshopAR* can provide tools to identify and manage aspects like:

- The participants in the work session.
- Status of the session.
- Current objectives and the status of those objectives.
- The resources at hand for the session.

Scenario 1 is about giving the room a digital representation that can be interacted with. Scenario 2 uses those elements to construct a digital representation of the work session, as shown in Figure 4.2. The participants will connect to an active room in which a session is

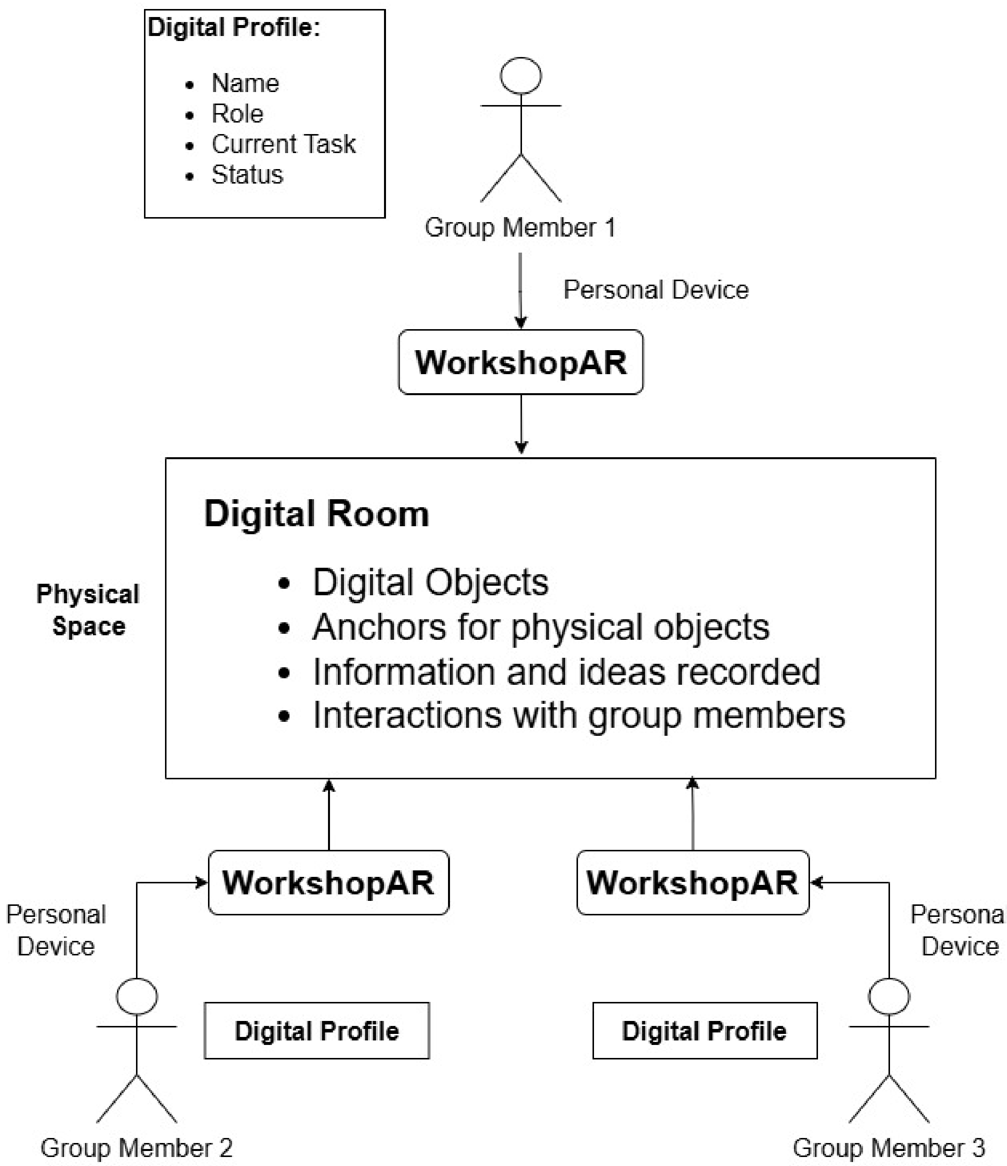

Figure 4.1: Team Management Scenario

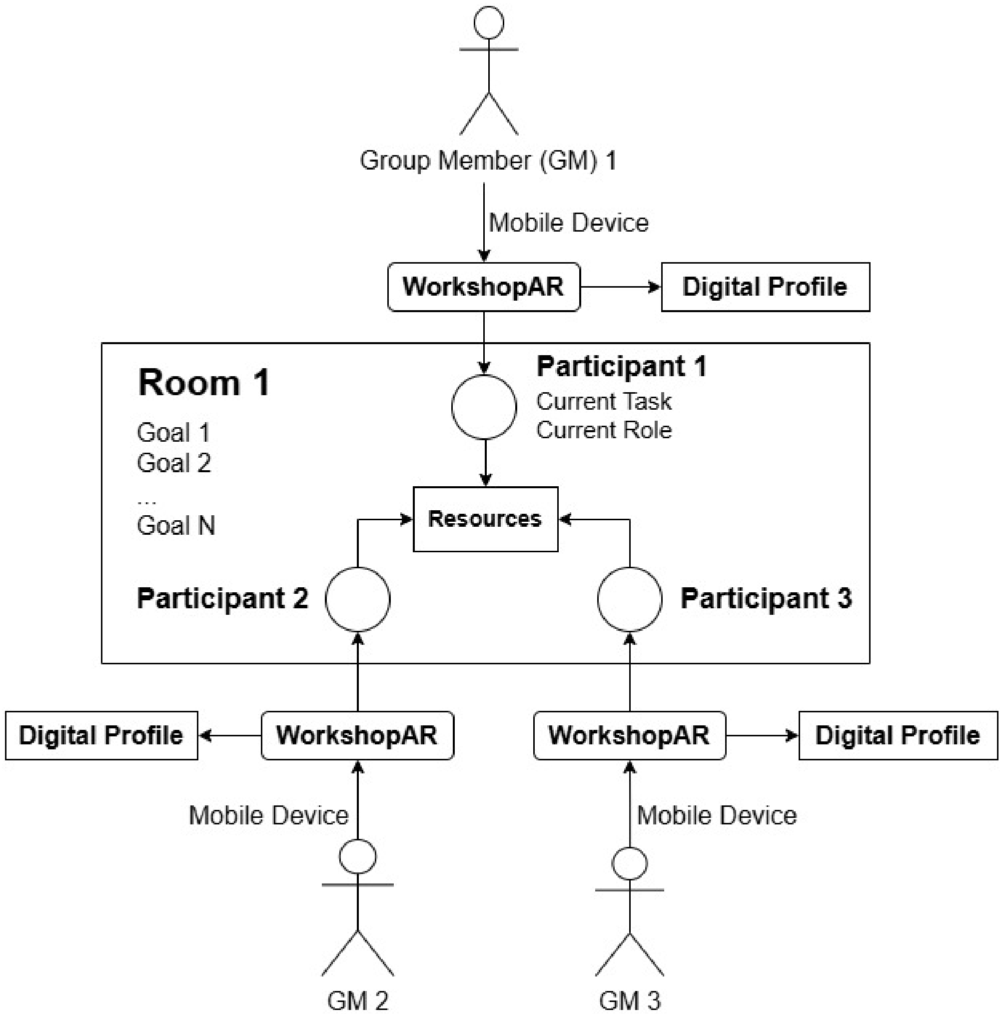

Figure 4.2: Work Session Management Scenario

taking place. The room will be associated with a set of goals, active resources and a physical space to which is anchored.

The second development constraint that can be analysed at this stage is related to resources. Ideally, WorkshopAR should be able to provide or interact with all the resources the group needs to perform their work. In just the context of ENBU 79X/89X, the list of available resources could comprise elements such as:

- Tools for creating documents and presentations.
- Tool for architectural design and analysis.
- Spreadsheets for calculations and project management.
- Visualisation of documents and design proposals.
- Tools for digital modelling and design.

It is not feasible for the scope of the development to consider all the possibilities. Seamless interaction between tools is a valuable quality functionality but would not add significant insight into the research goals at hand. In contrast, observing how the students interact and integrate all the technological tools at their disposal, and how they evaluate their usefulness and functionalities, does align with the observation objectives of the research, providing insights for questions SQ4 and SQ5, which are trying to identify the specific value of AR in learning scenarios, especially in comparison to other tools.

WorkshopAR will be limited to two resources that can be used as part of a working session:

1. Visualisation of and interaction with simple digital models for planification and discussion.
2. Anchoring data or information to a digital object or physical space.

These two resources offer general help for the most common activities of the course: proposing solutions to the spatial design of the project and organising discussions around such proposals. The purpose is to analyse how the students change (or not) their approach to collaborative work when they have access to more tangible representations of abstract concepts such as ideas, proposals and discussion, and how (if at all) the discussion is affected by the ideas being more visually anchored to the real world.

Scenario 3: Discussion and Debate Management

Scenario 3 is related to the activities that can be performed during a work session but that are not part of the team management. *WorkshopAR* aims to offer tools for discussions and debates around the work done by the group and the decisions being taken. The tool can give visualisation and structure to elements like:

- The object of debate or discussion.

- The structure and flow of the discussion.
- The responses and ideas generated.
- The active participation of different team members.

The proposal is that AR can offer a visual anchor or structure to the process, helping students in understanding it and managing its flow.

To summarise, the three scenarios presented above illustrate the main learning objectives guiding the design of *WorkshopAR*. The objective is to provide a tool that helps the students to identify and structure the most relevant aspects of their collaborative work such as team management, work management, communication and decision making.

4.1.3 Development goals

It is also possible to identify a set of points related to the technological and software development of *WorkshopAR* that are worth analysing and that can offer insights on how the technology influences the achievement of the learning goals. Some constraints in the development process can also be discussed related to issues highlighted in this section.

The first development goal is concerned with the multi-user AR experience. This requirement implies some form of network infrastructure to support several users interacting at the same time, sharing resources and visualisations. Thanks to the information provided on the descriptions of scenarios 1 through 3 in the previous section, it is possible to summarise the main extent of the networking development into the following objectives:

- Provide a lobby-like architecture for connectivity, allowing users to connect in and out of a work session in a seamless way.
- Share and synchronise information about the digital profiles of the team, the room and the resources being used.
- Share a synchronous interaction with the digital objects used and the augmentation created for the work session.

The main proposal of the research is not the software solution developed, rather the interaction design engineered to fulfil the networking requirements and the process followed to integrate the solution in the flow of a work session, considering some important requirements imposed by the context of the case study:

- Offering a seamless connection with minimal or high-level configuration.
- Disrupting the work of the students as little as possible, considering they are contributing to a graded project that is part of their professional education.

The second development goal is to provide a platform-independent solution that can support a Bring Your Own Device (BYOD) approach to the implementation of technology in the classroom (Disterer & Kleiner, 2013). This considers that, in theory, the main platform

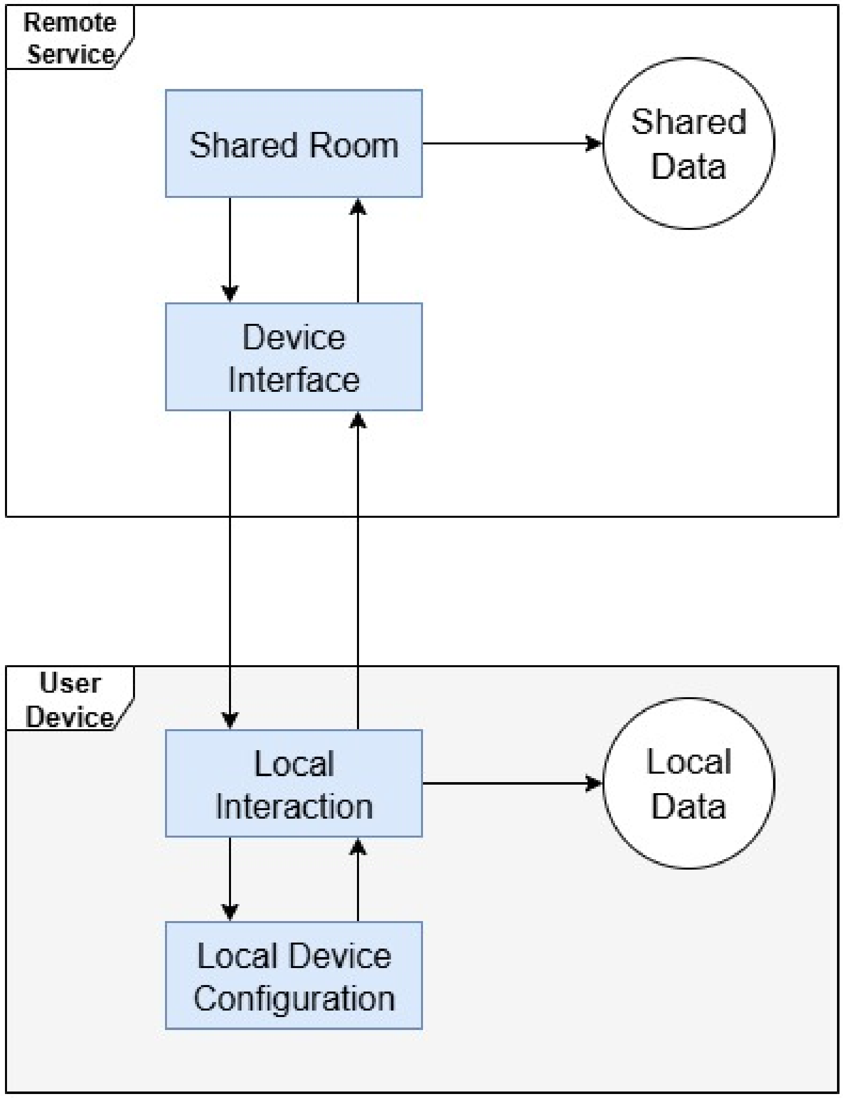

Figure 4.3: Concept of a Multi-Platform Architecture

for deployment of AR solutions are mobile technologies, an ecosystem that is ideal for BYOD initiatives (Imazeki, 2014; Väljataga et al., 2016) due to accessibility and ubiquity. Mobile devices also offer hardware solutions to several of the networking objectives proposed previously.

Additionally, a solution that is platform-independent can also aim to accommodate AR devices that are not based specifically on smartphones, such as specialised MR headsets like the Meta Quest 3, Magic Leap or the HoloLens. This would create a challenge in design to adapt the control and interactions of the experience specifications that each participant has access throughout a session. Figure 4.3 shows a high-level architecture of a possible solution for this scenario, which is similar to other systems that offer cross-platform functionalities, commonly across a web deployment (Medvidovic, 2002).

Similar to the way in which systems like Netflix balance access to content depending on the device used for access ("System Design Netflix | A Complete Architecture", 2023), a general architectural proposal for *WorkshopAR* would consider an interface between each local participant and the shared interaction in the room. The interface can manage input interactions, data types and visualisation types before handling control to the shared room, and vice versa, handle responses and transform them to data that can be properly processed by every device connected to the session, akin to a system like Andrés and Pérez (2017).

With that analysis in mind, Chapter **??** will describe in more detail the reasoning behind not pursuing a full cross-platform architecture. In summary, the research value did not align properly with the value gained by this type of development. The BYOD approach was instead tested by focusing on mobile devices which, paired with other technologies selected for the development, already covered any mobile architecture using Android or iOS, offering a wide spectrum of devices available to the users.

The one element left behind by not pursuing a full cross-platform architecture was testing the integration of different means of interaction in the same work session. It was decided that, for the scope of the research, it was worth to focus first on other aspects of the development and probably relegate this aspect for future work. Nonetheless, the question of cross-platform will resurface in the analysis of the case study and during the user-centred design described in section **??**.

In conclusion, the development of *WorkshopAR* proposes architectural and implementation solutions for a multiuser AR application deployed over cross-platform mobile technologies. These objectives provide opportunities to understand the technological challenges of implementing multiuser AR applications and how they affect and are affected by the learning goals proposed previously.

4.2 Use Cases

The scenarios described in the previous section represent the general goals for which WorkshopAR will be used as an intervention in a learning setting. To start the software design process, the first step is to identify and describe the specific functional requirements to be developed. The following section will describe the most relevant requirements identified for the research intervention of ENBU 79X/89X. This information is going to be described in the form of use cases, or user stories, to emphasise three aspects:

- The general interactions performed by the users, not the particularities that have to be developed.
- The development goal being fulfilled by the use case, or at least any technological challenge that could be of interest for the analysis process.
- The learning goal being fulfilled and how it is related to the educational objectives of the research.

Before exposing the use cases, it is necessary to discuss some of the vocabulary that is going to be used in the descriptions. These are terms referring to common elements of the interaction with the app, the activities enabled by it and the resources used to create the experience.

A *user* is any student using the app. Every user has a personal profile that is shared between sessions. Although in the context of ENBU 79X/89X all students belong to a particular group for the development of the course's project, the app is not going to enforce that organisation, and a user can become a participant in any session available.

A *participant* is a user that has joined a *digital room* and now is part of a *work session* with other users. As a participant, a user has a role and a set of assigned tasks, which are only valid inside a particular session.

A *digital room* is the space created by the app and used by the participants to hold the session. The room doesn't need to be a closed space. The app creates a *workspace* for the room and associate the session information to a room anchor, a digital object situated in the space selected.

A *work session* is the work done by the participants that enter a *digital room*. The session has a general goal associated, which can be modified and updated.

A *workspace* is a digital object representing a space in which the participants can position and interact with digital objects. In conjunction with the *room anchor*, it also creates a visual representation of the *digital room* in which the *work session* is taking place.

4.2.1 Create a Shared Digital Room

Description: A user must be able to initiate a *work session* by creating a new *digital room* for the group. The *digital room* is identified by a unique name and described by a set of goals for the session. The user can configure all these elements before launching the session. Upon opening the *digital room*, any other user can search the name of the room and join it, adding themselves as participants to the *work session*.

A *room anchor* will be added to the *digital room* which will present information about the *work session* upon interaction.

Development goal: The *digital room* represents the main point for the multiuser experience. All the participants in the *digital room* will share the same visualisations and will be able to interact simultaneously with the same digital objects. Ideally, the connection to the *digital room* is seamless, requiring only the initial configuration for setup and run.

Educational goal: The *digital room* will be linked to the concept of a *work session*. When a room is active it means that the group is working, and all the participants share a current goal. The *digital room* provides the students the opportunity to have a structured *work session*, and the digital markers used provide a tangible and interactable representation of the goal and the room itself.

4.2.2 Create a Shared Workspace

Description: Upon creating a *digital room*, a participant needs to define a space in the physical environment that will be used as the shared *workspace* for the session. *WorkshopAR* will identify suitable horizontal spaces like tables, desks or the floor, and overlay a digital plane over them. The participant can select one of the detected planes and modify it to convenience to suit the preferences of the group or the needs of the current session.

Development goal: The *workspace* will be the base augmentation provided by *WorkshopAR* to give a visual representation to the *digital room*. The *room anchor* gives basic information about the session, while the *workspace* facilitates the interactions with other digital objects, constraining the movement and location of all the interactive 3D models and providing a reference point for any other digital constructs.
Educational goal: The *workspace* gives a location for the shared resources of the group, elements that are public to everyone and could be used as inputs in each task. The *workspace* anchors all the digital objects to a common point and the group itself, creating a visual representation of the *digital room* while keeping the constrains of the physical space used. These constraints are more malleable than in a normal room, and the group can use the digital elements of the app to explore or expand what is provided by the physical space.

4.2.3 Create a Profile

Description: A user can create a digital profile to use in a *digital room*. The profile is linked to the instance of *WorkshopAR* installed on their device. When not connected to a *digital room*, the profile consists of a preferred name and a colour that represents the user. When connected, a user can also be described by a role and a set of current tasks.
Development goal: The profile is a representation of the users in the when connected to a *digital room*. It also serves as an anchor for interactions with other participants and provides a form of digital avatar that carries the information of the profile.
Educational goal: The participant's anchor gives visual structure and prompts the discussion around different topics like team management, assigned roles and allocation of tasks.

4.2.4 Point to Something or Someone in the Digital Room

Description: This case is related to provide basic communication tools to the students, specially to convey information about the resources available and the users connected to the *digital room*. Participants should be able to specify the person or object (physical or digital) they are referring to in a conversation or instruction. This clarification in the communication process is going to be referred as *pointing*.
Development goal: This interaction is meant mostly for digital objects and has an important role in the multiuser experience. It is a simple tool to facilitate communication and helps to mitigate any ambiguity or confusion. It is also an interesting technical challenge to account for both digital objects and the physical context of the room.
Educational goal: This is a simple tool to promote and facilitate communication. It is one of many ideas to explore the concept of *augmented collaboration* by providing digital tools to visualise the communication process and analyse its impact in the behaviour of the students regarding learning, group dynamics and team management.

4.2.5 Check the Status of Other Participants

Description: A participant should be able to access information about everyone connected to the same *digital room*. This information is related to their profile and their status, described by their role, assigned tasks and participation state in the activity.

Development goal: The technical challenge of this use case is related to the way the information present in the profile can be transmitted and synchronised on the network, as well as the proper interaction methods to access it. These interactions can also be extended to accessing the information and current state of the *digital room*.
Educational goal: This is an opportunity to understand the role and influence of technology as part of the collaboration process. Ideally, the tool provides quick access to information about the current state of the session and the team members, but it could also become an unnecessary barrier for something that should be verbally communicated. It should be possible to observe how students use this functionality, how it affects different communication processes inside the group and how *WorkshopAR* compares to other IT tools that mediate in communication.

4.2.6 Share and Pin an Idea

Description: A participant should be able to type an idea and pin the text representing it somewhere in the *digital room*, making it public and accessible to all the other participants. Similarly, a user should be able to see and interact with all the ideas pinned by them and other users.
Development goal: This use case also belongs in the *augmented collaboration* category and offers the opportunity to record and visualise ideas and information in the same way a whiteboard would do. This is a way to experiment with the idea of offering a relationship between the digital representation of the information and the physical space (Y.-H. Wang, 2017; Zünd et al., 2015). The objective is to prompt discussion, associations and structure to a conversation while sharing and recording ideas between participants.
Educational goal: In the same spirit, it is another tool to visualise and interact with information, making communication visible. The idea is that, by adding a relationship between the space of the conversation and the ideas proposed by the group it will open the possibilities to get deeper, different or more structured types of discussions.

4.2.7 Add and Interact with a Model in the Digital Room

Description: This interaction is related to simple digital models either build in external tools and imported to *WorkshopAR* for visualisation or built in the tool using simple construction blocks. Since the main objective of ENBU 79X/89X is related to architectural design, this interaction with digital models is meant to facilitate communication and discussions related to this field. This communication can be centre d around collaborative creation of a design or discussing ideas, getting feedback or proposing changes. This use case considers all the interactions needed to add the models to the app, share them in the *digital room* and perform simple interactions with them like moving, selecting and transforming (e.g., moving, rotating, scaling) the models.
Development goal: More than a technical challenge, the focus of this use case is on the interaction design needed to propose a solution. For one, there is an interesting opportunity to analyse the design and usability of the interactions needed for this task, considering that the app will use a mobile phone deployment. A set of conventions in interaction and visual language are expected for both the selected hardware and the AR nature of the tool, and it

is important to analyse those conventions, implement them, expand them or revaluate them if needed.

Second, this interaction must consider the multiuser scenario of the project and the necessity to coordinate interactions of shared objects over the network. Although the technical implementations of these requirements are interesting, it is more important for the research goals to observe how the user experience proposed forces or promotes different behaviours in the users, and how those behaviours affect the learning and collaborative processes.

Educational goal: The interaction between participants is again the focus rather than the activity itself. Since the main objective of the course was design and construction planning, it was considered interesting to provide a flexible enough interaction that promoted discussions around different elements of the design process and the ideas proposed by the students around space and structures. The building blocks metaphor was chosen to promote a visual and physical anchor of those elements of the architectural design as part of the AR experience.

A more explicit goal for the development of this functionality is to organise and structure the work of the students around digital and physical tools that boost the idea-discussion-decision loop with AR and communication technologies. This is, again, guided by the principle of augmented collaboration.

4.2.8 Participate in a Debate or a Vote

Description: A participant must be able to propose or be part of a group vote or debate. Both cases will be defined as instances in the *work session* in which the input of the participants is needed to take a decision. A vote is a decision to be taken among a possible group of options, while a debate has an open input nature.

Development goal: This is an interaction design that allows the users to visualise in the augmented space all the elements taking place in the decision-making process. This idea explores possible implementations for the tool to manage a debate and to storage, compile and show the results of the activity.

Educational goal: This use case reinforces the most prominent activities in which communication skills are developed thanks to the classroom groupwork. *WorkshopAR* is not expected to be used as the sole manager for the communication and planning needs of the group, as other tools probably already fulfil those roles. The idea is to offer a different approach to the communication interactions that happen when taking decisions, specifically in face-to-face scenarios. This type of discussions is prone to be more informal, unstructured and with no archive of the activity itself. Once more, the aim is to offer a different type of solution based on visualisation and context augmentation.

4.2.9 Pause the Current Participation

Description: A participant should be able to signal the group that he/she is not actively participating in the session and that progress in the current tasks is in pause.

Development goal: The reasoning behind this use case is to create a distinction between the physical presence of a participant in the *digital room* and their activities in the group, giving more depth to the status of a participant, also helping to visualise and symbolise some

social cues that are hard to verbalise. In a more technical sense, it is also a function that helps in pausing the use of the device without finishing the session.
Educational goal: This option is the best example of a tool for group management that has more significance than simple actions like checking a list of participants or managing access to the digital room. This functionality aims at giving visual presence to the work of the user, to establish boundaries, proper communication habits and prompting discussions around time and space management.

4.2.10 Conclusions

The exposed use cases represent the most important functionalities that *WorkshopAR* should be able to fulfil to provide insights for the research goals of the project. These scenarios present the most common instances in which the students can develop collaborative skills or in which their collaborative work is the key aspect of their work. These actions include diverse aspects of activities such as team management, task planning, effective communication and decision making.

The main proposal that this software development is meant to explore is that through AR it is possible to create a visual representation of the work that the users are doing and of the resources in use. The plan is to design simple interactions that create a relationship between the ideas explored with the tool and the collaborative skills that the students need to develop.

The plan also relies on offering the proper environment to think or discuss issues around the structure used to work and plan, rather than to force a specific way of working. This approach can create the opportunity for students to reformulate their current way of working and find alternatives that incorporate new information into their processes, but it can also cause the students to completely ignore the tool.

This is an acceptable risk scenario. In first instance, it creates the least amount of friction with the teaching approach of the course, giving the students a technology that they can freely use to enhance their work, and not a forced experience they could resent or that can even inhibit their work. It was decided that a free exploration of the tool offered a better and healthier experience.

It is also acceptable in terms of the research goals. A scenario in which a group did not interact with the tool also offers interesting insights. Paired with the case study, a failure engagement with *WorkshopAR* can elicit questions regarding the tools that the group is using for communication and planning, how the group is structuring their work and managing their time, among other possibilities. Answers to these questions can help in constructing an interesting case study.

4.3 Architectural Design

The following section will discuss the main ideas guiding the architectural design of *WorkshopAR* and the reasoning behind the decisions taken. The whole process ill be described using the most important quality requirements identified for the development of the tool, the most important functional components designed, and the available frameworks and technologies selected to help in the construction of the tool. For this purpose, several architectural and structural diagrams were created to explain and visualise the design decisions taken and the software architecture discussed throughout this section.

A common visual language was used that loosely follows the structure and rules proposed by UML, but it was not strictly followed nor are any of the diagrams in this section UML-compliant. The main objective was to offer a quick visual representation of software components and behaviours that facilitated the communication of concepts and the discussion of the decisions taken, while keeping the diagrams simple and easy to read. The main rule was to keep consistency between diagrams, especially with colour coded information as follows:

- Grey elements represent hardware and devices, physical instances where software is deployed.
- Yellow elements represent software components, entities or a set of entities that have a common purpose and functionality. They could be zoomed-in for more detail, but that is not necessary to understand the idea conveyed by the diagram overall.
- Blue elements represent more abstract entities, either because they represent big complex functionalities which serve several purposes or because they represent a state or logical concept in the execution of the software that is achieved by the interaction of several other components. As with yellow elements, the component can be described in more detail, but it is not important or necessary for the purposes of the diagram.
- Green elements are specific instances of an object (a class or a game object in Unity) used to show the composition of an important scene or the flow of data in a use case scenario.
- **White** elements are used for data or information being used, created or transmitted between other software components.
- **Arrows** are used to signify relationships and associations between components, or messages sent from one component to another, depending on the context of the diagram.

4.3.1 Quality Requirements

WorkshopAR is a prototype that helps in promoting proper work habits in the students for communication and collaboration. Independent of the specific functionalities to be built, the software needs to achieve these objectives through the architectural design proposed:

1. **Multiuser interaction:** *WorkshopAR* is a cooperative experience for the synchronous interaction of small groups of 2 to 6 students. All the participants in the interaction will be present in the same physical space and will share the same digital augmentation of the environment, but with different points of view and instances of interaction. Digital objects in the space will also be shared and interactions with them must be synchronised and coordinated between users. The participants will have a digital presence in the shared environment that can also be interacted with, and that interaction also needs to be synchronised and coordinated.

2. **Multi-platform deployment:** To support a BYOD approach, it is necessary to build a solution with a multi-platform deployment. *WorkshopAR* must work independent of any specific mobile platform and aim for the most common AR capabilities of devices currently in the market. Ideally, the tool should also support any other AR capable device, specifically, headsets available in the market like HoloLens, Magic Leap or Meta Quest 3. This implies that the tool must consider all the diverse input methods available, display varieties and processing capabilities, and coordinate them within the shared experience.

   This idea can be also expanded to a multi-role interaction, in which the activities are taking place in a room, consider different roles that any participant can take, and tailor how the user sees information and participates in the experience in accordance with that role. Different roles also open the possibility to interact within the room through mediums that are not necessarily AR.

3. **Spatial interactivity:** Due to the multi-platform approach taken and the irregular nature of the AR capabilities found in current available devices, it was necessary to make decisions about the minimal capabilities needed for the AR experience to function across different devices. It was decided that the core experience was in the augmentation of the communication process, and the relationship built between the group and the *workspace*. It was necessary to identify minimum features common in most devices and decide which aspects of the experience could rely on the available technology and tool, and which needed a solution through a more complex development.

The following quality requirements are not directly related to the outcomes needed for the research goals but are equally important to support a good development flow or to accommodate the particularities found in the execution of the case study.

4. **Flexibility:** Given the interactive nature of the development proposed and the exploratory approach of the research, the general component architecture of the tool needs to be as flexible and decoupled as possible. Particularly, it is important for the input system and interaction methods and processes to be easily swappable without affecting the core functionality of the software. These are the most fluid elements of the design, and from the very beginning, they are meant to be part of the main experimentation with the software. A flexible approach is necessary to facilitate testing and different ideas and approaches.

   Solutions related to the network communication and the creation of a shared space also needs to be flexible. Ideally the main utility of the tool is independent of the networking architecture.

5. **Extendibility:** Independent of the differences between the design and the implementation, component development will aim for cohesion and extendibility. The purpose is to support a small and incremental development that is easy to extend into full functionality over time. This will also support the user-centred approach proposed, where the users can experiment with a simpler version of a functionality, provide feedback and incorporate that information in the extension and refinement of the functionality.

6. **Easy-to-learn:** One of the core issues identified in section 2.2.3 in the implementation of AR in classroom is the all the extra time needed for students to learn the proper use of the tool. Lessons and plans had to accommodate, sometimes unsuccessfully, enough time to teach how to use the tool, coordinate its use in the planned activity and solve issues appearing when mishandling the tool.

   This issue is a complicated task worthy of its own focus research. The design of *WorkshopAR* can leverage the common knowledge that students possess of their phones and the common interactions with them to propose interactions that emerge from the natural use of the technology. This can be achieved by exposing the users to small samples of the software and incorporate a systematic analysis of their feedback in the design process.

It was important to follow a formal approach to the analysis and development of the easy-to-learn design, taking decisions based on principles found in the literature and approaches proven successful in similar scenarios. The most relevant guiding principles are based on the work of Wigdor and Wixon (2011) who state that a design based on natural interaction is characterised by:

- The feel of a natural extension of the body.
- Offering the same natural experience for novice and for expert users.
- An experience tailored for the medium (mobile devices), not a mimicry of the real world.
- The proper use of context, previous knowledge, metaphors and input/output devices.
- Using paradigms when needed but not being afraid to ignore them.

Other ideas were summarised from works like Good et al. (1984), Macaranas et al. (2015), and Strijbos et al. (2004) that state general good approaches such as:

- Test different metaphoric mappings depending on the content of the interaction.
- High usability is achieved through careful analysis of feedback.
- Manage and consider user's expectations and knowledge.
- The changing expectations of the participants depending on the context of the experience and the tools available for the interaction.

4.3.2 Technological Frameworks

A set of tools were selected to implement current available technologies and frameworks to simplify the constructions of the basic functionalities of the application, support the solutions for more complex problems and provide a pipeline for testing and deployment. The tools selected helped in the implementation of basic AR functions and worked as stepping stones for more complicated interactions, for the networking development and for the multi-platform deployment.

The Unity engine was selected as the main building platform. Similar to other tools and environments like Unreal and Godot, the initial use case of Unity is in the development of videogames, but the tool can also be adapted to be the engine behind any type of multimedia interaction software (CodeMonkey, 2024; Hussain et al., 2020). Unity was selected due to convenience and to leverage previous expertise with the tool.

Unity was used to handle the implementation of digital content and the interactions with it. Unity also helped in the management of diverse input methods and the multi-platform deployment of the software. Some of the issues related to networking were also solved using the Netcode utility. This package allows for basic network configuration and provides a framework to communicate information or to coordinate states of the objects across a network.

Finally, one of the biggest advantages of using Unity (and other similar tools) is that it provides an implementation of the OpenXR framework ("OpenXR - High-performance access to AR and VR —collectively known as XR— platforms and devices", 2016). This framework is and open-source implementation that abstracts into a common language all the specifics of different VR, AR and MR devices in the market. OpenXR provides a platform-agnostic implementation of different features like tracking management, display, environment tracking, input handling, among other things. This is a useful tool that helps in maintaining a single code base for the application and easily deploys it on different platforms at the same time. The project can then focus on solving specific problems without having to deal with issues at the hardware level.

Additional to the tools provided by the Unity engine, two other frameworks were integrated to the development to help with functionalities related to AR and the shared interactions in the network, as described in the following sections.

ARFoundation

ARFoundation is an implementation by Unity that provides a similar level of abstraction as OpenXR but specifically tailored to the AR environment and the mobile devices ecosystem. This is an important tool that helps in dealing with the most important frameworks in the market for AR development: ARCore and ARkit for Android and iOS devices respectively. These two frameworks cover most of the mobile devices available in the market, while more specialised technology uses APIs or frameworks that are either a derivation of the two main ones or can be handled with the OpenXR framework.

ARFoundation strongly influences the architecture of *WorkshopAR*. ARFoundation provides a set of manager utilities that act as a front-end configuration for several common functionalities of AR such as image processing, feature detection, point-of-view tracking or image composition. The managers communicate with a series of subsystems that abstract the specific hardware implementations of ARCore, ARKit or other similar toolkits. This means that all the functionalities of *WorkshopAR* are built by coordinating the managers provided by ARFoundation with the proprietary code developed for the application. Additionally, any implementation communicates with the hardware layer via the subsystems available, allowing the creation of a single code base for every platform. Figure 4.4 shows the component architecture of *WorkshopAR* when deployed in a generic mobile device, differentiating between

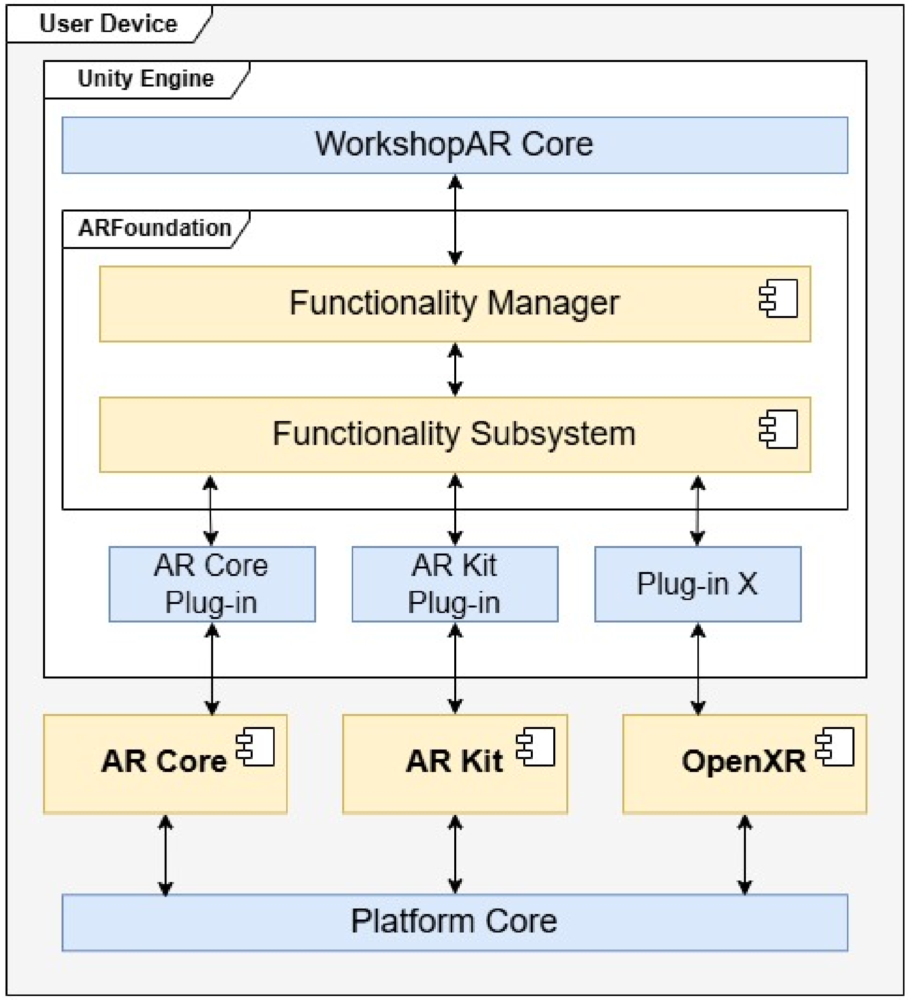

Figure 4.4: Integration with ARFoundation

the code found in the core Unity implementation vs the local toolkit used to provide AR functionalities in the device.

This solution facilitates a deployment that is, at least in theory, independent of any platform. In practice, it is not that simple. It is possible to support any mobile device that implements ARCore or ARkit, and there is also the possibility to deploy in any other AR-capable device if there is a subsystem in the ARFoundation toolkit that supports it or if the device supports the OpenXR framework. The problem is that all these implementations do not communicate smoothly, and to support a proper platform-agnostic single code base, it is necessary to identify all the intended platforms that are going to be used and coordinate all the low-level communication elements of the code to properly function. Other element to consider is that if used as-is, ARFoundation only offers functionalities exposed by the managers, which can obscure more updated versions of the lower-level systems or not show them at all.

Lightship

This component is part of the Niantic Spatial SDK suit ("Niantic Spatial", n.d.) and the set of solutions present in the Shared AR product ("Shared AR", n.d.). Niantic has pioneered the commercial use of AR in mobile environments through successful implementations in entertainment, educational and industrial contexts (BowTiedRobin, 2021; L. Lee, 2022). The SDK offered through Niantic Spatial is a set of solutions compatible with most development environments to create and manage complex AR interactions like life meshing, semantic detection, complex image recognition and networking.

For the development of *WorkshopAR*, the component integrated into the architecture is Lightship, a tool for the creation of shared AR experiences without the need of a complex backend network infrastructure. This tool is meant for small scale connectivity, supporting up to ten concurrent users. Lightship eases the development of the multiuser requirements and offers a scalable and flexible architecture that can be expanded and build upon in future work.

In isolation, Lightship works by using a simple client-server architecture. Any instance of Lightship can be connected to another one through an internet connection if they are using the same application ID. The architecture allows for the implementation of a dedicated server that controls the experience or for a distributed connection with one of the client instances acting as a host. At the network level, it is possible to use any tool at the developer's disposal, abstracting as much as possible the connection with the transport layer.

At the application layer, Lightship provides three important concepts to manage the shared interactions:

- **Room:** Parallel to the concept of a *digital room* provided in the use case definitions, Lightship defines a room as an abstraction for the main connection point for the users that are going to share an interaction. A room is identified by a unique ID within the rooms created for a unique application ID. Any client can see and connect to all the rooms available for that application. The rooms are created and managed by the service itself. They are hosted in an infrastructure provided by Niantic and can only be configured via the functionalities exposed in the SDK.

- **Shared Origin:** To create the functionality of a shared AR space, a host client first needs to create a room by providing an ID, configuration data and a point in space relative to the client that is going to be used as the origin point of the experience. This point can be obtained in different ways: image recognition, spatial features recognition, the relative position of the device or geospatial localisation. Upon connection with other clients, each instance of the shared space is coordinated through the network using the shared origin. If a new object is added to the interaction, its position is always relative to the shared origin and coordinated through the network.

- **Anchors:** As optional additions to the shared origin, the host or the clients can create (if the deployment platform has the capabilities) detected features in the real space as anchor points. An anchor can be used to provide visual reference points in the space to maintain a smooth AR experience, especially when a lot of movement and points of view changes are expected, or when the network connection is not stable. An anchor

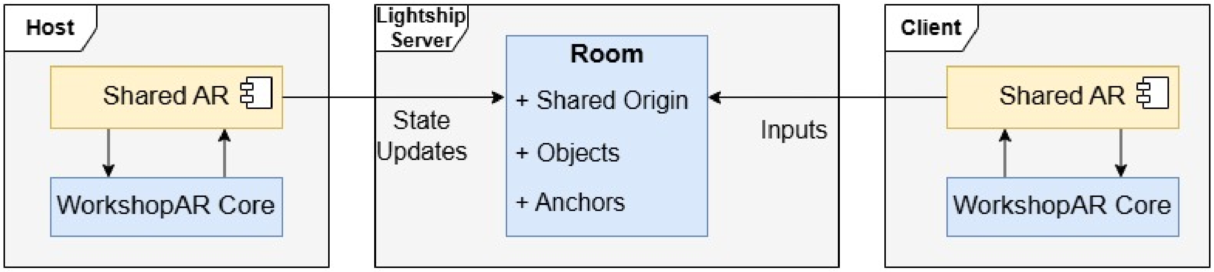

Figure 4.5: Integration with Lightship

is also shared through the network and helps in stabilising the room and the different points of view.

It is easy to integrate the main architectural points of Lightship into the architecture of *WorkshopAR*, mirroring the use of rooms to signify the connection of users to the group, the detection and analysis of features in the physical space and the coordination of data between interactions. The architecture proposed by Lightship also provides a simple and transparent networking component that allows a seamless connectivity on the side of the user with an easy process and minimal configuration, while also allowing the developer some control over the process. Figure 4.5 shows the architecture of the shared space as integrated with the core elements of *WorkshopAR*.

There is no necessity to go in deep detail about the general architecture of Unity as it is an extensive topic outside the scope of this document that offers little value to the research. Suffice to say that Unity encourages a modular component approach based around a core engine in charge of the coordination of resources and the communication with the local device. More information can be found in Messaoudi et al. (2015).

As for ARFoundation and Lightship, Figure 4.6 shows a relational model among their most important components, the core development of *WorkshopAR* and the external elements that interact with the frameworks in the execution of the experience.

4.3.3 Component Architecture

WorkshopAR uses two major components in its architecture: the *Session Manager*, in charge of the flow of activities in a single session for one participant, and the *Tracking Manager*, in charge of managing the AR interactions. Additional components are added to manage profile data and UI interactions. *WorkshopAR* relies on the Unity engine framework for deployment in mobile devices, on the ARFoundation package to communicate with the local implementation of AR and uses Lightship in coordination with some proprietary Netcode development to create the shared room and to coordinate interactions.

WorkshopAR follows the recommended architecture for general Unity projects, in which the experience follows the execution of one or more scenes which in turn are composed by a set of game objects. Each object has a specific behaviour determined by its interaction with the engine, other objects and the actions scripted by its modular components. Following this architecture, *WorkshopAR* creates a *work session* for the users and manages the tracking of all the AR elements through three main objects in a single scene:

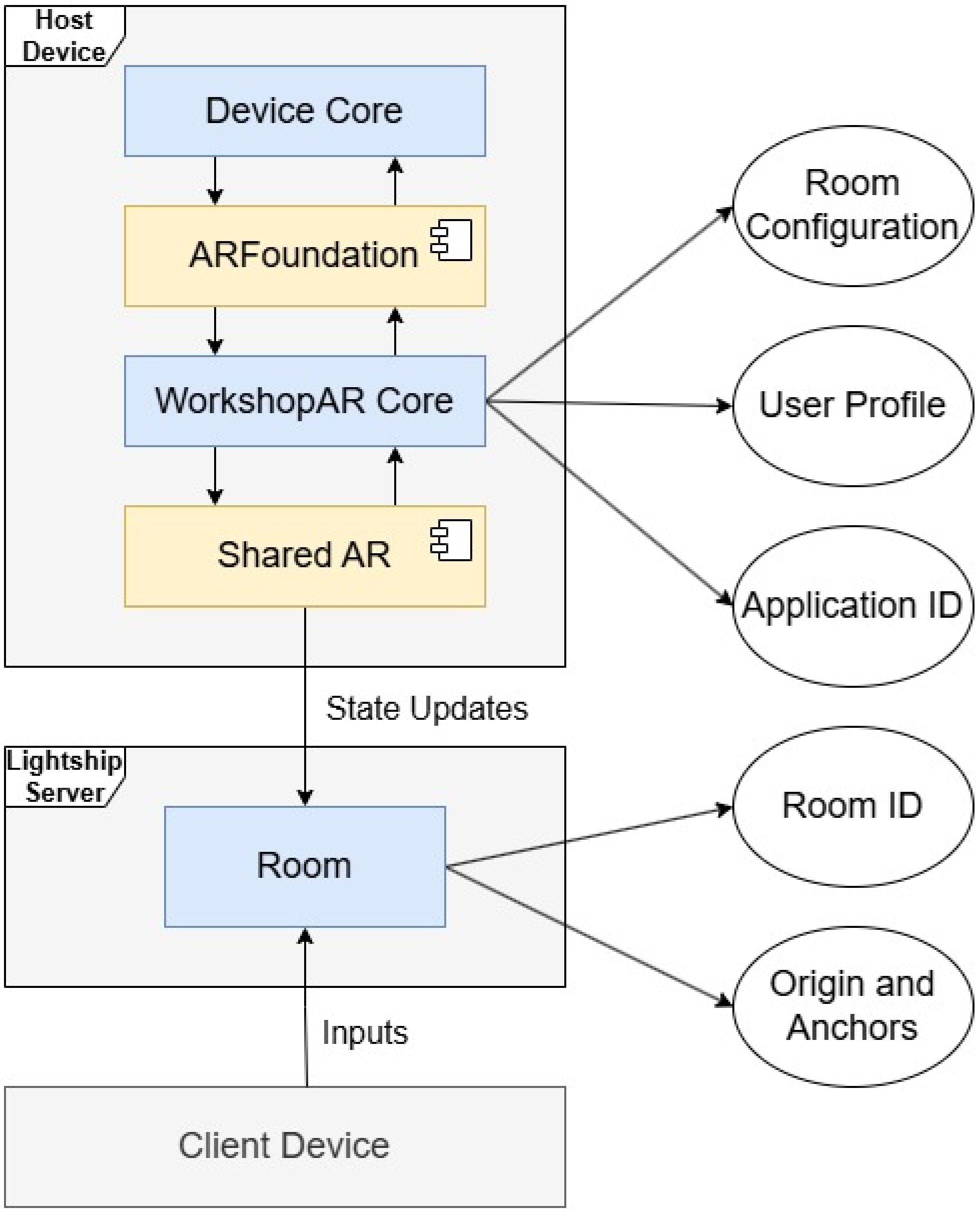

Figure 4.6: Relational Model with External Frameworks

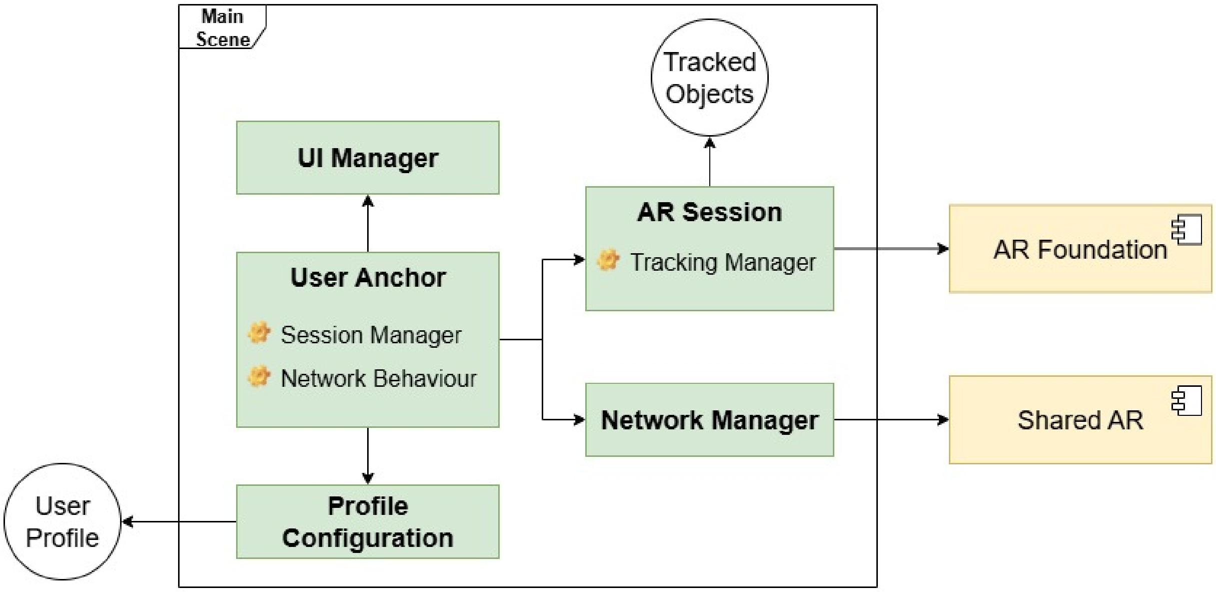

Figure 4.7: Unity Scene Relational Model

- The *User Anchor* acts as the visual representation of each participant in the *digital room* and the main control point of each individual action through the *Session Manager* component.
- The *AR Session* is part of the ARFoundation implementation and provides all the AR functionalities to integrate the scene with the camera image captured in the deployment device as well as the detection and management of features in the real space.
- The *Network Manager* helps with the implementation of Shared AR and manages all the Netcode used to create the shared experience.

Other objects are part to the scene to facilitate the visualisation of UI elements and to manage local data. Figure 4.7 shows a relational model of the previously described elements in the scene.

Session Manager

The architecture of *WorkshopAR* divides its functionality into two parts: the collaborative session and the AR tracking. The *Session Manager* is in charge of coordinating three elements of the experience: access to the local profile of the user, configuration of the shared room, and coordination of interactions with the room and with other participants through the network. Other less prominent components are used to provide networking functionalities not managed by Shared AR as well as the interactions with the UI. The component functionality can be overall described by the following process:

- Get the local user profile.
- Configure the connection to a room.

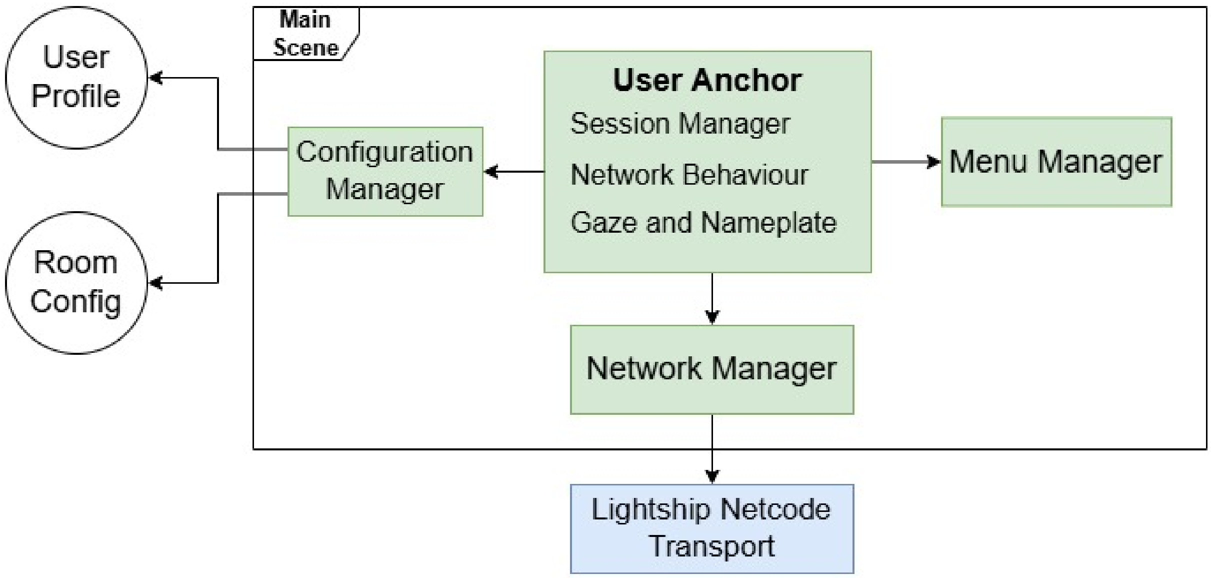

Figure 4.8: Session Manager Relational Model

- Coordinate network behaviours to fulfil the different interactions.
- Coordinate interaction calls through the network.
- Coordinate local interactions with the necessary data to show in the UI.

The *Session Manager* is also associated with the *User Anchor*. The Anchor is the visual and logical representation of the user in the room and provides different behaviours that show a visual representation of the user profile (position, name, current task, role) and social cues such as emotes and the user's gaze direction, all meant to be visual elements for the augmented collaboration design. The *Session Manager* coordinates the creation of *User Anchors* in the network, bridges local data between users and controls the flow of the session. Figure 4.8 shows a relational model of the most important components involved with the *Session Manager*.

Tracking Manager

The *Tracking Manager* is part of the AR Session game object in charge of controlling the visualisation of digital objects, their behaviour and life cycle. For purposes of the current prototype of *WorkshopAR*, the available digital objects are play models and annotations. Each object has a unique behaviour that will interact with the *Session Manager* or the *Tracking Manager* as needed.

The *Tracking Manager* is the bridge between *WorkshopAR* and the ARFoundation framework. It connects all the components needed to create an AR scene and configures the components that provide AR functionalities like image composition using the feed of the camera and plane detection in the real space. The manager is also associated with the XR Origin object, a component needed in the OpenXR framework to abstract the XR device in

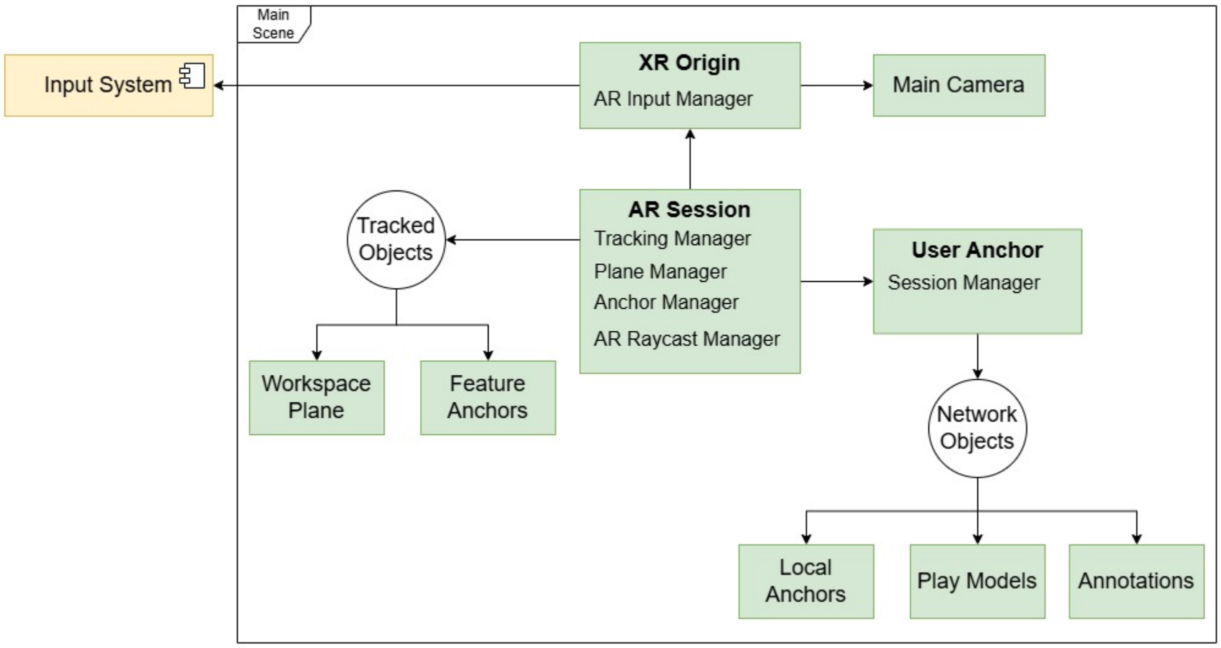

Figure 4.9: Tracking Manager Relational Model

which the app is going to be deployed. ARFoundation uses this component to communicate with the functionalities of the mobile device. In summary, the *Tracking Manager* is in charge of:

- Initializing the AR device.
- Managing the behaviour of the main camera.
- Managing the digital objects on the scene.
- Configuring the AR functionalities provided by ARFoundation.

The *Tracking Manager* specifically pulls the functionalities related to plane detection and the creation of visual anchors from the ARFoundation framework. The manager configures these components and keeps track of the game objects used for the visual composition of the scene with the information of the camera. A similar coordination is done with the *Session Manager*, which keeps track of the life cycle of the objects in the network and the data that needs to be coordinated between participants. Figure 4.9 shows a relational model of all the components related to the *Tracking Manager*.

Interaction Pipeline

The connection between the hardware layer of the device and the *digital room* created by *WorkshopAR* follows the architecture proposed by OpenXR. Figure 4.10 shows the interaction flow between components in case of a local interaction with a digital object, like a play model or an annotation.

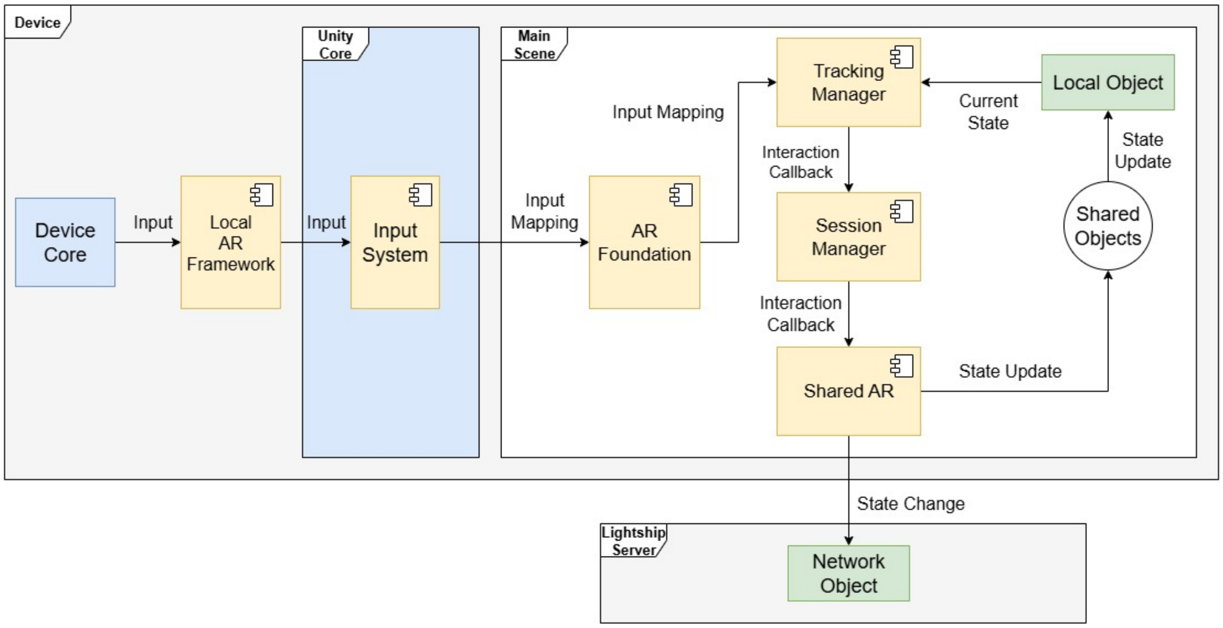

Figure 4.10: Device Interaction Flow

The local device is in charge of detecting interactions at a hardware level, most commonly inputs in the touch screen, the relative position of the device (accelerometer, gyroscope, etc.) and the image feed of the camera. All the input information is then related to the context of the AR application thanks to the local framework present in the device (ARCore or ARkit i.e.) which relays the raw data of the input to the Unity core engine. In here, the ARFoundation framework uses the Input System provided by Unity to read the raw information of the device and maps it to designated functions implemented by the *Tracking Manager*, which coordinates all the components, either from ARFoundation or from *WorkshopAR*, needed to respond to the input.

Independent of the components involved in the response, it is first necessary to determine if the interaction changes the state of a shared object. Any change to a shared object is handled by the *Session Manager*, who coordinates the remote network procedures that propagate the relevant information to all the users, and reacts locally when a change is received from other user. The *Session Manager* is in charge of updating the state of all the shared objects, while the *Tracking Manager* only reads the information for visualisation purposes and to manage interactions.

SharedAR supports both server-authoritative and client-authoritative network architectures, as well as the possibility to have dedicated servers or host/client dynamics. *WorkshopAR* follows a host/client architecture with no dedicated server controlling the experience. The users must decide who is going to act as the host. Then, the selected user needs to configure the *digital room* and open it, while the other users connect as clients. If the initial host leaves the *digital room*, it is easy to pass the host authority to any other connected user.

In terms of authority architecture, the host holds server authority for the configuration of the *digital room*, state of the *digital room*, the detected physical plane used as the shared *workspace*, and the physical features detected as visual anchors. Meanwhile, the *User Anchors*,

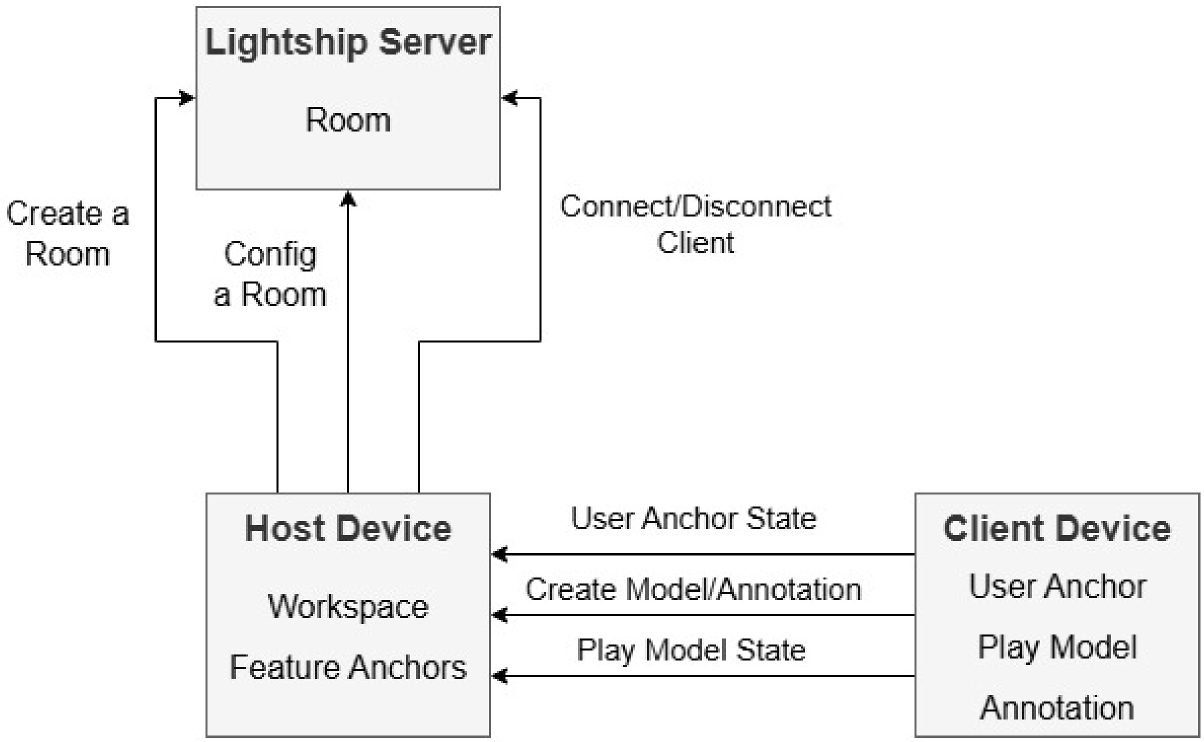

Figure 4.11: Network Authority Architecture

the play models and the annotations are client-authoritative, with ownership given to the user that initially spawned the object in the network.

A final consideration is that, despite being designed as a serverless architecture, *WorkshopAR* needs to follow the architecture enforced by SharedAR, which works as a central server that hosts the shared room and manage the connections of the clients to it. This element is outside the architecture of *WorkshopAR*, and for the end user, they just need to worry about the connection with the host. Figure 4.11 shows a diagram that represents the main components in the network architecture, the objects over which each component holds authority and the most important procedures being transmitted over the network.

In the case that an interaction does not involve digital objects, the pipeline maintains the same architecture. *WorkshopAR* relies on the services provided by ARFoundation for the detection of features using the camera present in the mobile device, as it is shown in Figure 4.4. The specific subsystems used by the prototype are the plane detector, the anchor manager and the AR raycast functionalities.

The plane detector is used to identify flat surfaces in the physical space, one of the steps in the configuration of a *digital room*. Similarly, the position of a digital object can be used to create an anchor in the digital space, which helps improve the quality and stability of the AR visualisation. A new anchor is created in the room every time a new user connects, and these anchors are managed by the host's *Session Manager*. The AR raycaster is needed to interact with digital objects in the AR space or with features in the AR space created by other subsystems. This is a local operation, so it does not need to be propagated in the

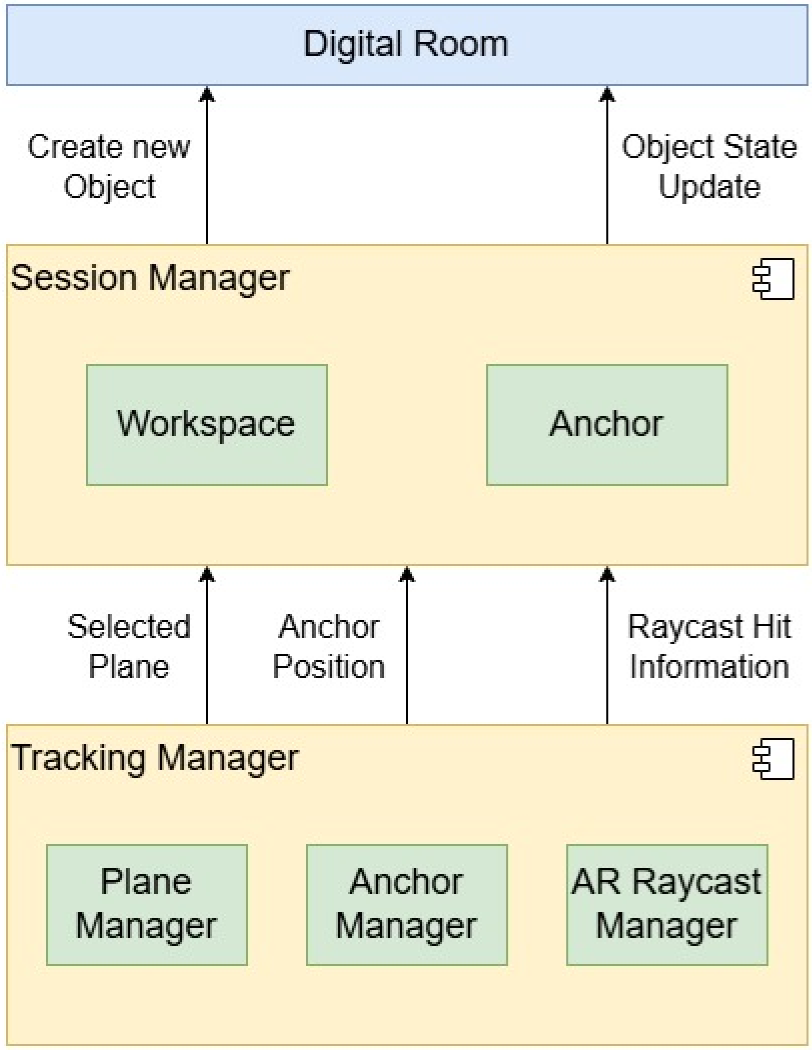

Figure 4.12: Interaction Pipeline with ARFoundation Subsystems

network. Figure 4.12 shows the interactions of the previously described components in the network architecture of *WorkshopAR*.

4.3.4 Data Architecture

Appendix A

Pedagogical Goals for Educational Technologies using Augmented Reality

A SLR was performed as part of the literature review process guiding this research project, and the results of the codification phase of the review can be found in the following repository:

https://github.com/speregil/ar-survey-annex.git

This SLR codified a total of 67 different articles related to the use of AR in educational settings. The analysis phase of the review focused on mapping what educational goals were more prevalent in the design proposed in each surveyed project. This information was codified using the BRT framework to understand the cognitive complexity of the goals identified, while the goals themselves were codified using the verb + statement approach proposed by Fuchs and Deno (1982).

The bibliographical information of all the surveyed articles was codified using the following structure shown in Table A.1.

Table A.1: Structure of the Coded Data

Field	**Description**
ID	Unique ID for the entry
Main Author	Last name of the main (first) author of the research
Year	Year of publication
Journal/Proceedings	Name of the journal or conference in which the article was published
Author's Discipline	Main subject affiliation of all the authors of the article as best as it could be extracted by the coders. The reported affiliation to universities and faculties and simple web searches were used to determine such subject affiliation

Continued on next page

Table A.1: Structure of the Coded Data (Continued)

Target Demographics	Intended target demography or educational level of the project or tool. Demographics were divided in primary school (6 - 12 yrs. approx.), secondary school (13 - 18 yrs. approx.), university education and vocational training
Specific Demographics	Specifics about the subjects using for the research as test or final users
Target Field	Intended field of knowledge in which the research or tool works
AR Technology	The main AR engine used to create the tool
Additional Technologies	Other notable technologies mentioned in the research
Custom Build	If the tool was built from zero, tailored for the research, or if some third-party tool was used or adapted to the needs of the research
Goal Statement	Educational goal guiding the research as stated by the article or inferred by the researchers. The goal was coded in the + format
Explicit Goal	If the goal was explicitly stated in the research or if the coders had to infer and formulate a goal based on the information provided by the research
Success	If a project could be categorized as a success or not based on the criteria stated in the systematic review, and as best judged by the coders
Comments	Notable comments of the coders about the research
Knowledge Dimension	Classification given by the coders in the knowledge dimension as stated in the systematic review
Cognitive Dimension	Classification given by the coders in the cognitive dimension as stated in the systematic review
Positive Experiences	List of all the coded positive experiences extracted from the research and using the Positive Experiences Code Book
Negative Experiences	List of all the coded negative experiences extracted from the research and using the Negative Experiences Code Book

The fields *Knowledge Dimension* and *Cognitive Dimension* use the BRT to classify the identified educational goals. Data was coded using the keys shown in Table A.2.

Finally, the *Positive* and Negative Outcomes fields codify the main themes and categories

Table A.2: Coding Keys for the *Knowledge* and emphCognitive Dimensions of the BRT

BRT Dimension	Key	Value	Description
Knowledge	A	Factual	Knowledge of simple facts without context or immediate application
	B	Conceptual	Knowledge of complex concepts, relationships between facts and the context around them
	C	Procedural	Knowledge on the components and execution of a tasks, the reasoning behind them and how they affect their context
	D	Metacognitive	Knowledge about learning, how to learn and your own learning abilities, limitations and approaches
Cognitive	1	Remember	Goals associated with correctly and consistently remembering facts and information
	2	Understand	Goals associated with recognizing concepts, relationships and compositions
	3	Apply	Goals associated with correctly using previous knowledge in a concrete context or scenario
	4	Analyse	Goals associated with inferring and extracting new knowledge from other concepts, context or the relationships between them
	5	Evaluate	Goals associated with giving a value judgement based on knowledge and analysis
	6	Create	Goals associated with creating new elements, procedures or knowledge

found on the recount of experiences extracted from the results, analysis and conclusions of the reviewed works. The categorisation process compiled and used the codebooks shown in table A.3 and table A.4.

Table A.3: Codebook for Positive Outcomes

Category	Key	Value	Description
Student Autonomy	1.1	Facilitates student autonomy	AR makes easy for students to work in an autonomous way.
	1.2	Offers student-controlled pace of the experience	The tool advances and can be used at different paces, controlled by the desired rhythm of the user.
	1.3	Offers a personalized experience	The tool changes content or presentation elements depending on the usage of the student
	1.4	Supports different types of learning methods	The tool presents the same content or activity in different ways to foster different types of learning, like visual, memory or other similar approaches

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Table A.3: Codebook for Positive Outcomes (Continued)

Accessibility	2.1	Easy access to the technology	Getting access to devices need to use the tool is easy
	2.2	Use of available technology	The tool uses available technology for the students, like their own smartphones or simple tech lab's devices
	2.3	Easy to learn	It is easy for the users to learn how to use the tool
	2.4	Unexpensive or easy implementation	The cost of building and deploying the tool is low or lower in comparison to other technologies
	2.5	Easy or intuitive interaction	The tool is easy and enjoyable to use

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Table A.3: Codebook for Positive Outcomes (Continued)

Performance	3.1	Improved academical performance is tested	Students gain academical performance using the tool and it was tested in the research
	3.2	Facilitates concentration/attention	Students or researchers reported improvements gain in concentration or attention
	3.3	Facilitates practice/training	The tool helped students to practice or train
	3.4	Improved long-term retention is tested	Students gain improved retention using the tool and it was tested in the research
	3.5	Boosts confidence in learning outcomes	Students report feeling more confident in what they learned and the long-term retention of it
	3.6	Faster/easier learning	Learning was achieved faster or students report doing it easier in comparison to other experiences
	3.7	Enhanced critical thinking is tested	Researchers tested gains in critical thinking
	3.8	Balanced learning outcomes for different students	Results for students at different levels were balanced or the gaps were minimized

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Table A.3: Codebook for Positive Outcomes (Continued)

Immersion and Motivation	4.1	Higher motivation for the learning subject/activity	Students report to be more motivated or invested in the learning subject
	4.2	Facilitates emotional investment in the subject/activity	Facilitates emotional investment in the subject/activity: Students report to be more invested with the activity, the tool or its content
	4.3	Content used encourages learning	Students feel motivated to engage with the content of the tool
	4.4	Fun or interesting tool	The tool or its content is reported to be fun for the students or presents an interesting approach to the learning material
Digital Literacy	5.1	Helps in digital literacy education	The tool can be used to educate in the use of digital devices
Visualization	6.1	Helps in the visualization of abstract/difficult concepts	The tool provides visual content to explain abstract or difficult subjects.

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Table A.3: Codebook for Positive Outcomes (Continued)

Tool Validation	7.1	Comparable or better than other VR/AR apps	Research shows better results in comparison to other tools or technologies
	7.2	General satisfaction of the subjects with the app	Users report high satisfaction with the tool
	7.3	Better tool to accomplish an objective in comparison to others	Better results in accomplishing a particular goal in comparison with similar tools
	7.4	Tool achieved some form of professional validation	The tool was evaluated and validated by professionals in the field
Multimedia Learning	8.1	Benefits of multimedia learning are noted	The use of multimedia content in the tool greatly contributes to its success
	8.2	Sets itself as an alternative tool for learning	Students see the tool as a good alternative to achieve their learning goals
	8.3	Is an alternative or better way to present a subject	The content used in the tool does a better job to present the learning subject
	8.4	Improvement when paired with other media or tool	The tool is better used paired with other content or tool
Cognitive Load	9.1	Reduce stress or cognitive load	Cognitive load on students was tested or reported to be reduced

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Table A.3: Codebook for Positive Outcomes (Continued)

Collaboration	10.1	Facilitates collaboration or socialization	The tool is used in collaborative scenarios

Table A.4: Codebook for Negative Outcomes

Category	**Key**	**Value**	**Description**
None reported	1.0	None reported	No negative outcome was reported

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Table A.4: Codebook for Negative Outcomes (Continued)

Technology Issues	2.1	General technology issues	Reported failures, problems or inconveniences related to technology usage or setup, mainly of the platform or the AR tool itself.
	2.2	Occlusion	The AR image recognition is occluded by other objects or the users themselves
	2.3	Illumination	Poor or excessive illumination
	2.4	Frame rate and computational power	The selected platform struggles to process or run the tool, creating frame rate issues or slowing down the experience
	2.5	Impractical markers or image recognition problems	The selected markers for image recognition are inconvenient, constants tracking lost or similar issues
	2.6	Connectivity problems	Connection was unstable, lost or the necessity for a connection was an inconvenience for the usage of the tool
	2.7	Budget/cost issues	The cost for creating or using the tool were too high or impossible to scale up
	2.8	Accuracy problems for geolocalization	AR is based on geolocalization, and it lacked precision or was to erratic

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Table A.4: Codebook for Negative Outcomes (Continued)

Pedagogical Design Issues	3.1	General pedagogical design issues	Issues related to the pedagogical design, the educational goals, the content presented or the activities around AR
	3.2	AR is not integrated well with other activities	The use of the AR is considered a distraction or harmful to the other elements of the activity
	3.3	AR is not correctly used as learning tool	The use of AR in the proposed project is minimal or is not related to learning
	3.4	Pacing and time consumption	Set up or usage of the AR tool consumes too much time and harms other activities
	3.5	Poor scalability of the activity	It is hard or non-viable to use the proposed tool with more users or in a more realistic scenario
Cognitive Issues	4.1	Easy to lose attention or AR takes attention from learning	The AR elements of the tool are distracting and take the attention from the learning core of the tool
	4.2	Increased Cognitive Load for students	Researchers reported an increase in the cognitive load of the students

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Table A.4: Codebook for Negative Outcomes (Continued)

Scope Issues	5.1	Scope of design too big for technology limitations	The technology used did not suited the pedagogical design
Isolation	6.1	AR promotes isolation	The use of AR prompts the students to isolate themselves and not pay attention to their peers or their context
Performance Issues	7.1	Lower or not significantly different academical results	Tests done by the research showed no academical performance increase or it was not statistically significant
	7.2	Results are not consistent between different types of users	Results show there is a significant difference in results between different types of users
Motivation Issues	8.1	No interest or motivation in the tool	Users reported or observed to have no interest in using the tool
	8.2	Novelty of the tool ended quickly	Initial interest in the tool lost quickly, before the activity ended or in a subsequent session
	8.3	Other tools are more reliable/interesting	Users found other tools more interesting or useful than the proposed tool

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Table A.4: Codebook for Negative Outcomes (Continued)

Harmful Effects	9.1	The tool harmed the development of other skills	The use of the tool had an undesired negative effect in the development or acquisition of other skills
Content Issues	10.1	Insufficient or lack-lustre content	Content created for the tool was reported as low quality by students or they would wish for more or different content
	10.2	Hard to create content for the tool	Difficulties in budget, time or resources to create or procure suitable content
Usability Issues	11.1	Hard to use or requires too much training	Hard to learn and/or use, or using the tool requires too much preparation and proper training
	11.2	Physical exhaustion	Using the tool is exhausting
	11.3	Limited access to technology	The platform chosen for deploying the AR tool is hard or expensive to come by
	11.4	Difficulties with multiple users or different types of users	The tool was not well suited or received by a percentage of the user population

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Table A.4: Codebook for Negative Outcomes (Continued)

Research Design Issues	12.1	Research design has problems	Some elements of the research designs are weird or harm the overall exposition of the tool in some way
	12.2	The AR tool used is not clear	The AR technology used is not clear
	12.3	Data presented by the study weas difficult to interpret	The research presented data in a format hard to interpret or that obscured aspects of the research
	12.4	Data presented contradicts results or presents some kind of problem	The data presented contradicts statements given in the discussions or conclusions sections
	12.5	Specific goals of the research were not met	The main hypothesis of the research was proven false or the educational goals stated for the tool were not met

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